

Lecture Note for Engineering Mathematics:

Probability and Statistics

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Preface

This lecture note is specially designed for undergraduate students majoring in science and engineering, and is tailored to meet the needs of English-taught or bilingual Probability and Mathematical Statistics courses. Aligned with the talent development goals of science and engineering programs, it maintains rigor in theoretical knowledge while emphasizing practical applications. We aim to help students establish a systematic knowledge framework of probability and mathematical statistics, master core concepts, theories, methods and practical skills, and meanwhile improve their ability to read, understand and apply academic English. This will lay a solid foundation for their future academic studies, research and engineering practice.

This lecture note is developed on the basis of years of teaching experience and practice in English-taught and bilingual Probability and Mathematical Statistics courses at the GuangZhou International Campus, South China University of Technology. The authorship is as follows:

- Chapters 1 and 6 were written by Ling Liming;
- Chapters 2 and 3 were written by Wu Wen;
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- Chapter 5 was written by Lin Yu.

Drawing on their teaching experience and research expertise, each author has carefully organized and clearly explained the content, striving to present solid materials with clear logic, distinct key points and difficulties, and a structure that fits the learning pace and cognitive characteristics of undergraduate science and engineering students. Probability and mathematical statistics is a discipline with both profound theory and wide applications, involving numerous knowledge points and a rigorous logical system. Given the special requirements of English-taught and bilingual education, this lecture note places higher demands on professionalism, accuracy and readability. Due to the limited knowledge, teaching

experience and research horizons of the authors, omissions and imperfections are inevitable in the manuscript. We sincerely welcome comments and suggestions from fellow researchers, instructors and students. We will continue to revise and improve the lecture note to provide a more practical teaching reference. We are deeply grateful for your understanding and support.

Chapter 1

Probability

1.1 Sample spaces and events

An experiment is a process whose outcome is uncertain. Although the word *experiment* generally suggests a planned or carefully controlled laboratory testing situation, we use it here in a much wider sense.

1.1.1 The sample space of an experiment

Definition 1.1.1. The **sample (or probability) space** of an experiment, denoted by $\mathcal{S}$, is the set of all possible outcomes of that experiment.

Example 1.1.1 (Tossing coins). The simplest experiment is to toss a coin once. The coin lands on either one or the other of its two sides. One is usually called head, the other tail. The sample space for this experiment can be denoted as $\mathcal{S} = \{H, T\}$, where H represents head and T represents tail.

Example 1.1.2 (Tossing a coin twice). If the experiment is tossing a coin twice, there are four distinct outcomes: (H, H) , (H, T) , (T, H) , (T, T) , which are often presented in a more concise form HH , HT , TH , TT . Thus, for this experiment $\mathcal{S} = \{HH, HT, TH, TT\}$.

Example 1.1.3 (Rolling a dice). A common dice is a small cube whose faces show numbers 1, 2, 3, 4, 5, 6 one way or another. These may be the real digits or arrangements of an appropriate number of dots. There are six possible outcomes and the sample space consists of six elements: $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$.

Example 1.1.4 (Infinite sample space). Consider the experiment of repeatedly tossing a coin until the first tail shows up. Possible outcomes are sequences of H

that, if finite, end with a single T , and an infinite sequence of H :

$$\mathcal{S} = \{T, HT, HHT, HHHT, \dots, HHH \dots\}$$

1.1.2 Events

Definition 1.1.2. An **event** is any subset of outcomes contained in the sample space $\mathcal{S}$. An event is said to be **simple** if it consists of exactly one outcome and compound if it consists of more than one outcome.

When an experiment is performed, an event is said to occur if it contains the actual outcome of that experiment.

Example 1.1.5. The sample space for rolling a dice is $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$. Let $B = \{3, 4, 5, 6\}$ and $A = \{2, 4, 6\}$. Here, A is the event of getting an even number, and B is the event of getting a number bigger than 2. Both A and B are compound events. If we roll a die and get a number 2, we can say the event A occurs.

Two special events

The whole set $\mathcal{S}$ contains all possible outcomes of an experiment and thus it is a compound event that would take place certainly. The empty set $\emptyset$ is an event that never occurs. We call it the null event. Since an event is a set, we can use set theory to study events.

Definition 1.1.3 (Events' operations and relationship). 1 The union of two events A and B , denoted by $A \cup B$ and read “ A or B ”, is the event consisting of all outcomes in A and B . So the event $A \cup B$ takes place whenever at least one of A and B occurs.

2 The intersection of two events A and B , denoted by $A \cap B$ and read “ A and B ”, is the event consisting of all outcomes that are both in A and B . So $A \cap B$ is the event that takes place only when both of A and B do.

3 The complement of an event A , denoted by A' (read “ A prime” or “ A complement”), is the set of all outcomes in $\mathcal{S}$ that are not in A . Obviously, $A \cap A' = \emptyset$ and $A \cup A' = \mathcal{S}$.

4 The difference $A - B$ is an event consisting of all outcomes that are in A but not in B . Clearly, $A - B = A \cap B'$.

Example 1.1.6. The sample space for rolling a dice is $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$. Let $A = \{2, 4, 6\}$, $B = \{4, 5, 6\}$, and $C = \{1, 3\}$. Then we have

$$A \cup B = \{2, 4, 5, 6\}, \quad A \cap B = \{4, 6\}$$

$$\begin{aligned} A' &= \{1, 3, 5\}, & (A \cup B)' &= \{1, 3\} \\ A \cup C &= \{1, 2, 3, 4, 6\}, & A \cap C &= \emptyset \end{aligned}$$

When A and C have no outcomes in common, they are said to be **mutually exclusive or disjoint events**. In other words, if $A \cap C = \emptyset$, then they are mutually exclusive. Obviously, A and A' are mutually exclusive.

In mathematics, events A and B are mutually exclusive if $A \cap B = \emptyset$. B is the complement of event A if $A \cap B = \emptyset$ and $A \cup B = \mathcal{S}$.

The operations of union and intersection can be extended to more than two events.

$$\begin{aligned} A \cup B \cup C &= \{1, 2, 3, 4, 5, 6\} \\ A \cap B \cap C &= \emptyset \end{aligned}$$

Properties of events's operations

1 Commutative laws

$$E \cup F = F \cup E, \quad E \cap F = F \cap E$$

2 Associative laws

$$(E \cup F) \cup G = E \cup (F \cup G), \quad (E \cap F) \cap G = E \cap (F \cap G)$$

3 Distributive laws

$$(A \cap B) \cup C = (A \cup C) \cap (B \cup C), \quad (A \cup B) \cap C = (A \cap C) \cup (B \cap C)$$

4 De Morgan's laws

$$(A \cup B)' = A' \cap B', \quad (A \cap B)' = A' \cup B'$$

Example 1.1.7. Let E , F , and G be three events. Find expressions for the events so that:

1. Only E occurs: $E \cap F' \cap G'$
2. Both E and G , but not F occur: $E \cap G \cap F'$
3. At least one of the events occurs: $E \cup F \cup G$
4. At least two of the events occur: $(E \cap F) \cup (E \cap G) \cup (F \cap G)$
5. All three events occur: $E \cap F \cap G$

6. None of the events occur: $E' \cap F' \cap G'$
7. At most one of the events occur: $(E' \cap F' \cap G) \cup (E' \cap F \cap G') \cup (E \cap F' \cap G') \cup (E' \cap F' \cap G')$
8. At most two of the events occur: $E' \cup F' \cup G' = (E \cap F \cap G)'$
9. Exactly two of the events occur: $(E \cap F \cap G') \cup (E \cap F' \cap G) \cup (E' \cap F \cap G)$

1.2 Axioms, Interpretations, and Properties of Probability

Given an experiment and a sample space $\mathcal{S}$, the objective of probability is to assign to each event A a real value $P(A)$, called the probability of the event A , which will give a precise measure of the chance that A will occur.

In other words, a probability is a real-value function P defined on the subsets of $\mathcal{S}$. To ensure that the probability assignments are consistent with intuitive notions of probability, it satisfies the following 3 axioms of probability:

Axiom 1 (Nonnegativity). *For any event A , $P(A) \geq 0$. It reflects the intuition that the chance of each event A occurring should be at least 0.*

Axiom 2 (Normalization). *$P(\mathcal{S}) = 1$. $\mathcal{S}$ contains all possible outcomes, so it is sure to take place and should have probability 1.*

Axiom 3 (Additivity). *If $A_1, A_2, A_3, \dots$ is an infinite collection of mutually exclusive events, then:*

$$P(A_1 \cup A_2 \cup A_3 \cup \dots) = \sum_{i=1}^{\infty} P(A_i)$$

The probability function is required to be additive.

A probability is a real-value function P defined on the subsets of $\mathcal{S}$ that satisfies the 3 probability axioms. Based on these 3 axioms, we can prove many useful properties of probability.

Question: How to determine the probability assigned to an event?

Example 1.2.1. In the experiment of tossing a coin (or any object with two distinct sides), $\mathcal{S} = \{H, T\}$. What is $P(H)$?

From the axioms:

$$P(\mathcal{S}) = 1, \quad P(\mathcal{S}) = P(H \cup T) = P(H) + P(T)$$

The axioms imply $P(T) = 1 - P(H)$, but they do not uniquely determine $P(H)$.

Probability and Frequency

If we toss a fair coin n times, H will occur in some replications and not in others. For a large number of tosses, H appears in approximately half the tosses. Intuitively, we define the chance of H occurring as $\frac{1}{2}$, so $P(H) = 0.5$.

Generally, the probability of an event A is defined by its frequency in a large number of experiments:

- Let n be the number of experiment replications.
- Let $n(A)$ denote the number of replications where A occurs.
- The *relative frequency* of A is $\frac{n(A)}{n}$.

As n becomes arbitrarily large, $\frac{n(A)}{n}$ approaches a limiting value called the *limiting (or long-run) relative frequency* of A . The *objective interpretation of probability* identifies this limit with $P(A)$ (based on the Law of Large Numbers, to be shown in the following chapter).

Example 1.2.2. Let A be the event that a package sent within California for 2nd-day delivery arrives within 1 day. The results of sending 10 such packages are shown in the table below:

Package Number	Did A occur?	Relative Frequency of A
1	N	0
2	Y	0.5
3	Y	0.667
4	Y	0.75
5	N	0.6
6	N	0.5
7	Y	0.571
8	Y	0.625
9	N	0.556
10	N	0.5

The relative frequency of A as the number of packages increases is visualized in Figure 1.1:

Properties of Probability (Deduced from the 3 Axioms)

Property 1.1. $P(\emptyset) = 0$, where $\emptyset$ is the null event.

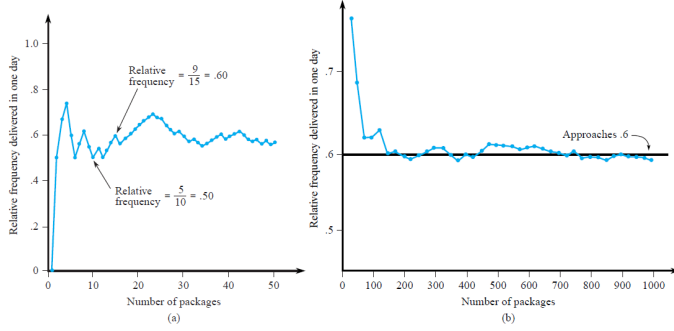


Figure 1.1: Relative frequency of event A vs. number of packages: (a) 0-50 packages; (b) 0-1000 packages

Proof. Let $A_1 = \emptyset, A_2 = \emptyset, A_3 = \emptyset, \dots$. Since $\emptyset \cap \emptyset = \emptyset$, these events are mutually exclusive. Moreover, $\bigcup_{i=1}^{\infty} A_i = \emptyset$. By Axiom 3:

$$P(\emptyset) = P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) = \sum_{i=1}^{\infty} P(\emptyset)$$

The only real number satisfying $x = \sum_{i=1}^{\infty} x$ is $x = 0$, so $P(\emptyset) = 0$. $\square$

Property 1.2. If $A_1, A_2, \dots, A_k$ is a finite collection of mutually exclusive events, then:

$$P(A_1 \cup A_2 \cup \dots \cup A_k) = \sum_{i=1}^k P(A_i)$$

Proof. Define an infinite sequence of events: $A_{k+1} = \emptyset, A_{k+2} = \emptyset, \dots$. Since $P(\emptyset) = 0$ (Property 1.1) and $A_i \cap A_j = \emptyset$ for $i \neq j$, by Axiom 3:

$$P\left(\bigcup_{i=1}^k A_i\right) = P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) = \sum_{i=1}^k P(A_i) + \sum_{i=k+1}^{\infty} P(\emptyset) = \sum_{i=1}^k P(A_i)$$

$\square$

Property 1.3. For any event A , $P(A) = 1 - P(A')$ (where A' denotes the complement of A).

Proof. Since $A \cup A' = S$ (all outcomes are in A or its complement) and $A \cap A' = \emptyset$ (mutually exclusive), by Axiom 2 and Property 1.2:

$$1 = P(S) = P(A \cup A') = P(A) + P(A')$$

Rearranging gives $P(A) = 1 - P(A')$. $\square$

Corollary: $P(A) \leq 1$ for any event A (follows directly from $P(A') \geq 0$).

Usefulness: If calculating $P(A)$ directly is difficult, compute $P(A')$ first and subtract from 1.

Example 1.2.3. Consider tossing a fair coin 5 times. Let A be the event that “at least one head appears”. Find $P(A)$.

Solution. - The sample space has $2^5 = 32$ outcomes, so A contains $32 - 1 = 31$ outcomes (all except the all-tails outcome).

- The complement $A' = \{TTTTT\}$ (only one outcome).

- For a fair coin, $P(A') = 0.5^5 = 0.03125$.

- By Property 1.3, we have

$$P(A) = 1 - P(A') = 1 - 0.03125 = 0.96875 \approx 0.969$$

$\square$

Property 1.4. If A and B are mutually exclusive, then $P(A \cap B) = 0$.

Proof 1. Mutually exclusive events satisfy $A \cap B = \emptyset$. By Property 1.1, $P(A \cap B) = P(\emptyset) = 0$. $\square$

Proof 2. Since $A \cap B = \emptyset$, its complement $(A \cap B)' = S$. By Property 1.3:

$$P(A \cap B) = 1 - P((A \cap B)') = 1 - P(S) = 1 - 1 = 0$$

$\square$

Property 1.5. For any two events A and B (not necessarily mutually exclusive):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Remark: This holds for mutually exclusive events (since $P(A \cap B) = 0$, reducing to Property 1.2).

Proof. From the Venn diagram, $A \cup B = A \cup (B \cap A')$, and A and $B \cap A'$ are mutually exclusive. By Property 1.2:

$$P(A \cup B) = P(A) + P(B \cap A') \tag{1-1}$$

Additionally, $B = (B \cap A) \cup (B \cap A')$, and $B \cap A$ and $B \cap A'$ are mutually exclusive. By Property 1.2:

$$P(B) = P(B \cap A) + P(B \cap A') \implies P(B \cap A') = P(B) - P(A \cap B) \quad (1-2)$$

Substitute (1-2) into (1-1), we get

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

□

Example 1.2.4. Randomly select a student from the class. Let:

- A : "the student can play basketball" ($P(A) = 0.6$),
- B : "the student can play football" ($P(B) = 0.35$),
- $P(A \cap B) = 0.2$.

Find

- a. The probability that the student can play at least one sport.
- b. The probability that the student can play neither sport.

Solution. a. "At least one sport" = $A \cup B$. By Property 1.5, we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.6 + 0.35 - 0.2 = 0.75$$

- b. "Neither sport" = $A' \cap B'$. By De Morgan's Law, $A' \cap B' = (A \cup B)'$. By Property 1.3, we have

$$P((A \cup B)') = 1 - P(A \cup B) = 1 - 0.75 = 0.25$$

□

Determining Probabilities Systematically

Suppose the sample space is $\mathcal{S} = \{E_1, E_2, \dots, E_N\}$ (finite number of simple events). To find probabilities of compound events:

- 1 First assign probabilities to simple events: $P(E_i) \geq 0$ and $\sum_{i=1}^N P(E_i) = 1$.
- 2 For any compound event A , sum the probabilities of simple events in A :

$$P(A) = \sum_{\text{all } E_i \in A} P(E_i)$$

Example 1.2.5. Rolling a single die, $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$. Suppose even outcomes are twice as likely as odd outcomes. Find:

- a. $P(E_i)$ for each E_i ,
- b. $P(A)$ where $A = \{\text{outcome is even}\}$,
- c. $P(B)$ where $B = \{\text{outcome} \leq 3\}$.

Solution. a. Let $P(1) = P(3) = P(5) = p$ (probability of odd outcomes). Then $P(2) = P(4) = P(6) = 2p$ (even outcomes). Since $\sum P(E_i) = 1$, we have

$$3p + 3(2p) = 1 \implies 9p = 1 \implies p = \frac{1}{9}$$

Thus:

$$P(1) = P(3) = P(5) = \frac{1}{9}, \quad P(2) = P(4) = P(6) = \frac{2}{9}$$

- b. $A = \{2, 4, 6\}$, so:

$$P(A) = P(2) + P(4) + P(6) = \frac{2}{9} + \frac{2}{9} + \frac{2}{9} = \frac{6}{9} = \frac{2}{3}$$

- c. $B = \{1, 2, 3\}$, so:

$$P(B) = P(1) + P(2) + P(3) = \frac{1}{9} + \frac{2}{9} + \frac{1}{9} = \frac{4}{9}$$

□

Special Case: Equally Likely Outcomes

If all outcomes in $\mathcal{S} = \{E_1, \dots, E_N\}$ are equally likely, let $P(E_i) = p$ for all i . By Axiom 2:

$$\sum_{i=1}^N P(E_i) = Np = 1 \implies p = \frac{1}{N}$$

For any event A with $N(A)$ outcomes (size of A), the probability is:

$$P(A) = \sum_{E_i \in A} \frac{1}{N} = \frac{N(A)}{N}$$

Key Insight: When outcomes are equally likely, computing probabilities reduces to *counting* the number of favorable outcomes ($N(A)$) and total outcomes (N).

Example 1.2.6. Roll two fair dice (one after the other). Let

$$A = \{\text{sum of the two numbers} = 7\}.$$

Find $P(A)$.

Solution. - Total outcomes: Each die has 6 outcomes, so $N = 6 \times 6 = 36$ (all equally likely).

- Favorable outcomes (sum = 7): $A = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$, so $N(A) = 6$.

- Probability:

$$P(A) = \frac{N(A)}{N} = \frac{6}{36} = \frac{1}{6}$$

□

1.3 Counting Techniques

When the number of outcomes in the sample space is large, exploiting general counting rules is critical for calculating probabilities. This section introduces core counting methods: product rules, permutations, combinations, and geometric probability.

The Product Rule for Ordered Pairs and k -tuples

The product rule applies to counting ordered collections (pairs, triples, or k -tuples) where each element's selection is independent of previous choices (up to dependence on prior selections).

Proposition 1.3.1 (Product Rule for Ordered Pairs). *If the first element of an ordered pair can be selected in n_1 ways, and for each of these n_1 ways the second element can be selected in n_2 ways, then the total number of ordered pairs is $n_1 \cdot n_2$.*

Example 1.3.1 (Rolling Two Dice). Consider rolling two dice sequentially. Each outcome is an ordered pair (d_1, d_2) , where:

- d_1 (first die) has 6 choices,
- d_2 (second die) has 6 choices for any d_1 .

Total outcomes: $6 \times 6 = 36$.

Let $A = \{\text{second die shows an odd number}\}$. For d_2 , there are 3 odd choices (1, 3, 5), so:

$$N(A) = 6 \times 3 = 18, \quad P(A) = \frac{18}{36} = 0.5$$

Proposition 1.3.2 (General Product Rule for k -tuples). *A k -tuple is an ordered collection of k objects. If:*

- *The 1st element has n_1 choices,*
- *The 2nd element has n_2 choices (for any 1st element),*
- *...*
- *The k -th element has n_k choices (for any $(k-1)$ -th element), then the total number of k -tuples is $n_1 \cdot n_2 \cdot \dots \cdot n_k$.*

Example 1.3.2 (Rolling a Die k Times). Each outcome is a k -tuple $(d_1, d_2, \dots, d_k)$, where each $d_i \in \{1, 2, 3, 4, 5, 6\}$ (6 choices per element).

Total outcomes: $6 \times 6 \times \dots \times 6 = 6^k$ (repetition of elements is allowed).

Permutations

A *permutation* of size k from n distinct objects is an ordered sequence of k distinct objects selected from the n objects (no repetition of elements).

The number of such permutations is denoted $P_{k,n}$, with the formula:

$$P_{k,n} = n(n-1)(n-2) \cdots (n-k+1)$$

Using *factorial* notation ($m! = m(m-1) \cdots 2 \cdot 1$, with $0! = 1$), this simplifies to:

$$P_{k,n} = \frac{n!}{(n-k)!}$$

Example 1.3.3 (Balls into Colored Boxes). Given 4 distinct balls (red, blue, yellow, green) and 4 distinct boxes (matching the ball colors), what is the probability the red ball is in the red box?

- Total outcomes: Each arrangement is a permutation of 4 balls (1 ball per box), so $N = 4! = 24$.
- Let $R = \{\text{red ball in red box}\}$: Fix the red ball in the red box; the remaining 3 balls can be permuted freely, so $N(R) = 3! = 6$.

Probability:

$$P(R) = \frac{N(R)}{N} = \frac{3!}{4!} = \frac{6}{24} = \frac{1}{4}$$

Combinations

A *combination* of size k from n distinct objects is an unordered subset of k objects (order does not matter).

The number of such combinations is denoted $\binom{n}{k}$. Since each combination corresponds to $k!$ permutations (all orderings of the subset), we have:

$$\binom{n}{k} = \frac{P_{k,n}}{k!} = \frac{n!}{k!(n-k)!}$$

Example 1.3.4 (Selecting Balls with Red). From 4 distinct balls (red, blue, yellow, green), what is the probability of selecting 2 balls that include the red ball?

- Total outcomes: Unordered subsets of size 2, so $N = \binom{4}{2} = 6$.
- Favorable outcomes: Subsets containing one red ball and one non-red ball, so $N(\text{favorable}) = \binom{3}{1} = 3$ (choose 1 from the 3 non-red balls).

Probability:

$$P = \frac{\binom{3}{1}}{\binom{4}{2}} = \frac{3}{6} = 0.5$$

Example 1.3.5 (Selecting Printers). A warehouse has 25 printers: 10 laser (L) and 15 inkjet (I). 6 printers are selected at random.

1 Probability of exactly 3 laser printers (D_3):

- Favorable outcomes: Choose 3 laser from 10 and 3 inkjet from 15: $N(D_3) = \binom{10}{3} \cdot \binom{15}{3}$.
- Total outcomes: Choose 6 from 25: $N = \binom{25}{6}$.

$$P(D_3) = \frac{\binom{10}{3} \cdot \binom{15}{3}}{\binom{25}{6}} \approx 0.3083$$

2 Probability of at least 3 laser printers ($D_3 \cup D_4 \cup D_5 \cup D_6$): Use the complement rule (at most 2 laser printers: $D_0 \cup D_1 \cup D_2$):

$$P(D_0 \cup D_1 \cup D_2) = \frac{\binom{10}{0}\binom{15}{6} + \binom{10}{1}\binom{15}{5} + \binom{10}{2}\binom{15}{4}}{\binom{25}{6}} \approx 0.147$$

$$P(\text{at least 3}) = 1 - 0.147 = 0.853$$

3 Probability of needing at least 3 selections to find a laser printer: This requires the first 2 selections are inkjet:

$$P = \frac{\binom{15}{2}}{\binom{25}{2}} = \frac{105}{300} = 0.35$$

Extended Counting: Replacement and Partitioning

We extend counting to scenarios with repetition or partitioning:

1. *Permutations with Replacement*: Draw k elements from n (replace after each draw). Total permutations: n^k (each draw has n choices).
2. *Combinations with Replacement*: Draw k elements from n (replace after each draw, unordered). This is equivalent to solving $x_1 + x_2 + \cdots + x_n = k$ (where $x_i \geq 0$ is the count of the i -th element). Using the "stars and bars" method:

$$\text{Total combinations} = \binom{n+k-1}{k}$$

3. *Partitioning into Groups*: Split n distinct objects into m groups with sizes $n_1, n_2, \dots, n_m$ (where $n_1 + n_2 + \cdots + n_m = n$). Total ways:

$$\frac{n!}{n_1!n_2!\cdots n_m!}$$

More Counting Examples

We apply the above rules to common problems:

Example 1.3.6 (Dice Roll (Even and ≥ 3)). Toss a fair die. Find $P(\text{even and } \geq 3)$:

- Sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$, $N = 6$.
- Event $A = \{4, 6\}$, $N(A) = 2$.

$$P(A) = \frac{2}{6} = \frac{1}{3}$$

Example 1.3.7 (Good/Bad Products). 100 products: 4 bad (B), 96 good (G).

- $P(\text{pick 3 good}) = \frac{\binom{96}{3}}{\binom{100}{3}}$
- $P(\text{pick 2 good, 1 bad}) = \frac{\binom{96}{2} \cdot \binom{4}{1}}{\binom{100}{3}}$

Example 1.3.8 (Sampling with/without Replacement). 10 products: 2 bad, 8 good. Find $P(\text{first good, second bad})$:

- *With Replacement*: Total outcomes $10 \times 10 = 100$; favorable $8 \times 2 = 16$:

$$P = \frac{16}{100} = 0.16$$

- *Without Replacement*: Total outcomes $10 \times 9 = 90$; favorable $8 \times 2 = 16$:

$$P = \frac{16}{90} \approx 0.1778$$

Example 1.3.9 (Balls into Boxes). 3 distinct balls, 5 distinct boxes.

- $A = \{\text{given 3 boxes each have 1 ball}\}$: Favorable outcomes $3! = 6$; total outcomes $5^3 = 125$:

$$P(A) = \frac{3!}{5^3} = \frac{6}{125} = 0.048$$

- $B = \{\text{any 3 boxes each have 1 ball}\}$: Favorable outcomes $\binom{5}{3} \cdot 3! = 10 \times 6 = 60$:

$$P(B) = \frac{\binom{5}{3} \cdot 3!}{5^3} = \frac{60}{125} = 0.48$$

Example 1.3.10 (Birthday Problem). 50 students: Find the probability that they all have distinct birthdays $P(\text{all distinct birthdays})$ (365 days/year).

- Total outcomes: 365^{50} (each student has 365 choices).
- Favorable outcomes: Choose 50 distinct days and permute them: $\binom{365}{50} \cdot 50!$.

$$P = \frac{\binom{365}{50} \cdot 50!}{365^{50}} \approx 0.03$$

Example 1.3.11 (Lottery Draw). 10 students, 3 concert tickets (3 "winning" lots, 7 "losing" lots). Find $P(\text{5th student wins})$:

- Total outcomes: $10!$ (all permutations of lots).
- Favorable outcomes: Assign 1 winning lot to the 5th student, permute the rest: $\binom{3}{1} \cdot 9!$.

$$P = \frac{\binom{3}{1} \cdot 9!}{10!} = \frac{3}{10} = 0.3$$

Example 1.3.12 (3-Digit Numbers from 1 – 5). Use digits 1 – 5 to form 3-digit numbers.

- $P(\text{all distinct digits}) = \frac{P_{3,5}}{5^3} = \frac{5 \times 4 \times 3}{125} = \frac{60}{125} = 0.48$
- $P(\text{even and distinct}) = \frac{P_{2,4 \times 2}}{5^3} = \frac{(4 \times 3) \times 2}{125} = \frac{24}{125} = 0.192$ (2 even digits: 2, 4; permute first 2 digits from remaining 4).

Example 1.3.13. Three molecules of type A, three of type B, three of type C, and three of type D are linked to form a chain molecule. Examples include ABCDABCDABCD and BCDDAAABDBCC.

- a How many such chain molecules are there?
- b What is the probability that all three molecules of each type are adjacent (e.g., BBBAADDCC)?

Solution. (a): We provide two equivalent methods:

Method 1: Adjust for indistinct molecules

- If all 12 molecules were distinguishable (e.g., $A_1, A_2, A_3, B_1, B_2, B_3$, etc.), the total number of chains is $12!$ (full permutation of 12 elements).
- Since molecules of the same type are indistinct, we divide by the permutations of each type ($3!$ for A, $3!$ for B, $3!$ for C, $3!$ for D):

$$\text{Total chains} = \frac{12!}{(3!)^4} = \frac{12!}{6^4} \quad (\text{since } 3! = 6)$$

Method 2: Select positions sequentially

- Step 1: Choose 3 positions out of 12 for A: $\binom{12}{3}$ ways.
- Step 2: Choose 3 positions out of the remaining 9 for B: $\binom{9}{3}$ ways.
- Step 3: Choose 3 positions out of the remaining 6 for C: $\binom{6}{3}$ ways.
- Step 4: The last 3 positions are for D: $\binom{3}{3} = 1$ way.

Total chains:

$$\binom{12}{3} \times \binom{9}{3} \times \binom{6}{3} = \frac{12!}{3!9!} \times \frac{9!}{3!6!} \times \frac{6!}{3!3!} = \frac{12!}{(3!)^4} = \frac{12!}{6^4}$$

(b):

- *Total possible chains:* $\frac{12!}{6^4}$ (from part a).
- *Favorable chains:* Treat each type as a single "block" (A-block, B-block, C-block, D-block). There are $4!$ ways to arrange these 4 blocks. Within each block, the 3 molecules are indistinct, so no additional permutations are needed.

Thus, the number of favorable chains is $4!$.

Probability:

$$\mathbb{P}(\text{all type-adjacent}) = \frac{\text{Favorable chains}}{\text{Total chains}} = \frac{4!}{\frac{12!}{6^4}} = \frac{24 \times 1296}{479001600} = \frac{1}{15400}$$

□

Geometric Probability

For sample spaces that are continuous (e.g., intervals, areas), probability is the ratio of the "measure" (length, area, volume) of the event to the measure of the sample space:

$$P(A) = \frac{\text{measure}(A)}{\text{measure}(S)}$$

Example 1.3.14 (Dating Problem). Two people agree to meet between 7-8 PM. The first arrival waits 20 minutes, then leaves. Find $P(\text{successful meeting})$.

Solution. - Let x = arrival time of person 1 (minutes after 7 PM), y = arrival time of person 2.

- Sample space $\mathcal{S} = \{(x, y) \mid 0 \leq x \leq 60, 0 \leq y \leq 60\}$ (a square with area $60^2 = 3600$).
- Successful meeting: $|x - y| \leq 20$ (the region between lines $y = x + 20$ and $y = x - 20$).

The *unsuccessful* region consists of two right triangles (top-left and bottom-right), each with legs of length $60 - 20 = 40$. Total area of unsuccessful regions:

$$2 \times \left(\frac{1}{2} \times 40 \times 40 \right) = 1600$$

Probability of success:

$$P(A) = \frac{\text{Area of } A}{\text{Area of } S} = \frac{3600 - 1600}{3600} = \frac{2000}{3600} = \frac{5}{9}$$

□

Example 1.3.15 (Buffon's Needle). A needle of length l is dropped onto a floor with parallel boards of width a ($a \geq l$). Find $P(\text{needle crosses a seam})$.

Solution. - Let x = distance from needle's center to the nearest seam, ϕ = angle between needle and seams.

- Sample space $G = \{(\phi, x) \mid 0 \leq \phi \leq \pi, 0 \leq x \leq \frac{a}{2}\}$ (a rectangle with area $\text{Area}(G) = \pi \times \frac{a}{2} = \frac{a\pi}{2}$).
- Needle crosses a seam iff $x \leq \frac{l}{2} \sin \phi$ (the projection of the needle onto the direction perpendicular to seams is $\frac{l}{2} \sin \phi$).

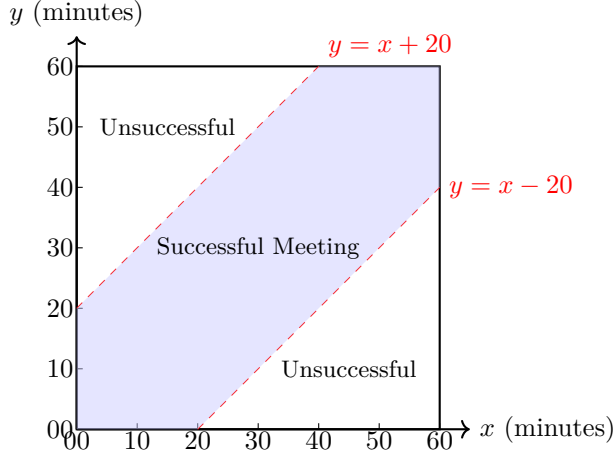


Figure 1.2: Sample Space and Successful Meeting Region for the Dating Problem

The area of the event $A = \{\text{needle crosses a seam}\}$ is the integral of $\frac{l}{2} \sin \phi$ over $\phi \in [0, \pi]$:

$$\text{Area}(A) = \int_0^\pi \frac{l}{2} \sin \phi \, d\phi = \frac{l}{2} [-\cos \phi]_0^\pi = \frac{l}{2} (-\cos \pi + \cos 0) = \frac{l}{2} (1 + 1) = l$$

Probability of crossing a seam (see the figure 1.3):

$$P(A) = \frac{\text{Area}(A)}{\text{Area}(G)} = \frac{l}{\frac{a\pi}{2}} = \frac{2l}{a\pi}$$

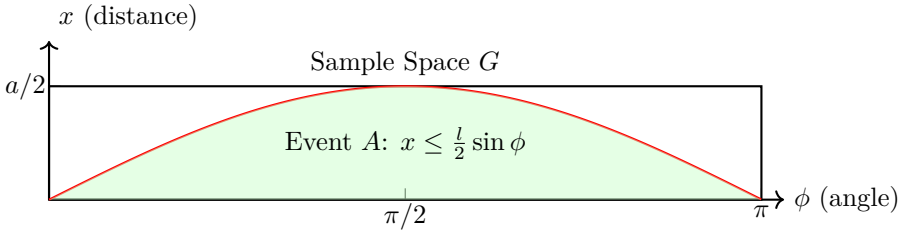


Figure 1.3: Sample Space G and Event A for Buffon's Needle Problem (with $a = 1, l = 1$)

□

Example 1.3.16. Split segment AB into three segments AC , CD , DB . Find the probability that the three segments can form a triangle.

Solution. - Let $AB = 1$, $AC = x$, $CD = y$. Then $DB = 1 - x - y$ (since $x + y + (1 - x - y) = 1$).

- *Sample space constraints:* $x \geq 0$, $y \geq 0$, $x + y \leq 1$ (all segments non-negative). This defines a right triangle in the xy -plane with area $\frac{1}{2}$.

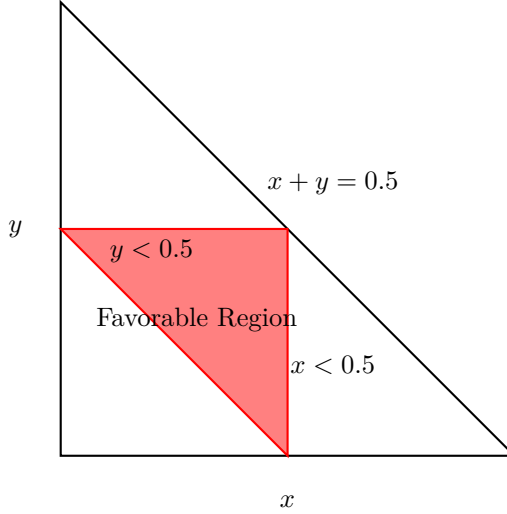


Figure 1.4: Sample Space (white) and Favorable Region (red) for Triangle Formation

- *Triangle formation condition:* For three segments a, b, c to form a triangle, the sum of any two sides must exceed the third. Here, $a = x$, $b = y$, $c = 1 - x - y$, so:

$$\begin{cases} x + y > 1 - x - y & (\text{sum of } a, b > c) \\ x + (1 - x - y) > y & (\text{sum of } a, c > b) \\ y + (1 - x - y) > x & (\text{sum of } b, c > a) \end{cases}$$

Simplify the inequalities:

$$\begin{cases} x + y > 0.5 \\ 1 - y > y \implies y < 0.5 \\ 1 - x > x \implies x < 0.5 \end{cases}$$

- *Area of favorable region:* The favorable region is a smaller right triangle with vertices $(0, 0.5)$, $(0.5, 0.5)$, $(0.5, 0)$. Its area is $\frac{1}{2} \times 0.5 \times 0.5 = \frac{1}{8}$.
- Probability:

$$\mathbb{P}(\text{form triangle}) = \frac{\text{Area of favorable region}}{\text{Area of sample space}} = \frac{\frac{1}{8}}{\frac{1}{2}} = 0.25$$

□

Example 1.3.17. Suppose n people are arranged in a row. Find the probability that there are exactly r people between A and B. Repeat for a circular arrangement (clockwise from A to B).

Solution. **Linear Arrangement**

- Total permutations of n people: $n!$;
- Favorable permutations:
 - Choose 2 positions with exactly r people between them: There are $n - r - 1$ such position pairs (e.g., positions $(1, 1 + r + 1)$, $(2, 2 + r + 1)$, ..., $(n - r - 1, n)$);
 - A and B can be ordered in 2 ways (A first or B first);
 - The remaining $n - 2$ people can be permuted freely: $(n - 2)!$.

Probability:

$$\mathbb{P}(\text{linear}) = \frac{2(n - r - 1)(n - 2)!}{n!} = \frac{2(n - r - 1)}{n(n - 1)}$$

Circular Arrangement

- Total circular permutations of n people: $(n - 1)!$ (fix one person to account for rotations);
- Favorable permutations:
 - In a circle, the number of ways to place A and B with exactly r people between them (clockwise) is n (one for each starting position of A);
 - Fix A's position, B has exactly 1 valid position out of $n - 1$ possible;
 - The remaining $n - 2$ people can be permuted freely: $(n - 2)!$.

Probability:

$$\mathbb{P}(\text{circular}) = \frac{n \times (n - 2)!}{(n - 1)!} = \frac{1}{n - 1}$$

□

1.4 Conditional Probability

When we acquire information that an event has occurred, the probability of another event may change. This adjusted probability is called a *conditional probability*, which is the core of this section. We also introduce related concepts: multiplication rule, law of total probability, and Bayes' theorem.

Motivating Example and Definition

We start with a concrete example to intuitively understand conditional probability.

Example 1.4.1 (Rolling Two Fair Dice). Consider rolling two dices. Define:

- A : “Sum of the two dices is 7” ($A = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$),
- B : “At least one die shows a 2” ($B = \{(2, 1), (2, 2), \dots, (2, 6), (1, 2), (3, 2), \dots, (6, 2)\}$).

First, calculate unconditional probabilities:

- Sample space size $N(\mathcal{S}) = 36$, so $\mathbb{P}(A) = \frac{N(A)}{N(\mathcal{S})} = \frac{6}{36} = \frac{1}{6}$,
- $N(B) = 11$ (11 outcomes in B), so $\mathbb{P}(B) = \frac{11}{36}$.

Now, suppose we know B has occurred (at least one die is 2). What is the probability of A now?

- The effective sample space shrinks to B (11 outcomes). Only 2 outcomes in B are also in A : $(2, 5)$ and $(5, 2)$, so $N(A \cap B) = 2$.
- Intuitively, $\mathbb{P}(A | B) = \frac{N(A \cap B)}{N(B)} = \frac{2}{11}$.

Notice this can be rewritten using unconditional probabilities:

$$\mathbb{P}(A | B) = \frac{N(A \cap B)/N(\mathcal{S})}{N(B)/N(\mathcal{S})} = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

Definition 1.4.1 (Conditional Probability). For any two events A and B with $\mathbb{P}(B) > 0$, the *conditional probability of A given B* is defined as:

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

Key Properties of Conditional Probability

1. $\mathbb{P}(A | B) \leq 1$: Since $\mathbb{P}(A \cap B) \leq \mathbb{P}(B)$ (as $A \cap B \subseteq B$), the ratio is at most 1.
2. $\mathbb{P}(A | B)$ and $\mathbb{P}(A)$ can be equal, greater, or smaller (depends on whether B affects A).

Conditional Probability Calculations with Tables

For categorical data, tables simplify conditional probability calculations.

Example 1.4.2 (Magazine Reading Habits). A news magazine has 3 columns: “Art” (A), “Books” (B), “Cinema” (C). The table below shows the reading probabilities of a random reader:

Read Regularly	A	B	C	$A \cap B$	$A \cap C$	$B \cap C$	$A \cap B \cap C$
Probability	0.14	0.23	0.37	0.08	0.09	0.13	0.05

Calculate:

1. $\mathbb{P}(A \mid B \cup C)$
2. $\mathbb{P}(A \mid A \cup B \cup C)$

Solution 1. $\mathbb{P}(A \mid B \cup C)$ By definition:

$$\mathbb{P}(A \mid B \cup C) = \frac{\mathbb{P}(A \cap (B \cup C))}{\mathbb{P}(B \cup C)}$$

- Step 1: Compute $\mathbb{P}(B \cup C)$ (addition rule):

$$\mathbb{P}(B \cup C) = \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(B \cap C) = 0.23 + 0.37 - 0.13 = 0.47$$

- Step 2: Compute $\mathbb{P}(A \cap (B \cup C))$ (distributive law: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$):

$$\begin{aligned} \mathbb{P}((A \cap B) \cup (A \cap C)) &= \mathbb{P}(A \cap B) + \mathbb{P}(A \cap C) - \mathbb{P}(A \cap B \cap C) \\ &= 0.08 + 0.09 - 0.05 = 0.12 \end{aligned}$$

- Step 3: Conditional probability:

$$\mathbb{P}(A \mid B \cup C) = \frac{0.12}{0.47} \approx 0.255$$

□

Solution 2. $\mathbb{P}(A \mid A \cup B \cup C)$ By definition:

$$\mathbb{P}(A \mid A \cup B \cup C) = \frac{\mathbb{P}(A)}{\mathbb{P}(A \cup B \cup C)}$$

- Step 1: Compute $\mathbb{P}(A \cup B \cup C)$ (addition rule for three events):

$$\begin{aligned}\mathbb{P}(A \cup B \cup C) &= \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) \\ &\quad - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C) \\ &= 0.14 + 0.23 + 0.37 - 0.08 - 0.09 - 0.13 + 0.05 = 0.49\end{aligned}$$

- Step 2: Conditional probability:

$$\mathbb{P}(A \mid A \cup B \cup C) = \frac{0.14}{0.49} \approx 0.286$$

□

Multiplication Rule for Joint Probability

Rearranging the conditional probability definition gives the *multiplication rule*, which calculates joint probabilities ($\mathbb{P}(A \cap B)$).

Proposition 1.4.1 (Multiplication Rule). *For any two events A and B:*

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B) \cdot \mathbb{P}(B) = \mathbb{P}(B \mid A) \cdot \mathbb{P}(A)$$

The rule extends to k events $A_1, A_2, \dots, A_k$:

$$\begin{aligned}&\mathbb{P}(A_1 \cap A_2 \cap \dots \cap A_k) \\ &= \mathbb{P}(A_k \mid A_1 \cap \dots \cap A_{k-1}) \cdot \mathbb{P}(A_{k-1} \mid A_1 \cap \dots \cap A_{k-2}) \dots \mathbb{P}(A_2 \mid A_1) \cdot \mathbb{P}(A_1)\end{aligned}$$

Example 1.4.3 (Blood Typing). Four people donate blood; only 1 has type O. Donors are typed in random order. Find $\mathbb{P}$ (at least 3 donors must be typed to find O).

Interpretation: “At least 3 donors” means the first 2 are *not* type O. Define:

- A : “First donor is not O”,
- B : “Second donor is not O”.

We need $\mathbb{P}(A \cap B)$.

Solution. - $\mathbb{P}(A) = \frac{3}{4}$ (3 non-O out of 4),

- Given A (1 non-O used), 2 non-O left out of 3: $\mathbb{P}(B \mid A) = \frac{2}{3}$,
- By multiplication rule:

$$\mathbb{P}(A \cap B) = \mathbb{P}(B \mid A) \cdot \mathbb{P}(A) = \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{2}$$

□

Example 1.4.4 (Bulb Selection). A box has 4(40W), 5(60W), 6(75W) bulbs (total 15). Bulbs are selected one by one until a 75W bulb is found.

Find $\mathbb{P}$ (at least 6 bulbs must be examined).

Interpretation: “At least 6 bulbs” means the first 5 are *not* 75W. Define A_i : “ i -th bulb is not 75W” (for $i = 1, \dots, 5$). We need $\mathbb{P}(A_1 \cap A_2 \cap A_3 \cap A_4 \cap A_5)$.

Solution. Extended multiplication rule

- $\mathbb{P}(A_1) = \frac{9}{15}$ (9 non-75W out of 15),
- $\mathbb{P}(A_2 \mid A_1) = \frac{8}{14}$ (8 non-75W left out of 14),
- $\mathbb{P}(A_3 \mid A_1 \cap A_2) = \frac{7}{13}$,
- $\mathbb{P}(A_4 \mid A_1 \cap A_2 \cap A_3) = \frac{6}{12}$,
- $\mathbb{P}(A_5 \mid A_1 \cap A_2 \cap A_3 \cap A_4) = \frac{5}{11}$.

Multiply these probabilities:

$$\mathbb{P}(A_1 \cap \dots \cap A_5) = \frac{9}{15} \cdot \frac{8}{14} \cdot \frac{7}{13} \cdot \frac{6}{12} \cdot \frac{5}{11} = \frac{6}{143} \approx 0.042$$

□

Law of Total Probability

For a set of “mutually exclusive and exhaustive” events (a *partition* of the sample space), we can decompose the probability of any event into a sum of conditional probabilities.

Theorem 1.4.1 (Law of Total Probability). *Let $A_1, A_2, \dots, A_k$ be a partition of $\mathcal{S}$ (i.e., $A_i \cap A_j = \emptyset$ for $i \neq j$, and $A_1 \cup \dots \cup A_k = \mathcal{S}$). For any event B :*

$$\mathbb{P}(B) = \sum_{i=1}^k \mathbb{P}(B \mid A_i) \cdot \mathbb{P}(A_i)$$

Proof. Because the A_i ’s are mutually exclusive and exhaustive, if B occurs it must be in conjunction with exactly one of the A_i ’s. That is,

$$B = (A_1 \cap B) \cup \dots \cup (A_k \cap B),$$

where the events $(A_i \cap B)$ are mutually exclusive. Thus

$$P(B) = \sum_{i=1}^k P(A_i \cap B) = \sum_{i=1}^k P(B \mid A_i)P(A_i)$$

is desired. □

Example 1.4.5 (Ball Transfer Between Boxes). Suppose that

- Box 1 : 6 red (R), 4 green (G),
- Box 2 : 7 red (R), 3 green (G).

A ball is randomly moved from Box 1 to Box 2, then a ball is drawn from Box 2. Find $\mathbb{P}(\text{drawn ball is red})$ (denoted $\mathbb{P}(R_2)$).

Solution. Define R_1 : “Ball moved from Box 1 is red” (so R'_1 : “green”). $\{R_1, R'_1\}$ is a partition of $\mathcal{S}$.

- $\mathbb{P}(R_1) = \frac{6}{10} = \frac{3}{5}$, $\mathbb{P}(R'_1) = \frac{4}{10} = \frac{2}{5}$,
- $\mathbb{P}(R_2 \mid R_1) = \frac{8}{11}$ (Box 2 has 8 R after adding 1 R),
- $\mathbb{P}(R_2 \mid R'_1) = \frac{7}{11}$ (Box 2 has 7 R after adding 1 G).

By total probability:

$$\mathbb{P}(R_2) = \mathbb{P}(R_2 \mid R_1) \cdot \mathbb{P}(R_1) + \mathbb{P}(R_2 \mid R'_1) \cdot \mathbb{P}(R'_1) = \frac{8}{11} \cdot \frac{3}{5} + \frac{7}{11} \cdot \frac{2}{5} = \frac{38}{55} \approx 0.691$$

□

Bayes' Theorem

Bayes' theorem relates *prior probabilities* (before observing data) to *posterior probabilities* (after observing data), using the law of total probability.

Theorem 1.4.2 (Bayes' Theorem). *Let $A_1, A_2, \dots, A_k$ be a partition of $\mathcal{S}$ with prior probabilities $\mathbb{P}(A_i) > 0$. For any event B with $\mathbb{P}(B) > 0$, the posterior probability of A_j given B is:*

$$\mathbb{P}(A_j \mid B) = \frac{\mathbb{P}(B \mid A_j) \cdot \mathbb{P}(A_j)}{\sum_{i=1}^k \mathbb{P}(B \mid A_i) \cdot \mathbb{P}(A_i)} \quad (j = 1, 2, \dots, k)$$

Example 1.4.6 (Sick Child Diagnosis). A doctor knows:

- 90% of sick children have flu (F), 10% have measles (M) ($F \cup M = \mathcal{S}$),
- Measles causes a rash (R) in 95% of cases: $\mathbb{P}(R \mid M) = 0.95$,
- Flu causes a rash in 8% of cases: $\mathbb{P}(R \mid F) = 0.08$.

The child has a rash. Find $\mathbb{P}(M \mid R)$ (probability of measles given rash).

Solution. By Bayes' theorem:

$$\mathbb{P}(M | R) = \frac{\mathbb{P}(R | M) \cdot \mathbb{P}(M)}{\mathbb{P}(R | M) \cdot \mathbb{P}(M) + \mathbb{P}(R | F) \cdot \mathbb{P}(F)}$$

- Prior probabilities: $\mathbb{P}(M) = 0.1$, $\mathbb{P}(F) = 0.9$,
- Denominator (total probability of rash):

$$\mathbb{P}(R) = 0.95 \cdot 0.1 + 0.08 \cdot 0.9 = 0.095 + 0.072 = 0.167$$

- Posterior probability:

$$\mathbb{P}(M | R) = \frac{0.95 \cdot 0.1}{0.167} \approx 0.57$$

Similarly, $\mathbb{P}(F | R) = 1 - 0.57 = 0.43$ (consistent with the partition property: $\sum \mathbb{P}(A_i | B) = 1$). $\square$

Additional Properties of Conditional Probability

1. For any events A, B, C :

$$\mathbb{P}(A \cup B | C) = \mathbb{P}(A | C) + \mathbb{P}(B | C) - \mathbb{P}(A \cap B | C)$$

(addition rule for conditional probability)

2. For any event A and partition $\{B, B'\}$:

$$\mathbb{P}(A | B) = 1 - \mathbb{P}(A' | B)$$

(complement rule for conditional probability)

Example 1.4.7. A transmitter sends signals "." (dot) and "-" (dash) with probabilities 0.6 and 0.4, respectively. Due to interference:

- When sending ".", the receiver gets "." with 0.8 and "-" with 0.2.
- When sending "-", the receiver gets "-" with 0.9 and "." with 0.1.

Find:

1. Probability the receiver gets ".";
2. Probability the transmitter sent "." given the receiver got ".".

Solution. Define events:

- A : Transmitter sent ".", $\mathbb{P}(A) = 0.6$, $\mathbb{P}(\bar{A}) = 0.4$ (sent "-");
- B : Receiver got ".", $\mathbb{P}(B|A) = 0.8$, $\mathbb{P}(B|\bar{A}) = 0.1$.

1. Probability of receiving ”” ($\mathbb{P}(B)$)

Use the law of total probability:

$$\mathbb{P}(B) = \mathbb{P}(B|A)\mathbb{P}(A) + \mathbb{P}(B|\bar{A})\mathbb{P}(\bar{A}) = 0.8 \times 0.6 + 0.1 \times 0.4 = 0.48 + 0.04 = 0.52$$

2. Probability of sending ”” given receiving ”” ($\mathbb{P}(A|B)$)

Use Bayes' theorem:

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)} = \frac{0.48}{0.52} = \frac{12}{13} \approx 0.923$$

□

Example 1.4.8. A box has 15 ping-pong balls, 9 of which are new. For the first competition, 3 balls are drawn with replacement; for the second competition, 3 balls are drawn again. Find the probability that all 3 balls in the second competition are new.

Solution. Define partition events (first draw results):

- A_1 : 3 new balls drawn first, $\mathbb{P}(A_1) = \left(\frac{9}{15}\right)^3 = \left(\frac{3}{5}\right)^3 = \frac{27}{125}$;
- A_2 : 2 new, 1 old drawn first, $\mathbb{P}(A_2) = \binom{3}{2} \times \frac{9}{15} \times \frac{9}{15} \times \frac{6}{15} = 3 \times \frac{486}{3375} = \frac{1458}{3375} = \frac{54}{125}$;
- A_3 : 1 new, 2 old drawn first, $\mathbb{P}(A_3) = \binom{3}{1} \times \frac{9}{15} \times \frac{6}{15} \times \frac{6}{15} = 3 \times \frac{324}{3375} = \frac{972}{3375} = \frac{36}{125}$;
- A_4 : 3 old balls drawn first, $\mathbb{P}(A_4) = \left(\frac{6}{15}\right)^3 = \left(\frac{2}{5}\right)^3 = \frac{8}{125}$.

Conditional probabilities (second draw all new):

- $\mathbb{P}(B|A_1) = \left(\frac{9}{15}\right)^3 = \frac{27}{125}$ (replacement, new ball count unchanged);
- $\mathbb{P}(B|A_2) = \left(\frac{9}{15}\right)^3 = \frac{27}{125}$ (replacement, new ball count unchanged);
- $\mathbb{P}(B|A_3) = \left(\frac{9}{15}\right)^3 = \frac{27}{125}$ (replacement, new ball count unchanged);
- $\mathbb{P}(B|A_4) = \left(\frac{9}{15}\right)^3 = \frac{27}{125}$ (replacement, new ball count unchanged).

By total probability:

$$\begin{aligned} \mathbb{P}(B) &= \sum_{i=1}^4 \mathbb{P}(A_i)\mathbb{P}(B|A_i) \\ &= \left(\frac{27 + 54 + 36 + 8}{125}\right) \times \frac{27}{125} = \frac{125}{125} \times \frac{27}{125} = \frac{27}{125} = 0.216 \end{aligned}$$

□

Example 1.4.9 (Two-Box Ball Problem). There are two boxes:

- Box 1: 5 red (R), 4 white (W);
- Box 2: 4 red, 5 white.

Draw 2 balls from Box 1 and transfer to Box 2, then draw 1 ball from Box 2. Find the probability the drawn ball is white.

Solution. We provide two methods:

Method 1: Partition by red balls transferred

Define A_i : "Transfer i red balls from Box 1" ($i = 0, 1, 2$):

- A_0 : 0 red, 2 white transferred: $\mathbb{P}(A_0) = \frac{\binom{4}{2}}{\binom{9}{2}} = \frac{6}{36} = \frac{1}{6}$; Box 2 after transfer: 4R, 7W $\rightarrow \mathbb{P}(W|A_0) = \frac{7}{11}$;
- A_1 : 1 red, 1 white transferred: $\mathbb{P}(A_1) = \frac{\binom{5}{1}\binom{4}{1}}{\binom{9}{2}} = \frac{20}{36} = \frac{5}{9}$; Box 2 after transfer: 5R, 6W $\rightarrow \mathbb{P}(W|A_1) = \frac{6}{11}$;
- A_2 : 2 red, 0 white transferred: $\mathbb{P}(A_2) = \frac{\binom{5}{2}}{\binom{9}{2}} = \frac{10}{36} = \frac{5}{18}$; Box 2 after transfer: 6R, 5W $\rightarrow \mathbb{P}(W|A_2) = \frac{5}{11}$.

Total probability:

$$\mathbb{P}(W) = \frac{1}{6} \times \frac{7}{11} + \frac{5}{9} \times \frac{6}{11} + \frac{5}{18} \times \frac{5}{11} = \frac{21 + 60 + 25}{198} = \frac{106}{198} = \frac{53}{99} \approx 0.535$$

Method 2: Partition by ball origin

Define B : "Drawn ball is from Box 1 (transferred)"; $\overline{B}$: "Drawn ball is from Box 2 (original)":

- $\mathbb{P}(B) = \frac{2}{11}$ (2 transferred balls in Box 2's 11 balls); $\mathbb{P}(W|B) = \frac{4}{9}$ (probability a transferred ball is white, from Box 1's 4W out of 9);
- $\mathbb{P}(\overline{B}) = \frac{9}{11}$ (9 original balls in Box 2's 11 balls); $\mathbb{P}(W|\overline{B}) = \frac{5}{9}$ (original white balls in Box 2).

Total probability:

$$\mathbb{P}(W) = \mathbb{P}(W|B)\mathbb{P}(B) + \mathbb{P}(W|\overline{B})\mathbb{P}(\overline{B}) = \frac{4}{9} \times \frac{2}{11} + \frac{5}{9} \times \frac{9}{11} = \frac{8 + 45}{99} = \frac{53}{99}$$

□

Example 1.4.10 (Sack Ball Problem). A sack has 15 balls: 10 red (R), 5 yellow (Y). Draw twice without replacement:

- First draw: 5 balls;
- Second draw: 6 balls.

Find the probability of the following events:

1. In the second draw, the 5th ball is red;
2. First draw has 2 yellow balls, second draw has 4 red balls;
3. First draw has 3 red balls $\cup$ second draw has 2 yellow balls.

Solution. **1. Probability the 5th ball in the second draw is red**

- The 5th ball in the second draw is the $5 + 5 = 10$ -th ball drawn overall (5 first draw + 4 from second draw before the 5th).
- For any specific position in a random sequence of draws without replacement, the probability that the position is red is equal to the proportion of red balls in the sack.

Thus:

$$\mathbb{P}(\text{10th ball red}) = \frac{10}{15} = \frac{2}{3}$$

2. First draw: 2Y, second draw: 4R

- Define A : First draw has 2Y (3R), B : Second draw has 4R (2Y).
- Total ways to draw first 5 and next 6 balls: $\binom{15}{5} \times \binom{10}{6}$;
- Favorable ways:
 - Choose 2Y, 3R for first draw: $\binom{5}{2} \times \binom{10}{3}$;
 - From remaining 3Y, 7R, choose 4R, 2Y for second draw: $\binom{7}{4} \times \binom{3}{2}$.

Probability:

$$\mathbb{P}(A \cap B) = \frac{\binom{5}{2} \binom{10}{3} \binom{7}{4} \binom{3}{2}}{\binom{15}{5} \binom{10}{6}} = \frac{10 \times 120 \times 35 \times 3}{3003 \times 210} = \frac{126000}{630630} \approx 0.2$$

3. $A \cup B$ (A: first draw 3R; B: second draw 2Y)

Use the addition rule:

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

- $\mathbb{P}(A) = \frac{\binom{10}{3}\binom{5}{2}}{\binom{15}{5}} = \frac{120 \times 10}{3003} = \frac{1200}{3003};$
- $\mathbb{P}(B) = \frac{\binom{5}{2}\binom{10}{4}}{\binom{15}{6}} = \frac{10 \times 210}{5005} = \frac{2100}{5005} = \frac{360}{858} \text{ (simplified);}$
- $\mathbb{P}(A \cap B) \approx 0.2 = \frac{200}{1001} \text{ (from part 2).}$

After unifying denominators and simplifying:

$$\mathbb{P}(A \cup B) \approx 0.62$$

□

Example 1.4.11. Factories A, B, C manufacture the same product:

- Yield: A (25%), B (35%), C (40%);
- Defective rate: A (5%), B (4%), C (2%).

All products are mixed. Find:

1. Total defective rate;
2. Probability a defective product is from Factory A.

Solution. Define D : Defective product, A, B, C : Products from Factory A, B, C (partition of sample space).

1. Total Defective Rate

Use the law of total probability:

$$\begin{aligned} \mathbb{P}(D) &= \mathbb{P}(D|A)\mathbb{P}(A) + \mathbb{P}(D|B)\mathbb{P}(B) + \mathbb{P}(D|C)\mathbb{P}(C) \\ &= 0.05 \times 0.25 + 0.04 \times 0.35 + 0.02 \times 0.40 = 0.0125 + 0.014 + 0.008 = 0.0345 \end{aligned}$$

2. Probability Defective Product is from A

Use Bayes' theorem:

$$\mathbb{P}(A|D) = \frac{\mathbb{P}(D|A)\mathbb{P}(A)}{\mathbb{P}(D)} = \frac{0.0125}{0.0345} \approx 0.36$$

□

1.5 Independence

The definition of conditional probability enables us to revise the probability $P(A)$ originally assigned to A when we are subsequently informed that another event B has occurred; the new probability of A is $P(A|B)$. In our examples, it was frequently the case that $P(A|B)$ differed from the unconditional probability $P(A)$, indicating that the information “ B has occurred” resulted in a change in the chance of A occurring. Often the chance that A will occur or has occurred is not affected by knowledge that B has occurred, so that $P(A|B) = P(A)$. It is then natural to regard A and B as independent events, meaning that the occurrence or nonoccurrence of one event has no bearing on the chance that the other will occur.

Independence of Two Events

If $\mathbb{P}(A | B) = \mathbb{P}(A)$, then knowing B occurs does not change the probability of A . We call such events *independent*.

Definition 1.5.1 (Independence). Two events A and B are *independent* if $\mathbb{P}(A | B) = \mathbb{P}(A)$ (with $\mathbb{P}(B) > 0$) or $\mathbb{P}(B | A) = \mathbb{P}(B)$ (with $\mathbb{P}(A) > 0$). If not, they are *dependent*.

Example 1.5.1 (8-Face Die Independence). Sample space $\mathcal{S} = \{1, 2, 3, 4, 5, 6, 7, 8\}$ (equally likely outcomes). Define:

- A : “Outcome is odd” ($A = \{1, 3, 5, 7\}$),
- B : “Outcome is at least 7” ($B = \{7, 8\}$).

Calculate probabilities:

- $\mathbb{P}(A) = \frac{4}{8} = \frac{1}{2}$, $\mathbb{P}(B) = \frac{2}{8} = \frac{1}{4}$,
- $A \cap B = \{7\}$, so $\mathbb{P}(A \cap B) = \frac{1}{8}$,
- Conditional probability: $\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{1/8}{1/4} = \frac{1}{2} = \mathbb{P}(A)$.

Thus, A and B are independent.

Example 1.5.2 (Dependent Events). For the same 8-face die, define:

- $C = \{1, 2, 3, 5, 7\}$, $D = \{3, 5, 7\}$.

Calculate conditional probabilities:

- $\mathbb{P}(B) = \frac{1}{4}$, $\mathbb{P}(B \cap C) = \frac{1}{8}$ (only $7 \in B \cap C$), so $\mathbb{P}(B | C) = \frac{1/8}{5/8} = \frac{1}{5} < \frac{1}{4} = \mathbb{P}(B)$,

- $\mathbb{P}(B \cap D) = \frac{1}{8}$ (only 7 $\in B \cap D$), so $\mathbb{P}(B \mid D) = \frac{1/8}{3/8} = \frac{1}{3} > \frac{1}{4} = \mathbb{P}(B)$.

Thus, B is dependent on both C and D .

From the multiplication rule, we derive an equivalent condition for independence — this is often easier to verify:

Proposition 1.5.1 (Equivalent Condition for Independence). *Events A and B are independent if and only if:*

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$$

Proof. - If A and B are independent, then $\mathbb{P}(A \mid B) = \mathbb{P}(A)$. Substitute into the multiplication rule:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B) \cdot \mathbb{P}(B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$$

- Conversely, if $\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$ and $\mathbb{P}(B) > 0$, then:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(A) \cdot \mathbb{P}(B)}{\mathbb{P}(B)} = \mathbb{P}(A)$$

Thus, the two conditions are equivalent. □

Property 1.6 (Complementary Events and Independence). *If A and B are independent, then the following pairs are also independent:*

1. A' and B

2. A and B'

3. A' and B'

Proof. We prove Case 1 (others follow similarly): Since A and B are independent, $\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$. By the complement rule for conditional probability:

$$\mathbb{P}(A' \mid B) = 1 - \mathbb{P}(A \mid B)$$

Since A and B are independent, $\mathbb{P}(A \mid B) = \mathbb{P}(A)$, so:

$$\mathbb{P}(A' \mid B) = 1 - \mathbb{P}(A) = \mathbb{P}(A')$$

Thus, A' and B are independent. □

Key Distinction: Mutually Exclusive vs. Independent

A common misconception is confusing "mutually exclusive" and "independent." We clarify this with a critical result:

Proposition 1.5.2 (Mutually Exclusive $\nRightarrow$ Independent). *If $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$, and A and B are mutually exclusive, then A and B are dependent.*

Proof. - Mutually exclusive events satisfy $A \cap B = \emptyset$, so $\mathbb{P}(A \cap B) = 0$.

- If A and B were independent, then $\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$.
- But $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$ imply $\mathbb{P}(A) \cdot \mathbb{P}(B) > 0$, which contradicts $\mathbb{P}(A \cap B) = 0$.
- Thus, A and B cannot be independent.

□

Practice Question: Suppose A and B are mutually exclusive. Which of the following is true?

- A. $\mathbb{P}(A' \cap B') = 0$;
- B. $\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$;
- C. $\mathbb{P}(A) = 1 - \mathbb{P}(B)$;
- D. $\mathbb{P}(A' \cup B') = 1$.

Answer: D. *Explanation:* By De Morgan's Law, $A' \cup B' = (A \cap B)'$. Since A and B are mutually exclusive, $A \cap B = \emptyset$, so $(A \cap B)' = \mathcal{S}$. Thus, $\mathbb{P}(A' \cup B') = \mathbb{P}(\mathcal{S}) = 1$.

Simplified Probability Rules for Independence/Mutual Exclusivity

We summarize how independence and mutual exclusivity simplify the multiplication and addition rules:

Two properties that are commonly used:

1. Multiplication rule: $P(A \cap B) = P(A|B)P(B)$
2. Addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

If A and B are **independent**, then the multiplication rule can be simplified: $P(A \cap B) = P(A)P(B)$

If A and B are **mutually exclusive**, then the addition rule can be simplified: $P(A \cup B) = P(A) + P(B)$.

Applications of Independence

We apply independence to real-world problems, including product reliability and system probability.

Example 1.5.3 (Washing Machine and Dryer Warranty). A company's washing machines (W) require warranty service with probability 0.3, and dryers (D) with probability 0.1. Assume W and D function independently. Find:

1. Probability both need service.
2. Probability neither needs service.

Solution. 1. By independence, $\mathbb{P}(W \cap D) = \mathbb{P}(W) \cdot \mathbb{P}(D) = 0.3 \times 0.1 = 0.03$.

2. By the complement property of independence, W' and D' are independent. Thus:

$$\mathbb{P}(W' \cap D') = \mathbb{P}(W') \cdot \mathbb{P}(D') = (1 - 0.3) \times (1 - 0.1) = 0.7 \times 0.9 = 0.63$$

□

Example 1.5.4 (System Component Reliability). Consider a system with 4 components connected as shown in Figure 1.5:

- Components 1 and 2 : Parallel (subsystem works if *at least one* works).
- Components 3 and 4 : Series (subsystem works if *both* work). All components work independently, with $\mathbb{P}(\text{component } i \text{ works}) = 0.9$.

Find $\mathbb{P}(\text{system works})$.

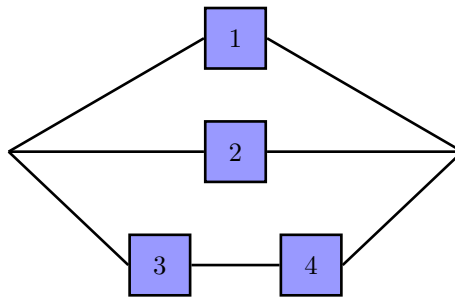


Figure 1.5: System Component Configuration (1 – 2 Parallel, 3 – 4 Series)

Solution. Let $A_i = \{\text{component } i \text{ works}\}$. The system works if *either the parallel subsystem (1 – 2) works or the series subsystem (3 – 4) works* (i.e., $(A_3 \cap A_4) \cup (A_1 \cup A_2)$).

By the addition rule:

$$\mathbb{P}(\text{system works}) = \mathbb{P}(A_3 \cap A_4) + \mathbb{P}(A_1 \cup A_2) - \mathbb{P}((A_3 \cap A_4) \cap (A_1 \cup A_2))$$

We calculate each term using independence:

1. *Series subsystem (3 – 4):* Works iff both 3 and 4 work.

$$\mathbb{P}(A_3 \cap A_4) = \mathbb{P}(A_3) \cdot \mathbb{P}(A_4) = 0.9 \times 0.9 = 0.81$$

2. *Parallel subsystem (1 – 2):* Works iff at least one of 1 or 2 works.

$$\mathbb{P}(A_1 \cup A_2) = \mathbb{P}(A_1) + \mathbb{P}(A_2) - \mathbb{P}(A_1 \cap A_2) = 0.9 + 0.9 - (0.9 \times 0.9) = 0.99$$

3. *Intersection of the two subsystems:* By independence of all components, $A_3 \cap A_4$ and $A_1 \cup A_2$ are independent (proof below). Thus:

$$\mathbb{P}((A_3 \cap A_4) \cap (A_1 \cup A_2)) = \mathbb{P}(A_3 \cap A_4) \cdot \mathbb{P}(A_1 \cup A_2) = 0.81 \times 0.99 = 0.8019$$

Substitute back into the addition rule:

$$\mathbb{P}(\text{system works}) = 0.81 + 0.99 - 0.8019 = 0.9981$$

□

Independence of More Than Two Events

The concept of independence extends to collections of n events, but requires a stronger condition than pairwise independence.

Definition 1.5.2 (Mutual Independence of n Events). Events $A_1, A_2, \dots, A_n$ are *mutually independent* if for every $k = 2, 3, \dots, n$ and every subset of indices $i_1, i_2, \dots, i_k$:

$$\mathbb{P}(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) = \mathbb{P}(A_{i_1}) \cdot \mathbb{P}(A_{i_2}) \cdot \dots \cdot \mathbb{P}(A_{i_k})$$

Example for 4 Events: For A, B, C, D to be mutually independent, all of the following must hold:

- Pairwise: $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B), \mathbb{P}(A \cap C) = \mathbb{P}(A)\mathbb{P}(C), \dots, \mathbb{P}(C \cap D) = \mathbb{P}(C)\mathbb{P}(D)$ (6 conditions),
- Triplewise: $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C), \dots, \mathbb{P}(B \cap C \cap D) = \mathbb{P}(B)\mathbb{P}(C)\mathbb{P}(D)$ (4 conditions),
- 4-way: $\mathbb{P}(A \cap B \cap C \cap D) = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C)\mathbb{P}(D)$ (1 condition).

Key Remark: Pairwise Independence $\neq$ Mutual Independence Mutually independent events are always pairwise independent, but the reverse is not true. We illustrate this with an example:

Example 1.5.5 (Pairwise Independent but Not Mutually Independent). A box contains 4 balls labeled 110, 101, 011, 000. One ball is selected at random. Define $A_k = \{\text{ball has 1 in the } k\text{-th position}\}$ (for $k = 1, 2, 3$). Are A_1, A_2, A_3 mutually independent?

Solution. 1. *Define Events:*

- $A_1 = \{110, 101\}$, $A_2 = \{110, 011\}$, $A_3 = \{101, 011\}$,
- Sample space size $N(\mathcal{S}) = 4$, so all single-event probabilities: $\mathbb{P}(A_1) = \mathbb{P}(A_2) = \mathbb{P}(A_3) = \frac{2}{4} = \frac{1}{2}$.

2. *Pairwise Independence Check:*

- $A_1 \cap A_2 = \{110\}$, so $\mathbb{P}(A_1 \cap A_2) = \frac{1}{4} = \frac{1}{2} \times \frac{1}{2} = \mathbb{P}(A_1)\mathbb{P}(A_2)$,
- Similarly, $\mathbb{P}(A_1 \cap A_3) = \frac{1}{4} = \mathbb{P}(A_1)\mathbb{P}(A_3)$ and $\mathbb{P}(A_2 \cap A_3) = \frac{1}{4} = \mathbb{P}(A_2)\mathbb{P}(A_3)$. Thus, A_1, A_2, A_3 are pairwise independent.

3. *Mutual Independence Check:*

- $A_1 \cap A_2 \cap A_3 = \emptyset$ (no ball has 1s in all 3 positions), so $\mathbb{P}(A_1 \cap A_2 \cap A_3) = 0$,
- But $\mathbb{P}(A_1)\mathbb{P}(A_2)\mathbb{P}(A_3) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8} \neq 0$. Thus, A_1, A_2, A_3 are *not* mutually independent.

□

Summary of Probability Calculation Rules

When calculating probabilities for complex events, use the following rules (extended to n events as needed):

1 *Subtraction Rule:* $\mathbb{P}(A) = 1 - \mathbb{P}(A')$ (simplifies calculations for "at least one" events).

2 *Multiplication Rule:*

- General: $\mathbb{P}(A_1 \cap A_2 \cap \dots \cap A_n) = \prod_{i=1}^n \mathbb{P}(A_i \mid A_1 \cap \dots \cap A_{i-1})$
- Independent Events: $\mathbb{P}(A_1 \cap \dots \cap A_n) = \prod_{i=1}^n \mathbb{P}(A_i)$

3 *Addition Rule:*

- General:

$$\begin{aligned} & \mathbb{P}(A_1 \cup \dots \cup A_n) \\ &= \sum \mathbb{P}(A_i) - \sum \mathbb{P}(A_i \cap A_j) \\ & \quad + \sum \mathbb{P}(A_i \cap A_j \cap A_k) - \dots + (-1)^{n+1} \mathbb{P}(A_1 \cap \dots \cap A_n) \end{aligned}$$

- Mutually Exclusive Events: $\mathbb{P}(A_1 \cup \dots \cup A_n) = \sum_{i=1}^n \mathbb{P}(A_i)$

4 *Law of Total Probability*: Decompose $\mathbb{P}(B)$ using a partition of $\mathcal{S}$.

Exercise

1.1

1. Four universities-1, 2, 3, and 4-are participating in a holiday basketball tournament. In the first round, 1 will play 2 and 3 will play 4. Then the two winners will play for the championship, and the two losers will also play. One possible outcome can be denoted by 1324 (1 beats 2 and 3 beats 4 in first-round games, and then 1 beats 3 and 2 beats 4).

 - (a) List all outcomes in $\mathcal{S}$.
 - (b) Let A denote the event that 1 wins the tournament. List outcomes in A .
 - (c) Let B denote the event that 2 gets into the championship game. List outcomes in B .
 - (d) What are the outcomes in $A \cup B$ and in $A \cap B$? What are the outcomes in A' ?

2. Three components are connected to form a system as shown in the accompanying diagram. Because the components in the 2-3 subsystem are connected in parallel, that subsystem will function if at least one of the two individual components functions. For the entire system to function, component 1 must function and so must the 2-3 subsystem. The experiment consists of determining the condition of each component [S (success) for a functioning component and F (failure) for a nonfunctioning component].



- (a) Which outcomes are contained in the event A that exactly two out of the three components function?
 - (b) Which outcomes are contained in the event B that at least two of the components function?
 - (c) Which outcomes are contained in the event C that the system functions?
 - (d) List outcomes in C' , $A \cup C$, $A \cap C$, $B \cup C$, and $B \cap C$.
3. Use Venn diagrams to verify the following two relationships for any events A and B (these are called De Morgan's laws):
- (a) $(A \cup B)' = A' \cap B'$
 - (b) $(A \cap B)' = A' \cup B'$

[Hint: In each part, draw a diagram corresponding to the left side and another corresponding to the right side.]

1.2

4. Let A denote the event that the next request for assistance from a statistical software consultant relates to the SPSS package, and let B be the event that the next request is for help with SAS. Suppose that $P(A) = .30$ and $P(B) = .50$.
- (a) Why is it not the case that $P(A) + P(B) = 1$?
 - (b) Calculate $P(A')$.
 - (c) Calculate $P(A \cup B)$.
 - (d) Calculate $P(A' \cap B')$.
5. Human visual inspection of solder joints on printed circuit boards can be very subjective. Part of the problem stems from the numerous types of solder defects (e.g., pad nonwetting, knee visibility, voids) and even the degree to which a joint possesses one or more of these defects. Consequently, even highly trained inspectors can disagree on the disposition of a particular joint. In one batch of 10,000 joints, inspector A found 724 that were judged defective, inspector B found 751 such joints, and 1159 of the joints were judged defective by at least one of the inspectors. Suppose that one of the 10,000 joints is randomly selected.
- (a) What is the probability that the selected joint was judged to be defective by neither of the two inspectors?

- (b) What is the probability that the selected joint was judged to be defective by inspector B but not by inspector A?
6. The three most popular options on a certain type of new car are a built-in GPS (A), a sunroof (B), and an automatic transmission (C). If 40% of all purchasers request A , 55% request B , 70% request C , 63% request A or B , 77% request A or C , 80% request B or C , and 85% request A or B or C , determine the probabilities of the following events.
- [Hint: “ A or B ” is the event that at least one of the two options is requested; try drawing a Venn diagram and labeling all regions.]
- (a) The next purchaser will request at least one of the three options.
- (b) The next purchaser will select none of the three options.
- (c) The next purchaser will request only an automatic transmission and not either of the other two options.
- (d) The next purchaser will select exactly one of these three options.

1.3

7. The composer Beethoven wrote 9 symphonies, 5 piano concertos (music for piano and orchestra), and 32 piano sonatas (music for solo piano).
- (a) How many ways are there to play first a Beethoven symphony and then a Beethoven piano concerto?
- (b) The manager of a radio station decides that on each successive evening (7 days per week), a Beethoven symphony will be played followed by a Beethoven piano concerto followed by a Beethoven piano sonata. For how many years could this policy be continued before exactly the same program would have to be repeated?
8. A production facility employs 10 workers on the day shift, 8 workers on the swing shift, and 6 workers on the graveyard shift. A quality control consultant is to select 5 of these workers for in-depth interviews. Suppose the selection is made in such a way that any particular group of 5 workers has the same chance of being selected as does any other group (drawing 5 slips without replacement from among 24).
- (a) How many selections result in all 5 workers coming from the day shift? What is the probability that all 5 selected workers will be from the day shift?

- (b) What is the probability that all 5 selected workers will be from the same shift?
 - (c) What is the probability that at least two different shifts will be represented among the selected workers?
 - (d) What is the probability that at least one of the shifts will be unrepresented in the sample of workers?
9. A box in a supply room contains 15 compact fluorescent lightbulbs, of which 5 are rated 13-watt, 6 are rated 18-watt, and 4 are rated 23-watt. Suppose that three of these bulbs are randomly selected.
- (a) What is the probability that exactly two of the selected bulbs are rated 23-watt?
 - (b) What is the probability that all three of the bulbs have the same rating?
 - (c) What is the probability that one bulb of each type is selected?
 - (d) If bulbs are selected one by one until a 23-watt bulb is obtained, what is the probability that it is necessary to examine at least 6 bulbs?
10. In five-card poker, a straight consists of five cards with adjacent denominations (e.g., 9 of clubs, 10 of hearts, jack of hearts, queen of spades, and king of clubs). Assuming that aces can be high or low, if you are dealt a five-card hand, what is the probability that it will be a straight with high card 10? What is the probability that it will be a straight? What is the probability that it will be a straight flush (all cards in the same suit)?

1.4

11. The population of a particular country consists of three ethnic groups. Each individual belongs to one of the four major blood groups. The accompanying *joint probability table* gives the proportions of individuals in the various ethnic group-blood group combinations.

		Blood Group			
		O	A	B	AB
Ethnic Group	1	.082	.106	.008	.004
	2	.135	.141	.018	.006
	3	.215	.200	.065	.020

Suppose that an individual is randomly selected from the population, and define events by $A = \{\text{type A selected}\}$, $B = \{\text{type B selected}\}$, and $C = \{\text{ethnic group 3 selected}\}$.

- (a) Calculate $P(A)$, $P(C)$, and $P(A \cap C)$.
 - (b) Calculate both $P(A|C)$ and $P(C|A)$, and explain in context what each of these probabilities represents.
 - (c) If the selected individual does not have type B blood, what is the probability that he or she is from ethnic group 1?
12. According to a July 31, 2013, posting on [cnn.com](http://www.cnn.com) subsequent to the death of a child who bit into a peanut, a 2010 study in the journal *Pediatrics* found that 8% of children younger than 18 in the United States have at least one food allergy. Among those with food allergies, about 39% had a history of severe reaction.
- (a) If a child younger than 18 is randomly selected, what is the probability that he or she has at least one food allergy and a history of severe reaction?
 - (b) It was also reported that 30% of those with an allergy in fact are allergic to multiple foods. If a child younger than 18 is randomly selected, what is the probability that he or she is allergic to multiple foods?
13. Deer ticks can be carriers of either Lyme disease or human granulocytic ehrlichiosis (HGE). Based on a recent study, suppose that 16% of all ticks in a certain location carry Lyme disease, 10% carry HGE, and 10% of the ticks that carry at least one of these diseases in fact carry both of them. If a randomly selected tick is found to have carried HGE, what is the probability that the selected tick is also a carrier of Lyme disease?
14. If $P(B|A) > P(B)$, show that $P(B'|A) < P(B')$.
15. Components of a certain type are shipped to a supplier in batches of ten. Suppose that 50% of all such batches contain no defective components, 30% contain one defective component, and 20% contain two defective components. Two components from a batch are randomly selected and tested. What are the probabilities associated with 0, 1, and 2 defective components being in the batch under each of the following conditions?
- (a) Neither tested component is defective.
 - (b) One of the two tested components is defective. [Hint: Draw a tree diagram with three first-generation branches for the three different types of batches.]

1.5

16. An aircraft seam requires 25 rivets. The seam will have to be reworked if any of these rivets is defective. Suppose rivets are defective independently of one another, each with the same probability.
- (a) If 15% of all seams need reworking, what is the probability that a rivet is defective?
 - (b) How small should the probability of a defective rivet be to ensure that only 10% of all seams need reworking?
17. Components arriving at a distributor are checked for defects by two different inspectors (each component is checked by both inspectors). The first inspector detects 90% of all defectives that are present, and the second inspector does likewise. At least one inspector does not detect a defect on 20% of all defective components. What is the probability that the following occur?
- (a) A defective component will be detected only by the first inspector? By exactly one of the two inspectors?
 - (b) All three defective components in a batch escape detection by both inspectors (assuming inspections of different components are independent of one another)?
18. Suppose identical tags are placed on both the left ear and the right ear of a fox. The fox is then let loose for a period of time. Consider the two events $C_1 = \{ \text{left ear tag is lost} \}$ and $C_2 = \{ \text{right ear tag is lost} \}$. Let $\theta = P(C_1) = P(C_2)$, and assume C_1 and C_2 are independent events. Derive an expression (involving θ) for the probability that exactly one tag is lost, given that at most one is lost ([“Ear Tag Loss in Red Foxes”, J. Wildlife Mgmt., 1976: 164-167](#)).

Supplementary Exercises

19. One satellite is scheduled to be launched from Cape Canaveral in Florida, and another launching is scheduled for Vandenberg Air Force Base in California. Let A denote the event that the Vandenberg launch goes off on schedule, and let B represent the event that the Cape Canaveral launch goes off on schedule. If A and B are independent events with $P(A) > P(B)$, $P(A \cup B) = .626$, and $P(A \cap B) = .144$, determine the values of $P(A)$ and $P(B)$.

20. Individual A has a circle of five close friends (B, C, D, E, and F). A has heard a certain rumor from outside the circle and has invited the five friends to a party to circulate the rumor. To begin, A selects one of the five at random and tells the rumor to the chosen individual. That individual then selects at random one of the four remaining individuals and repeats the rumor. Continuing, a new individual is selected from those not already having heard the rumor by the individual who has just heard it, until everyone has been told.
- What is the probability that the rumor is repeated in the order B, C, D, E, and F?
 - What is the probability that F is the third person at the party to be told the rumor?
 - What is the probability that F is the last person to hear the rumor?
 - If at each stage the person who currently has the rumor does not know who has already heard it and selects the next recipient at random from all five possible individuals, what is the probability that F has still not heard the rumor after it has been told ten times at the party?
21. Disregarding the possibility of a February 29 birthday, suppose a randomly selected individual is equally likely to have been born on any one of the other 365 days.
- If ten people are randomly selected, what is the probability that all have different birthdays? That at least two have the same birthday?
 - With k replacing ten in part (a), what is the smallest k for which there is at least a 50-50 chance that two or more people will have the same birthday?
 - If ten people are randomly selected, what is the probability that either at least two have the same birthday or at least two have the same last three digits of their Social Security numbers? [Note: The article “Methods for Studying Coincidences” (F. Mosteller and P. Diaconis, *J. Amer. Stat. Assoc.*, 1989: 853-861) discusses problems of this type.]
22. A box contains the following four slips of paper, each having exactly the same dimensions: (1) win prize 1; (2) win prize 2; (3) win prize 3; (4) win prizes 1, 2, and 3. One slip will be randomly selected. Let $A_1 = \{\text{win prize 1}\}$, $A_2 = \{\text{win prize 2}\}$, and $A_3 = \{\text{win prize 3}\}$. Show that A_1 and A_2 are independent, that A_1 and A_3 are independent, and that A_2 and A_3 are also independent (this is pairwise independence). However, show that $P(A_1 \cap A_2 \cap A_3) \neq P(A_1) \cdot P(A_2) \cdot P(A_3)$, so the three events are not mutually independent. [Note: A_1 =draw slip 1 or 4.]

Chapter 2

Random variables and distributions

Whether in scientific research or engineering practice, there is a **critical** need for the quantitative prediction of uncertainty. Random events are often expressed as abstract verbal descriptions (e.g., “rolling an even number”, “rainfall tomorrow”) or complex set-theoretic expressions, which are only suitable for **qualitative descriptions**. To conduct quantitative research on the uncertainty inherent in random phenomena, it is necessary to convert random events into numerical values.

Random variables serve as a “translation tool” — converting abstract, non-numerical random phenomena (such as dice roll outcomes or weather conditions) into specific real numbers, thereby enabling us to analyze uncertainty using quantitative tools, **such as** Calculus and Linear Algebra.

2.1 Random variable

2.1.1 Random variable

Definition 2.1.1 (Random variable). Let Ω be a sample space. A real-valued function X defined on Ω is called a *random variable*.

We remark that the “randomness” comes from the fact that we do not know which outcome ω will occur (and thus which value x the random variable X will take) *before* the experiment. However, the mapping rule of X is fixed. Thus

“Random” = Uncertain Outcome, Not Uncertain Rule

Example 2.1.1. Let $\Omega = \{1, 2, 3, 4, 5, 6\}$ be the sample space of rolling a die.

- Define X = “value of the die roll”. Then $X(1) = 1, X(2) = 2, \dots, X(6) = 6$.
- If we focus on parity and define $Y = 1$ if the roll is even, $Y = 0$ if odd. Then $Y(1) = 0, Y(2) = 1, Y(3) = 0, Y(4) = 1, Y(5) = 0, Y(6) = 1$.

Example 2.1.2. A class is purchasing custom uniforms, and the clothing sizes are determined by students’ heights. The available sizes are S (Small), M (Medium), L (Large), and XL (Extra Large). The class first measures the height of each student (all measurements are in centimeters, cm), and then maps each height to a corresponding size based on the following predefined criteria:

- Size S: Height < 155 cm;
- Size M: $155 \text{ cm} \leq \text{Height} < 165$ cm;
- Size L: $165 \text{ cm} \leq \text{Height} < 175$ cm;
- Size XL: Height ≥ 175 cm.

Let the sample space Ω be the set of all students in the class. Let X be a (*categorical*) random variable representing the uniform size assigned to a randomly selected student. For example, for student A with height 153 cm, $X(A) = S$; for student C with 160 cm, $X(C) = M$. In practical scenarios, we often care about the probability that X takes a specific value, which helps the class estimate the number of uniforms of each size to purchase. For instance, the probability that a randomly selected student needs size M (i.e. $X = M$).

Example 2.1.3. Let Ω be the sample space of tossing five fair coins. Let X denote the number of heads in an outcome. Then $X(\text{HTHTH}) = 3$ and $X(\text{TTTHH}) = 2$. We are interested in the probability of X taking certain values. For example, consider the probability of exactly 1 heads occur (i.e. $X = 1$). Formally,

$$\begin{aligned} \{X = 1\} &\triangleq \{\omega \in \Omega: X(\omega) = 1\} \\ &= \{\text{HTTTT}, \text{THTTT}, \text{TTHTT}, \text{TTTHT}, \text{TTTTH}\}. \end{aligned}$$

If there is a probability on the sample space Ω , then X brings the probability to $\mathbb{R}$ in the following way: for any subset $B \subseteq \mathbb{R}$, the probability of that X takes values in B (denoted as $P_X(B)$) is defined as the probability of the preimage $X^{-1}(B)$ on Ω :

$$P_X(B) \triangleq P(X^{-1}(B)).$$

For instance, consider the sample space Ω corresponding to tossing five fair coins, which consists of all possible outcomes of 5 coin flips with cardinality $|\Omega| = 2^5 =$

32. Since the coins are fair, each outcome $\omega \in \Omega$ is equally likely to occur. Thus, we have a well-defined probability on Ω : for every $\omega \in \Omega$, $\mathbb{P}(\{\omega\}) = \frac{1}{32}$. For any event $A \subseteq \Omega$, its probability is $\mathbb{P}(A) = \frac{|A|}{32}$. Taking $B = \{1\} \subseteq \mathbb{R}$, its preimage is exactly the event given in the example:

$$X^{-1}(\{1\}) = \{X = 1\} = \{\text{HTTTT}, \text{THTTT}, \text{TTHTT}, \text{TTTHT}, \text{TTTTH}\}.$$

The number of basic outcomes in this preimage is $|X^{-1}(\{1\})| = 5$, so the induced probability on $\mathbb{R}$ is:

$$P_X(\{1\}) = P(X^{-1}(\{1\})) = \frac{5}{32}.$$

In this manner, the probability originally defined on the abstract sample space Ω is successfully “transported” to the real number set $\mathbb{R}$ via the random variable X . This enables us to directly analyze the probabilistic behavior of X through concrete real numbers, simplifying the study of random phenomena.

2.1.2 Distribution function

The randomness of a random variable X lies essentially in the fact that the event of X taking values in any set of real numbers C is a random event. Therefore, to fully characterize the randomness of X , it is crucial to determine the probability of the event $X \in C$.

However, due to the diversity of real number sets, it is impractical to calculate the probabilities corresponding to all sets one by one. Notably, the probabilities of all real number sets can be derived from the probability of the simplest interval $(-\infty, x]$ (where $x \in \mathbb{R}$). Based on this, we introduce the core tool for characterizing random variables—the *cumulative distribution function* of X .

Definition 2.1.2 (Distribution Function). Let Ω be a sample space and let X be a random variable over Ω . The *cumulative distribution function* (cdf) of X is formally defined as follows: for all $x \in \mathbb{R}$,

$$F(x) \triangleq \mathbb{P}(X \leq x). \quad (2.1)$$

Directly from Definition 2.1.2, two fundamental inferences can be drawn. First, the cdf is bounded between 0 and 1, i.e., $0 \leq F(x) \leq 1$ for all $x \in \mathbb{R}$. Second, for any two real numbers satisfying $a < b$, the probability of X falling within the half-open interval $(a, b]$ can be computed via the difference of the cdf values:

$$\mathbb{P}(a < X \leq b) = \mathbb{P}(X \leq b) - \mathbb{P}(X \leq a) = F(b) - F(a).$$

The cumulative distribution function $F(x)$ satisfies three core defining properties, as listed below:

Proposition 2.1.1. *Let X be a random variable on Ω and let F be its distribution function defined as in Eq. (2.1).*

1. **Monotonically Non-decreasing:** *For any real numbers $x < y$, it holds that $F(x) \leq F(y)$;*
2. **Boundary Conditions at Infinity:** *The limits of $F(x)$ as x tends to positive and negative infinity are*

$$\lim_{x \rightarrow -\infty} F(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow +\infty} F(x) = 1;$$

3. **Right-Continuity:** *For any fixed real number a , the right-hand limit of $F(x)$ at a equals the function value at a , i.e.,*

$$\lim_{x \rightarrow a^+} F(x) = F(a).$$

Proof. 1. Note that $(-\infty, x] \subset (-\infty, y]$ for all $x < y$. We have

$$F(x) = \mathbb{P}((-\infty, x]) \leq \mathbb{P}((-\infty, y]) = F(y).$$

2. For $n = 1, 2, 3, \dots$, let $A_n = \{X \leq -n\}$. Then $A_1 \supset A_2 \supset A_3 \supset \dots$ and $\bigcap_{n=1}^{\infty} A_n = \emptyset$. Applying the continuity of the probability $\mathbb{P}$, we have

$$\lim_{n \rightarrow \infty} \mathbb{P}(A_n) = \mathbb{P}(\bigcap_{n=1}^{\infty} A_n) = \mathbb{P}(\emptyset) = 0.$$

For any $x \in \mathbb{R}$, there exists an integer $n(x)$ such that $-n(x) - 1 < x \leq -n(x)$. Then $F(x) = \mathbb{P}((-\infty, x]) \leq \mathbb{P}((-\infty, -n(x)]) = \mathbb{P}(A_{n(x)})$. When $x \rightarrow -\infty$, we have $n \rightarrow +\infty$ and

$$0 \leq \lim_{x \rightarrow -\infty} F(x) \leq \lim_{n \rightarrow \infty} \mathbb{P}(A_n) = 0.$$

Similarly, one has $\lim_{x \rightarrow +\infty} F(x) = 1$.

3. For any $x \in [a, a+1)$, there exists an integer n such that $x \leq a + \frac{1}{n}$. When $x \rightarrow a^+$, we have $n \rightarrow \infty$ and

$$\begin{aligned} F(a) &\leq \lim_{x \rightarrow a^+} F(x) \leq \lim_{n \rightarrow \infty} F(a + \frac{1}{n}) \\ &= \lim_{n \rightarrow \infty} \mathbb{P}((-\infty, a + \frac{1}{n}]) \\ &= \mathbb{P}(\bigcap_{n=1}^{\infty} (-\infty, a + \frac{1}{n}]) \\ &= \mathbb{P}((-\infty, a]) = F(a). \end{aligned}$$

Therefore, $\lim_{x \rightarrow a^+} F(x) = F(a)$. □

Example 2.1.4. Let t be a real number and let $X = t$ be the random variable that takes constant value. Then $\mathbb{P}(X = t) = 1$ and $\mathbb{P}(X = a) = 0$ for all $a \neq t$. The cumulative distribution function (cdf) of X is

$$F(x) = \delta_t(x) := \begin{cases} 0, & \text{for } x < t; \\ 1, & \text{for } x \geq t. \end{cases}$$

We call the function δ_t *the point mass of t* (see Figure 2.1).

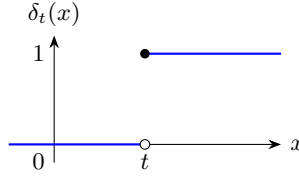


Figure 2.1: The point mass of t in Example 2.1.4

Example 2.1.5. Consider a coin toss experiment. Let X be a random variable defined as follows: $X = 1$ if the toss results in heads, and $X = 0$ if it results in tails. Suppose the probability of heads is p (where $0 < p < 1$), so that

$$\mathbb{P}(X = 1) = p \quad \text{and} \quad \mathbb{P}(X = 0) = 1 - p.$$

The cdf of X is a step function (see Figure 2.2), explicitly given by

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - p, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

Then $F(x) = (1 - p)\delta_0(x) + p\delta_1(x)$.

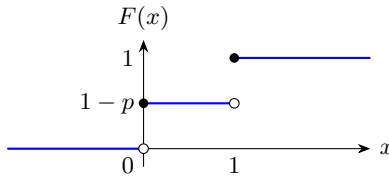


Figure 2.2: Step function (The cdf in Example 2.1.5)

Example 2.1.6. The engineers found that the lifetime X (in hours) of electronic equipment (e.g., resistors) has the cumulative distribution function F (see Figure 2.3):

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-\lambda x}, & x \geq 0, \end{cases}$$

where $\lambda = 0.5$ is failure rate. Here $F(x)$ is continuous.

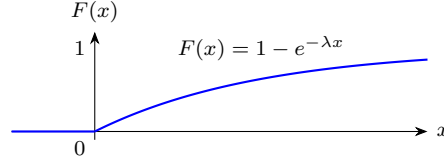


Figure 2.3: Non-step cdf (in Example 2.1.6)

Example 2.1.7. Lithium-ion batteries produced by a production line have a 80% pass rate, and the qualified units have lifetimes that follow a uniform distribution between 100 and 200 hours. Let X denote the lifetime (in hours) of the battery. The cumulative distribution function $F(x) = \mathbb{P}(X \leq x)$ (see Figure 2.4) is:

$$F(x) = \begin{cases} 0, & x < 0, \\ 0.2, & 0 \leq x < 100, \\ 0.2 + 0.8 \cdot \frac{x-100}{100}, & 100 \leq x < 200, \\ 1, & x \geq 200. \end{cases}$$

Let $F_c(x) = \frac{1}{0.8}(F(x) - 0.2\delta_0(x))$. Then $F_c(x)$ is continuous and

$$F(x) = 0.2\delta_0(x) + 0.8F_c(x).$$

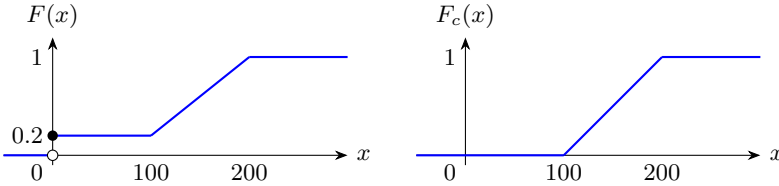


Figure 2.4: The cdf in Example 2.1.7

From the previous examples, we have a feeling that besides the ‘jumps’, the cdf’s are continuous. Indeed, it is true.

Definition 2.1.3 (Distribution Function). A real-valued function F with domain $(-\infty, +\infty)$ that is increasing and right continuous with $F(-\infty) = 0$, $F(+\infty) = 1$ is called a *distribution function* (in abbreviation d.f.). A d.f. F that can be represented in the form

$$F(x) = \sum_j b_j \delta_j(x)$$

where $\{a_j\}$ is a countable set of real numbers, $b_j > 0$ for every j and $\sum_j b_j = 1$ is called a *discrete* d.f. A d.f. that is continuous everywhere is called a *continuous*¹ d.f.

Theorem 2.1.1 (Theorem 1.2.3 in [1]). *Every d.f. can be written as the convex combination of a discrete and a continuous one. Such a decomposition is unique.*

The classification of random variables is inherently determined by the type of their corresponding cumulative distribution functions (cdf’s).

1. A random variable is *discrete* if and only if its cdf is a step function that increases only at a finite or countably infinite set of points. (See Example 2.1.4 and 2.1.5)
2. A random variable is *continuous* if its cdf can be expressed as the integral of an integrable density function. (See Example 2.1.6)
3. A random variable is *mixed* if its cdf is a weighted combination of discrete and continuous components, i.e., $F(x) = \alpha F_d(x) + (1 - \alpha) F_c(x)$ where $0 < \alpha < 1$, $F_d(x)$ denotes a discrete cdf, and $F_c(x)$ denotes a continuous cdf. (See Example 2.1.7)

In this course, our discussion will focus exclusively on the first two types of random variables introduced above.

¹It is worth noting that continuity does not imply differentiability. A deeper analysis of distribution functions (d.f.s) necessitates the theory of Lebesgue measure. Specifically, continuous distribution functions can be partitioned into two categories: absolutely continuous ones and singular continuous ones (see [1, Section 1.3]). Given that this course restricts prerequisite knowledge to Calculus, when we refer to “continuous d.f.s” herein, we specifically mean **absolutely continuous distribution functions**—that is, those for which there exists a integrable positive function f such that for every $a < b$,

$$F(b) - F(a) = \int_a^b f(x) \, dx.$$

We call f the *density* of F .

2.1.3 Discrete random variables

Definition 2.1.4 (Discrete Random Variables). A random variable X is said to be *discrete* if it can take on only a finite set of distinct values or, at most, a countably infinite sequence of distinct values.

It is easy to see that X is discrete if and only if its d.f. is.

Definition 2.1.5 (p.m.f.). If X is a discrete r.v., the *probability mass function* (in abbreviation p.m.f.) is defined as the function that for every real number x ,

$$p(x) := \mathbb{P}(X = x).$$

The p.m.f. of a discrete r.v. X has the following fundamental properties:

1. $p(x) \geq 0$ for all $x \in \mathbb{R}$, and $p(x) = 0$ if x is not a possible value of X .
2. The total probability mass sums to 1. Specifically, if the distinct possible values of X are enumerated as $x_1, x_2, \dots$ (which may form a finite or countably infinite sequence), then $\sum_i p(x_i) = 1$.
3. The probability of each subset $C \subseteq \mathbb{R}$ can be determined by

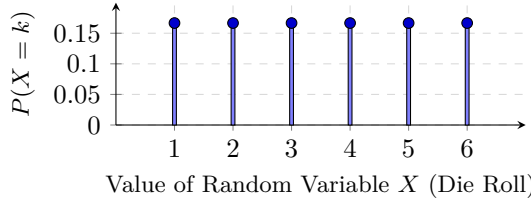
$$\mathbb{P}(X \in C) = \sum_{x \in C} p(x).$$

Thus, the p.m.f. is an alternative tool for describing the probability distribution of a discrete random variable, in addition to the c.d.f.

Example 2.1.8 (Rolling a Fair Six-Sided Die). Let X represent the *number showing on the die after a single roll*. The set of all possible values (sample space) for X is $\{1, 2, 3, 4, 5, 6\}$. This set is countable and finite. For a *fair die* (each outcome is equally likely), the probability of each value of X is:

$$P(X = k) = \frac{1}{6} \quad \text{for } k = 1, 2, 3, 4, 5, 6.$$

The probability histogram is the following:



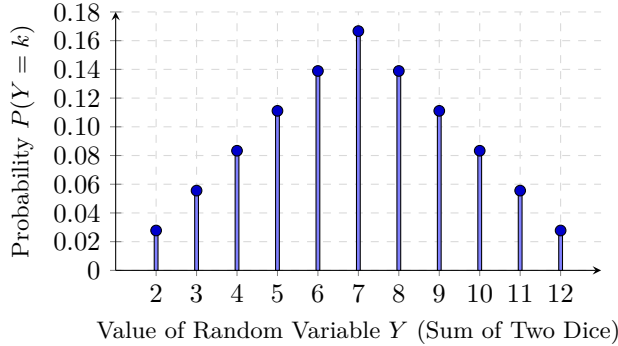
Example 2.1.9 (Sum of Two Six-Sided Dice Rolls). Roll two distinct fair six-sided dice. The outcomes are pair of numbers (d_1, d_2) where $d_1, d_2 \in \{1, 2, 3, 4, 5, 6\}$. Let $Y = d_1 + d_2$ represent the sum of the two dice. The minimum possible sum is 2, and the maximum is 12. Thus, the set of all possible values for Y is:

$$Y \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}.$$

For a fair die, each of the 36 outcome pairs has a probability of $\frac{1}{36}$. We calculate $P(Y = k)$ (the probability mass function, PMF) by counting how many pairs sum to k :

Value of Y	Favorable Outcomes	$P(Y = k)$
2	(1, 1)	$\frac{1}{36}$
3	(1, 2), (2, 1)	$\frac{2}{36}$
4	(1, 3), (2, 2), (3, 1)	$\frac{3}{36}$
5	(1, 4), (2, 3), (3, 2), (4, 1)	$\frac{4}{36}$
6	(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)	$\frac{5}{36}$
7	(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)	$\frac{6}{36}$
8	(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)	$\frac{5}{36}$
9	(3, 6), (4, 5), (5, 4), (6, 3)	$\frac{4}{36}$
10	(4, 6), (5, 5), (6, 4)	$\frac{3}{36}$
11	(5, 6), (6, 5)	$\frac{2}{36}$
12	(6, 6)	$\frac{1}{36}$

The probability histogram is



Example 2.1.10. Find $k \in \mathbb{Z}_{\geq 1}$ and $c \in \mathbb{R}$ so that the following function would be a legitimate pmf:

$$p(x) = \begin{cases} \frac{c}{x^k}, & x = 1, 2, 3, \dots, \\ 0, & \text{otherwise.} \end{cases}$$

Solution. For $p(x)$ to be a legitimate probability mass function (pmf), two conditions must be satisfied: $p(x) \geq 0$ for all x , and $\sum_x p(x) = 1$.

First, we require $c \geq 0$ (to ensure non-negativity of $p(x)$) and $\sum_{x=1}^{\infty} \frac{c}{x^k} = 1$. When $k = 1$, the harmonic series $\sum_{x=1}^{\infty} \frac{1}{x}$ diverges to infinity. Thus, $\sum_{x=1}^{\infty} \frac{c}{x^k}$ equals 0 if $c = 0$ and ∞ if $c > 0$. In either case, $\sum_{x=1}^{\infty} \frac{c}{x^k} \neq 1$, so $p(x)$ cannot be a valid pmf when $k = 1$.

When $k \geq 2$, the series² $\sum_{x=1}^{\infty} \frac{1}{x^k}$ is convergent (by the integral test):

$$\sum_{x=1}^{\infty} \frac{1}{x^k} \leq \int_1^{\infty} \frac{1}{x^k} dx = \frac{1}{k-1} < \infty.$$

By setting $c = (\sum_{x=1}^{\infty} \frac{1}{x^k})^{-1}$, we ensure $\sum_{x=1}^{\infty} p(x) = 1$ and $p(x) \geq 0$ (since $c > 0$). Therefore, $p(x)$ is a legitimate pmf for all integers $k \geq 2$. $\square$

2.1.4 Continuous random variables

We now turn to random variables with continuous distribution functions. For technical reasons, we restrict our attention to random variables whose distribution functions are expressible as the integral of an integrable function.

Definition 2.1.6 (Continuous Random variables). A random variable X is called *continuous* if there exists a nonnegative function f , defined on the real line, such that for any $a, b \in \mathbb{R}$ with $a \leq b$,

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x) dx.$$

Definition 2.1.7 (pdf & support). The function f (described in Definition 2.1.6) is called the *probability density function* (abbreviated pdf) of X . The closure of the set $\{x : f(x) > 0\}$ is called the *support* of (the distribution of) X .

Recall that the distribution function of a random variable X is defined as $F(x) = \mathbb{P}(X \leq x)$. In the continuous case, it is convenient to obtain the distribution function from the pdf and vice versa. In fact, if X is a continuous random variable with pdf $f(x)$, then by Definition 2.1.6, its distribution function is

$$F(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x f(t) dt,$$

²We remark that $\zeta(k) = \sum_{x=1}^{\infty} \frac{1}{x^k}$ is the Riemann zeta function. For even $k \geq 2$, $\zeta(k)$ has an explicit form in terms of Bernoulli numbers, while for odd $k \geq 3$, it has no elementary closed-form representation.

which is continuous. Using the fundamental theorem of calculus, we also see that

$$\frac{dF(x)}{dx} = f(x).$$

Any valid probability density function (p.d.f.) $f(x)$ must satisfy the following two fundamental properties:

1. **Nonnegativity:** $f(x) \geq 0$ for all $x \in \mathbb{R}$;
2. **Normalization (unit total density):** $\int_{-\infty}^{\infty} f(x) dx = 1$.

In the discrete case, positive probability (or intuitively, *probability mass*) can be assigned to or accumulated at a single point (see Examples 2.1.8 and 2.1.9). In contrast, for continuous random variables, no positive probability can be assigned to any individual real value. As follows from Definition 2.1.6, for any $a \in \mathbb{R}$,

$$\mathbb{P}(X = a) = \int_a^a f(x) dx = 0.$$

We remark that probability zero does not mean impossible. If $\mathbb{P}(X = a) = 0$ implies $X = a$ is impossible, then all values of X would be impossible and X could not assume any value (a contradiction). In the shooting target model, where a bullet lands uniformly at random on a circular target of radius R , the event that the bullet lands exactly at a specific point P has probability

$$\mathbb{P}(A) = \frac{\text{Area of point } P}{\text{Total area of target}} = \frac{0}{\pi R^2} = 0,$$

yet this event is still possible. When we simulate a real-world problem using a continuous random variable, the event $X \approx a$ is more meaningful than $X = a$ (since error is allowed in the physical world). For each a such that $f(a) > 0$ and for any $\varepsilon > 0$,

$$\mathbb{P}(a - \varepsilon \leq X \leq a + \varepsilon) = \int_{a-\varepsilon}^{a+\varepsilon} f(x) dx \approx 2\varepsilon f(a) > 0,$$

which is intuitively reasonable.

Finally, we remark that the pdf of a continuous random variable is not unique. Due to the subtle properties of the Riemann integral, the values of a pdf can be altered at finitely many points (or even at certain countably infinite sequences of points) without changing the value of its integral over any subset $A \subseteq \mathbb{R}$. This means the probability that X takes a value in A remains unchanged. Any such modified version of the original pdf is also a pdf of the random variable.

Example 2.1.11. Let X be a continuous random variable with the pdf

$$f(x) = \begin{cases} \frac{x-1}{4}, & 1 \leq x < 3, \\ \frac{5-x}{4}, & 3 \leq x \leq 5, \\ 0, & \text{otherwise.} \end{cases}$$

Then $f(x) \geq 0$ for all x and $\int_{-\infty}^{+\infty} f(x) dx = 1$. In particular,

$$\mathbb{P}(2 \leq X \leq 4) = \int_2^4 f(x) dx = \int_2^3 \frac{x-1}{4} dx + \int_3^4 \frac{5-x}{4} dx = \frac{3}{4}.$$

The distribution function of X is

$$F(x) = \int_{-\infty}^x f(t) dt = \begin{cases} 0, & \text{if } x < 1, \\ \frac{(x-1)^2}{8} x^2, & \text{if } 1 \leq x < 3, \\ 1 - \frac{(5-x)^2}{8}, & \text{if } 3 \leq x \leq 5, \\ 1, & \text{if } x > 5. \end{cases}$$

The graphs of $f(x)$ and $F(x)$ are given in Figure 2.5.

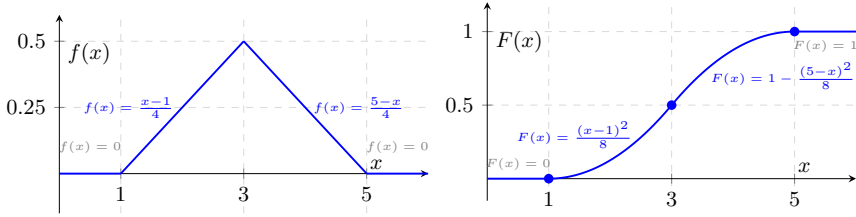


Figure 2.5: PDF and DF in Example 2.1.11.

Example 2.1.12. Let X be a continuous random variable with the following pdf:

$$f(x) = k \cdot x e^{-x^2/2}, \quad x > 0,$$

where k is an unknown constant. Find k and the df of X .

Solution. For f to be a valid probability density function (pdf), two conditions must hold: $f(x) \geq 0$ for all $x \in \mathbb{R}$, and $\int_{-\infty}^{+\infty} f(x) dx = 1$. The non-negativity condition $f(x) \geq 0$ implies $k \geq 0$. The normalization condition gives:

$$\int_{-\infty}^{+\infty} f(x) dx = \int_0^{+\infty} f(x) dx + \int_{-\infty}^0 0 dx = \int_0^{+\infty} k x e^{-x^2/2} dx = 1.$$

Solving this integral yields $k = 1$. The distribution function of X is

$$F(x) = \int_{-\infty}^x f(t) dt = \begin{cases} 0, & \text{if } x \leq 0, \\ 1 - e^{-x^2/2}, & \text{if } x > 0. \end{cases}$$

□

2.2 Bivariate distributions

On many occasions, a single random variable is not sufficient to model real-world problems. For example, in the shooting problem, the impact point should be represented as a pair of numbers (X, Y) —that is, as a vector—rather than as a single number. To characterize the randomness of impact points, we need to analyze the distribution of the random vector (X, Y) .

2.2.1 Random vectors

Bivariate distribution

Definition 2.2.1 (joint cdf). The *joint (cumulative) distribution function* (in abbreviation *joint cdf*) of two random variables X and Y is defined as

$$F(x, y) = \mathbb{P}(X \leq x, Y \leq y), \quad \forall x, y \in \mathbb{R}$$

One sees that $F(x, y)$ is monotone increasing in x for each fixed y and is monotone increasing in y for each fixed x .

As in the univariate case, Definition 2.2.1 allows us to characterize the probability that $(X, Y) \in C$ for any³ subset $C \subset \mathbb{R}^2$. In particular, one can find the probability of the pair (X, Y) lies in a rectangle $(a, b] \times (c, d]$:

$$\begin{aligned} \mathbb{P}(a < X \leq b, c < Y \leq d) &= \mathbb{P}(a < X \leq b, Y \leq d) - \mathbb{P}(a < X \leq b, Y \leq c) \\ &= [\mathbb{P}(X \leq b, Y \leq d) - \mathbb{P}(X \leq a, Y \leq d)] \\ &\quad - [\mathbb{P}(X \leq b, Y \leq c) - \mathbb{P}(X \leq a, Y \leq c)] \\ &= F(b, d) - F(a, d) - F(b, c) + F(a, c). \end{aligned}$$

Discrete Joint Distribution

Definition 2.2.2. Let X and Y be random variables. We say that X and Y have a *discrete joint distribution* if the pair (X, Y) takes only finitely or countably many distinct values (x, y) .

³In fact, it is impossible to compute the probability that $(X, Y) \in C$ for **literally all subsets** C . In real analysis or measure theory, C may range over all Borel sets. This is sufficient for most applications in science and engineering.

It is easy to see that if X and Y have a discrete joint distribution, then X and Y are both discrete random variables and vice versa.

Definition 2.2.3. The *joint probability mass function* (joint pmf) of X and Y is defined as the function $p(x, y)$ such that for all $(x, y) \in \mathbb{R}^2$,

$$p(x, y) = \mathbb{P}(X = x, Y = y).$$

The joint pmf has the following properties:

- (Nonnegativity) $p(x, y) \geq 0$ for all x and y . In particular, if (x, y) is not a possible value, then $p(x, y) = 0$.
- (Normalization) $\sum_{\text{All } (x, y)} p(x, y) = 1$.
- For any subset $C \in \mathbb{R}^2$, $\mathbb{P}((X, Y) \in C) = \sum_{(x, y) \in C} p(x, y)$.

Example 2.2.1 (Two Dice Rolls). Consider the experiment of rolling two fair six-sided dice. Let X be the outcome of the first die. Let Y be the sum of the outcomes of the two dice. Let $p(x, y) = \mathbb{P}(X = x, Y = y)$ denote the joint pmf of X and Y . The non-zero values of $p(x, y)$ are:

$y \setminus x$	1	2	3	4	5	6
2	1/36	0	0	0	0	0
3	1/36	1/36	0	0	0	0
4	1/36	1/36	1/36	0	0	0
5	1/36	1/36	1/36	1/36	0	0
6	1/36	1/36	1/36	1/36	1/36	0
7	1/36	1/36	1/36	1/36	1/36	1/36
8	0	1/36	1/36	1/36	1/36	1/36
9	0	0	1/36	1/36	1/36	1/36
10	0	0	0	1/36	1/36	1/36
11	0	0	0	0	1/36	1/36
12	0	0	0	0	0	1/36

We can see that $p(x, y) \geq 0$ and $\sum p(x, y) = 1$. In particular,

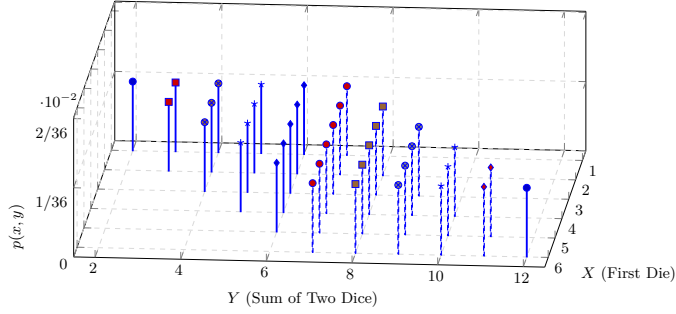
$$\mathbb{P}(X \geq 5, Y \geq 10) = p(5, 10) + p(6, 10) + p(5, 11) + p(6, 11) + p(6, 12) = \frac{5}{36}.$$

Joint Continuous Distribution

Definition 2.2.4. Two random variables X and Y have a continuous joint distribution if there exists a nonnegative function f defined on $\mathbb{R}^2$ such that for any subset $C \subset \mathbb{R}^2$,

$$\mathbb{P}((X, Y) \in C) = \iint_{(x, y) \in C} f(x, y) dx dy,$$

provided that the integral exists. The function f is called the *joint probability density function* (abbreviated as *joint pdf*) of X and Y .


 Figure 2.6: Joint pmf of X and Y in Example 2.2.1

Similar to the univariate case, the joint pdf has the following properties:

1. (Nonnegativity) $f(x, y) \geq 0$ for all $(x, y) \in \mathbb{R}^2$;
2. (Normalization) $\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy = 1$.
3. For any $(a, b) \in \mathbb{R}^2$,

$$\mathbb{P}((X, Y) = (a, b)) = \iint_{(x, y) = (a, b)} f(x, y) dx dy = 0.$$

4. The joint df can be derived from the joint pdf and vice versa:

$$F(x, y) = \int_{-\infty}^x \int_{-\infty}^y f(u, v) du dv$$

and

$$f(x, y) = \frac{\partial^2 F(x, y)}{\partial x \partial y} = \frac{\partial^2 F(x, y)}{\partial y \partial x}.$$

Example 2.2.2. Consider two random variables X and Y whose joint pdf $f(x, y)$ is defined as follows:

$$f(x, y) = \begin{cases} 8xy \cdot e^{-(2x^2+4y^2)}, & \text{if } x > 0, y > 0, \text{ and } x^2 < y < 2x^2, \\ 0, & \text{otherwise.} \end{cases}$$

We see that $f \geq 0$ and

$$\iint_{\mathbb{R}^2} f(x, y) dx dy = \int_0^{+\infty} \int_{x^2}^{2x^2} 8xy \cdot e^{-(2x^2+4y^2)} dy dx$$

$$\begin{aligned}
 &= \int_0^{+\infty} x e^{-2x^2} \cdot \frac{1}{8} \left[-e^{-4y^2} \right]_{y=x^2}^{2x^2} dx \\
 &= \int_0^{+\infty} x e^{-2x^2} \left(e^{-4x^4} - e^{-16x^4} \right) dx = 1.
 \end{aligned}$$

Example 2.2.3. Let X and Y be two jointly continuous random variables with the following joint probability density function:

$$f(x, y) = \begin{cases} 2(x + y), & 0 \leq x \leq 1, 0 \leq y \leq x, \\ 0, & \text{otherwise.} \end{cases}$$

Compute $\mathbb{P}(Y \leq \frac{1}{2}, X \leq \frac{1}{2})$ and $\mathbb{P}(X + Y \leq 1)$.

Solution. (1) The region of integration is defined by:

$$0 \leq x \leq \frac{1}{2}, \quad 0 \leq y \leq x.$$

This is a triangular subregion of the support of (X, Y) . Then

$$\begin{aligned}
 \mathbb{P}\left(Y \leq \frac{1}{2}, X \leq \frac{1}{2}\right) &= \int_0^{\frac{1}{2}} \int_0^x 2(x + y) dy dx \\
 &= \int_0^{\frac{1}{2}} 2 \left[xy + \frac{y^2}{2} \right]_0^x dx \\
 &= \int_0^{\frac{1}{2}} 3x^2 dx \\
 &= 3 \left[\frac{x^3}{3} \right]_0^{\frac{1}{2}} = \frac{1}{8}.
 \end{aligned}$$

(2) Similarly, one has

$$\begin{aligned}
 \mathbb{P}(X + Y \leq 1) &= \int_0^{\frac{1}{2}} \int_0^x 2(x + y) dy dx + \int_{\frac{1}{2}}^1 \int_0^{1-x} 2(x + y) dy dx \\
 &= \frac{1}{8} + \int_{\frac{1}{2}}^1 \int_0^{1-x} 2(x + y) dy dx \\
 &= \frac{1}{8} + \int_{\frac{1}{2}}^1 (1 - x^2) dx = \frac{1}{8} + \frac{5}{24} = \frac{1}{3}.
 \end{aligned}$$

□

In the discrete case, X and Y have a joint discrete distribution if and only if X and Y are both discrete random variables. However, this equivalence does not hold in the continuous case.

Example 2.2.4. Let X be a continuous random variable and let $Y = X$. Then $\mathbb{P}(X = Y) = 1$. If X and Y were jointly continuous with joint pdf f , then

$$\mathbb{P}(X = Y) = \iint_L f(x, y) dx dy = 0,$$

where L is the line $y = x$. This contradicts the fact that $\mathbb{P}(X = Y) = 1$. Hence, X and Y cannot have a joint continuous distribution.

2.2.2 Independent random variables

Marginal distribution

The joint distribution function contains richer information compared to the distribution function of X and the distribution function of Y . One can find the distribution functions of X and Y from their joint distribution function.

Proposition 2.2.1. Let X and Y be random variables with joint cumulative distribution function F . Then the distribution function F_1 of X is $F_1(x) = \lim_{y \rightarrow +\infty} F(x, y)$ and the df F_2 of Y is $F_2(y) = \lim_{x \rightarrow +\infty} F(x, y)$.

Proof. We only prove for F_1 . Note that $\{Y < +\infty\}$ is the sample space and $\{Y < +\infty\} = \cup_{n \geq 1} \{Y \leq n\}$. Then

$$\begin{aligned} F_1(x) &= \mathbb{P}(X \leq x) = \mathbb{P}(X \leq x, Y < +\infty) \\ &= \mathbb{P}(X \leq x, \cup_{n \geq 1} \{Y \leq n\}) \\ &= \mathbb{P}(\cup_{n \geq 1} (\{X \leq x\} \cap \{Y \leq n\})) \\ &= \lim_{n \rightarrow \infty} \mathbb{P}(X \leq x, Y \leq n) \\ &= \lim_{n \rightarrow \infty} F(x, n) = \lim_{y \rightarrow +\infty} F(x, y). \end{aligned} \quad \square$$

Definition 2.2.5 (Marginal df). The functions F_1 and F_2 , obtained in Proposition 2.2.1, are called the *marginal distribution functions* of X and Y , respectively.

Definition 2.2.6 (Marginal pmf & pdf). If X and Y are jointly discrete (resp. continuous), then the pmfs p_1 and p_2 (resp. the pdfs f_1 and f_2) of X and Y are called the *marginal pmfs* (resp. *marginal pdfs*) of X and Y , respectively.

The marginal pmfs or pdfs can also be derived from the joint pmf or pdf.

Proposition 2.2.2. *Let X and Y be jointly discrete (or continuous) with joint pmf $p(x, y)$ (or joint pdf $f(x, y)$).*

1. *In the discrete case, the marginal pmfs p_1 and p_2 of X and Y can be derived by*

$$p_1(x) = \sum_{\text{all } y} p(x, y), \quad p_2(y) = \sum_{\text{all } x} p(x, y).$$

2. *In the continuous case, the marginal pdfs f_1 and f_2 of X and Y can be derived by*

$$f_1(x) = \int_{-\infty}^{+\infty} f(x, y) dy, \quad f_2(y) = \int_{-\infty}^{+\infty} f(x, y) dx.$$

Proof. (1) Note that the event $\{-\infty < Y < +\infty\}$ is the sample space, and $\{-\infty < Y < +\infty\}$ can be written as the disjoint union $\bigcup_{\text{all } y} \{Y = y\}$. Thus

$$\begin{aligned} p_1(x) &= \mathbb{P}(X = x) = \mathbb{P}(X = x, -\infty < Y < +\infty) \\ &= \mathbb{P}\left(\bigcup_{\text{all } y} \{X = x, Y = y\}\right) \\ &= \sum_{\text{all } y} \mathbb{P}(X = x, Y = y) = \sum_{\text{all } y} p(x, y). \end{aligned}$$

- (2) For any $a, b \in \mathbb{R}$ with $a < b$,

$$\begin{aligned} \mathbb{P}(a \leq X \leq b) &= \mathbb{P}(a \leq X \leq b, -\infty < Y < +\infty) \\ &= \int_{-\infty}^{+\infty} \int_a^b f(x, y) dx dy \\ &= \int_a^b \left(\int_{-\infty}^{+\infty} f(x, y) dy \right) dx. \end{aligned} \quad (\text{Fubini's Theorem})$$

From Definition 2.1.6, we see that $\int_{-\infty}^{+\infty} f(x, y) dy$ is the pdf of X . The same argument applies to Y . $\square$

Example 2.2.5 (Two Dice Rolls). Let's look at Example 2.2.1 again. Recall that X is the outcome of the first die and Y is the sum of the outcomes of the two dice. The joint pmf is given in Table 2.1. Applying Proposition 2.2.2 (1), we obtain $p_1(x)$ by taking column sums and $p_2(y)$ by taking row sums in Table 2.1. Moreover, $p_1(x)$ and $p_2(y)$ are presented in the margins of the table. This is the reason why they are called marginal pmfs.

One can verify that the marginal pmfs p_1 and p_2 of X and Y are exactly the pmfs given in Examples 2.1.8 and 2.1.9, respectively.

$y \setminus x$	1	2	3	4	5	6	$p_2(y)$
2	1/36	0	0	0	0	0	1/36
3	1/36	1/36	0	0	0	0	2/36
4	1/36	1/36	1/36	0	0	0	3/36
5	1/36	1/36	1/36	1/36	0	0	4/36
6	1/36	1/36	1/36	1/36	1/36	0	5/36
7	1/36	1/36	1/36	1/36	1/36	1/36	6/36
8	0	1/36	1/36	1/36	1/36	1/36	5/36
9	0	0	1/36	1/36	1/36	1/36	4/36
10	0	0	0	1/36	1/36	1/36	3/36
11	0	0	0	0	1/36	1/36	2/36
12	0	0	0	0	0	1/36	1/36
$p_1(x)$	6/36	6/36	6/36	6/36	6/36	6/36	

Table 2.1: Joint pmf and marginal pmfs of rolling two dice.

Example 2.2.6. Let X and Y be jointly continuous random variables with the following joint pdf:

$$f(x, y) = \begin{cases} k(x + y), & 0 \leq x \leq 1, 0 \leq y \leq x, \\ 0, & \text{otherwise.} \end{cases}$$

Determine the constant k and find the marginal pdfs f_1 and f_2 of X and Y .

Solution. (1) For f to be a valid joint pdf, we need $f \geq 0$ and $\iint f(x, y) dx dy = 1$. The non-negativity condition $f \geq 0$ implies $k \geq 0$. The normalization condition gives

$$1 = \iint_{\mathbb{R}^2} f(x, y) dx dy = \int_0^1 \int_0^x k(x + y) dy dx = \int_0^1 \frac{3k}{2} x^2 dx = \frac{k}{2}.$$

Thus $k = 2$.

(2) The marginal pdf of X is

$$f_1(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \begin{cases} \int_0^x 2(x + y) dy = 3x^2, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

The marginal pdf of Y is

$$f_2(y) = \int_{-\infty}^{+\infty} f(x, y) dx = \begin{cases} \int_y^1 2(x + y) dx = 1 + 2y - 3y^2, & 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

□

Independent random variables

In Examples 2.2.5 and 2.2.6, we see that the joint pmf $p(x, y)$ or joint pdf $f(x, y)$ cannot be recovered from the marginal pmfs p_1 and p_2 or the marginal pdfs f_1 and f_2 alone. Thus, one may ask: when is it possible to recover the joint distribution from the marginal distributions?

Definition 2.2.7 (Independent random variables). Two random variables X and Y are *independent* if, for every two subsets $A, B \subseteq \mathbb{R}$,

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B). \quad (2.2)$$

This means X and Y are independent (in the probabilistic sense) that the probability of X taking values in any given set does not affect the probability of Y taking values in any given set.

Definition 2.2.7 clearly states the independence of two random variables X and Y in probability. However, to apply this definition directly, one has to verify that Equation (2.2) holds for **all** subsets $A, B \subseteq \mathbb{R}$, which is practically impossible. According to measure theory, if Equation (2.2) holds for all sets of the form $A = (-\infty, a]$ and $B = (-\infty, b]$ with $a, b \in \mathbb{R}$, then it holds for all Borel sets A and B . In this case, Equation (2.2) reduces to

$$\mathbb{P}(X \leq a, Y \leq b) = \mathbb{P}(X \leq a)\mathbb{P}(Y \leq b), \quad \forall a, b \in \mathbb{R}.$$

This allows us to determine independence using the joint distribution function and marginal distribution functions, which leads to the following theorem. The proof is beyond the scope of this course. For curious readers, we refer to [cite].

Theorem 2.2.1 (Criterion for Independence). *Let X and Y be two random variables with joint distribution function $F(x, y)$ and marginal distribution functions $F_1(x)$ and $F_2(y)$, respectively. Then X and Y are independent if and only if*

$$F(x, y) = F_1(x)F_2(y) \quad \text{for all } x, y \in \mathbb{R}.$$

For the joint discrete or continuous cases, one can characterize the independence by using the joint pmf or pdf.

Theorem 2.2.2. *Let X and Y be two random variables with joint pmf or pdf $h(x, y)$. Then X and Y are independent if and only if h can be presented in the following form*

$$h(x, y) = h_1(x)h_2(y), \quad \text{for all } x, y \in \mathbb{R}. \quad (2.3)$$

Proof. We prove only for the joint continuous case. For the ‘if’ part, let $c_1 = \int_{-\infty}^{+\infty} h_2(y) dy$. By Eq. (2.3), the marginal pdf of X is

$$f_1(x) = \int_{-\infty}^{+\infty} h(x, y) dy = \left(\int_{-\infty}^{+\infty} h_2(y) dy \right) h_1(x) = c_1 h_1(x).$$

Since $\int_{-\infty}^{+\infty} f_1(x) dx = 1$, we have $\int_{-\infty}^{+\infty} h_1(x) dx = 1/c_1$. Then, by Eq. (2.3), the marginal pdf of Y is

$$f_2(y) = \int_{-\infty}^{+\infty} h(x, y) dx = \left(\int_{-\infty}^{+\infty} h_1(x) dx \right) h_2(y) = c_1^{-1} h_2(y).$$

Thus $h(x, y) = h_1(x)h_2(y) = (c_1 h_1(x)) \cdot (c_1^{-1} h_2(y)) = f_1(x)f_2(y)$. For every $A, B \subseteq \mathbb{R}$,

$$\begin{aligned} \mathbb{P}(X \in A, Y \in B) &= \int_A \int_B h(x, y) dy dx = \int_A \int_B f_1(x)f_2(y) dy dx \\ &= \left(\int_A f_1(x) dx \right) \left(\int_B f_2(y) dy \right) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B). \end{aligned}$$

By Definition 2.2.7, X and Y are independent.

For the ‘only if’ part, if X and Y are independent, then by Theorem 2.2.1,

$$h(x, y) = \frac{\partial^2 F(x, y)}{\partial x \partial y} = \frac{\partial^2 (F_1(x)F_2(y))}{\partial x \partial y} = \frac{\partial F_1(x)}{\partial x} \cdot \frac{\partial F_2(y)}{\partial y}.$$

Letting $h_1(x) = \frac{\partial F_1(x)}{\partial x}$ and $h_2(y) = \frac{\partial F_2(y)}{\partial y}$, Equation (2.3) follows. $\square$

From the proof of Theorem 2.2.2, we see that $h_1(x)$ is not necessarily the pmf or pdf of X , and the same holds for $h_2(y)$. Only when h_1 and h_2 are exactly the pmfs or pdfs of X and Y , respectively, can we recover their joint pmf or pdf.

Corollary 2.2.1. *Let X and Y be two random variables with joint pmf or pdf $f(x, y)$ and marginal pmfs or pdfs f_1 and f_2 . Then X and Y are independent if and only if their joint pmf or pdf is the product of their marginal pmfs or pdfs:*

$$f(x, y) = f_1(x)f_2(y) \quad \text{for all } x, y \in \mathbb{R}.$$

Example 2.2.7 (Dependent random variables). According to Corollary 2.2.1, the random variables given in Examples 2.2.5 and 2.2.6 are dependent.

Example 2.2.8. Consider an urn containing 4 balls where two of them are red (labeled R_1, R_2) and two of them are blue (labeled B_1, B_2). We perform two draws **without replacement**:

- Draw 1: Pick one ball at random from the urn;
- Draw 2: Pick a second ball from the remaining 3 balls.

Let X be the indicator for “even label” in Draw 1, that is $X = 1$ if R_2 or B_2 is drawn and $X = 0$ otherwise. Let Y be the indicator for “red color” in Draw 2, that is $Y = 1$ if R_1 or R_2 is drawn and $Y = 0$ otherwise.

The table below summarizes the joint pmf $p(x, y) = \mathbb{P}(X = x, Y = y)$ and marginal pmfs $p_X(x)$ (row totals) and $p_Y(y)$ (column totals):

X	Y		$p_X(x)$
	$Y = 0$	$Y = 1$	
$X = 0$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
$X = 1$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
$p_Y(y)$	$\frac{1}{2}$	$\frac{1}{2}$	

It is therefore clear that $p(x, y) = p_X(x)p_Y(y)$ for all x, y . This confirms that X and Y are independent (in the probabilistic sense). We note that although the sampling is performed without replacement (meaning the outcome of the first draw affects the second draw), the random variables X and Y remain independent.

Example 2.2.9. Let X and Y be independent random variables with pdfs

$$f_1(x) = 2x, \quad 0 < x < 1 \quad \text{and} \quad f_2(y) = \frac{3}{2}y, \quad -1 < y < 1.$$

Find $\mathbb{P}(X < Y)$.

Solution. Since X and Y are independent, their joint pdf is

$$f(x, y) = f_1(x)f_2(y) = 3xy, \quad 0 < x < 1, \quad -1 < y < 1.$$

Then

$$\begin{aligned} \mathbb{P}(X < Y) &= \iint_{x < y} f(x, y) dx dy = \int_0^1 \int_x^1 f_1(x)f_2(y) dy dx \\ &= \int_0^1 \int_x^1 3xy dy dx = \int_0^1 \frac{3}{2}x(1 - x^2) dx = \frac{3}{8}. \end{aligned}$$

□

Conditional distribution

Let us look at Example 2.2.5 once again. We roll two dice. The random variable X indicate the number of the first die and the random variable Y is the sum of the two dice. The joint pmf and marginal pmfs are given in the table below.

$y \setminus x$	1	2	3	4	5	6	$p_2(y)$
2	1/36	0	0	0	0	0	1/36
3	1/36	1/36	0	0	0	0	2/36
4	1/36	1/36	1/36	0	0	0	3/36
5	1/36	1/36	1/36	1/36	0	0	4/36
6	1/36	1/36	1/36	1/36	1/36	0	5/36
7	1/36	1/36	1/36	1/36	1/36	1/36	6/36
8	0	1/36	1/36	1/36	1/36	1/36	5/36
9	0	0	1/36	1/36	1/36	1/36	4/36
10	0	0	0	1/36	1/36	1/36	3/36
11	0	0	0	0	1/36	1/36	2/36
12	0	0	0	0	0	1/36	1/36
$p_1(x)$	6/36	6/36	6/36	6/36	6/36	6/36	

Suppose we know the sum is 10, namely $Y = 10$. What is the probability of $X = 4$? This is the conditional probability

$$\mathbb{P}(X = 4 \mid Y = 10) = \frac{\mathbb{P}(X = 4, Y = 10)}{\mathbb{P}(Y = 10)} = \frac{p(4, 10)}{p_2(10)} = \frac{1/36}{3/36} = \frac{1}{3}.$$

In general, the distribution of X given that $Y = 10$ is

$$\mathbb{P}(X = x \mid Y = 10) = \frac{\mathbb{P}(X = x, Y = 10)}{\mathbb{P}(Y = 10)} = \frac{p(x, 10)}{p_2(10)}.$$

Denote $p(x|10) = \mathbb{P}(X = x|Y = 10)$. The distribution is essentially normalized row in the joint pmf:

$y \setminus x$	1	2	3	4	5	6	$p_2(y)$
10	0	0	0	$1/36$	$1/36$	$1/36$	$3/36$

↓

$p(x 10)$	1	2	3	4	5	6
	$\frac{0}{3/36}$	$\frac{0}{3/36}$	$\frac{0}{3/36}$	$\frac{1/36}{3/36}$	$\frac{1/36}{3/36}$	$\frac{1/36}{3/36}$

Definition 2.2.8 (Conditional pmf). Suppose that X and Y are joint discrete with joint pmf $p(x, y)$. Let p_X and p_Y be the marginal pmfs of X and Y . For each y such that $p_Y(y) > 0$, define

$$p_{X|Y}(x|y) = \frac{p(x, y)}{p_Y(y)}.$$

Then $p_{X|Y}(\cdot|y)$ is called the *conditional probability mass function* (conditional pmf) of X given that $Y = y$. Similarly, for each x such that $p_X(x) > 0$, define

$$p_{Y|X}(y|x) = \frac{p(x, y)}{p_X(x)}.$$

Then $p_{Y|X}(\cdot|x)$ is called the *conditional probability mass function* (conditional pmf) of Y given that $X = x$.

Intuitively, in Example 2.2.5, the conditional discrete distributions correspond to the normalized rows or columns in Table 2.1. We encourage readers to calculate the conditional pmf $p_2(y \mid 4)$ and compare it with the column for $X = 4$ in Table 2.1.

For the joint continuous case, the conditional probability

$$\mathbb{P}(X = x \mid Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}$$

does not make sense, since $\mathbb{P}(Y = y) = 0$ for all y . In the continuous case, it is better to consider the event $Y \approx y$. For any $\varepsilon_1, \varepsilon_2 > 0$, by $X \approx x$ and $Y \approx y$ we mean the events $\{x - \varepsilon_1 \leq X \leq x + \varepsilon_1\}$ and $\{y - \varepsilon_2 \leq Y \leq y + \varepsilon_2\}$; then

$$\begin{aligned} \mathbb{P}(X \approx x \mid Y \approx y) &= \frac{\mathbb{P}(X \approx x, Y \approx y)}{\mathbb{P}(Y \approx y)} \\ &\approx \frac{f(x, y) \cdot 2\varepsilon_1 \cdot 2\varepsilon_2}{f_2(y) \cdot 2\varepsilon_2} \approx \frac{f(x, y)}{f_2(y)} \cdot 2\varepsilon_1. \end{aligned}$$

This leads us to the definition of conditional pdf.

Definition 2.2.9 (Conditional pdf). Let X and Y be jointly continuous random variables with joint probability density function (pdf) $f(x, y)$ and marginal pdfs $f_X(x)$ and $f_Y(y)$. For any y such that $f_Y(y) > 0$, we define

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}.$$

We refer to $f_{X|Y}(\cdot|y)$ as the *conditional probability density function* (conditional pdf) of X given $Y = y$. The conditional pdf of Y given $X = x$, denoted $f_{Y|X}(y|x)$, is defined analogously.

We remark that for y such that $f_Y(y) = 0$, we may define $f_{X|Y}(x|y) = 0$ (or any other reasonable value such that $f_{X|Y}(\cdot|y) \geq 0$ and $\int_{\mathbb{R}} f_{X|Y}(x|y) dx = 1$). Thus $f_{X|Y}(\cdot|y)$ is a valid pdf, and the same holds for $f_{Y|X}(\cdot|x)$.

Example 2.2.10 (Example 2.2.6: Revisit). Let X and Y be two jointly continuous random variables with the following joint pdf:

$$f(x, y) = \begin{cases} 2(x + y), & 0 \leq x \leq 1, 0 \leq y \leq x, \\ 0, & \text{otherwise.} \end{cases}$$

Compute the conditional pdf $f_{X|Y}(\cdot|y)$ and $f_{Y|X}(\cdot|x)$.

Solution. Recalling that in Example 2.2.6, we have the marginal pdfs of X and Y as below

$$f_1(x) = \begin{cases} 3x^2, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}, \quad f_2(y) = \begin{cases} 1 + 2y - 3y^2, & 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}.$$

For any $y \in [0, 1]$,

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_2(y)} = \begin{cases} \frac{2(x+y)}{1+2y-3y^2}, & y \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

For any $x \in [0, 1]$,

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_1(x)} = \begin{cases} \frac{2(x+y)}{3x^2}, & 0 \leq y \leq x, \\ 0, & \text{otherwise.} \end{cases}$$

□

To visualize the conditional density geometrically, consider the joint pdf $f(x, y)$ as a surface over the xy -plane. Fixing a value $Y = y$, we take a *vertical slice* of the joint density surface at that y . The conditional pdf $f_{X|Y}(x|y)$ is precisely this slice, *renormalized* so that its total integral over x equals 1; see Figure 2.7. In other words, the conditional density represents the shape of the joint distribution along the cross-section $Y = y$, scaled to form a valid probability density.

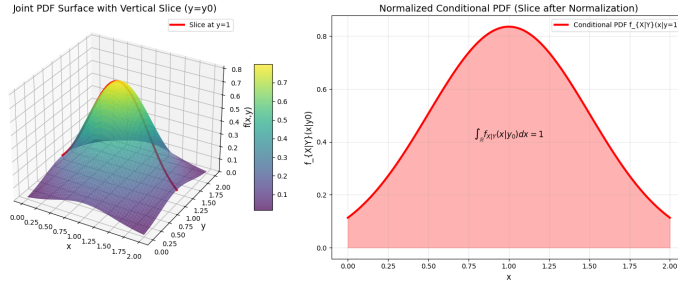


Figure 2.7: Illustration of the Conditional PDF

In addition, the conditional pmf/pdf leads a generalization of the product formula, the Bays' theorem and Law of total probability. Let X and Y be joint discrete/continuous with joint pmf/pdf $f(x, y)$. (To make the equation have a uniform form, we also use f to indicate the pmf.)

1. **Product rule:**

$$f(x, y) = f_Y(y)f_{X|Y}(x|y);$$

2. **Law of total probability:**

$$f_X(x) = \sum_y f(x, y) = \sum_y f_Y(y)f_{X|Y}(x|y);$$

(in the continuous case, ‘ $\sum$ ’ should be replaced by ‘ $\int$ ’)

3. Bays' Theorem:

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \frac{f_Y(y)f_{X|Y}(x|y)}{f_X(x)}.$$

For independent X and Y , $f(x, y) = f_Y(y)f_X(x)$ and $f_{X|Y}(x|y) = f_X(x)$, meaning the conditional pdf equals the marginal pdf when variables are independent.

Example 2.2.11. Let X be uniformly distributed over $[0, 1]$. Namely, the pdf of X is $f_X(x) = 1$ for $0 \leq x \leq 1$. After the value $X = x$ ($0 < x < 1$) is observed, we choose y uniformly from $[x, 1]$. That is the conditional pdf of Y given that $X = x$ is $f_{Y|X}(y|x) = \frac{1}{1-x}$ for $x \leq y \leq 1$.

1. Find the marginal pdf of Y .
2. Compute $\mathbb{P}(1/3 < Y < 1/2)$.
3. Find the conditional pdf of X given that $Y = y$.

Solution. (1) Using the product rule, the joint pdf is

$$f(x, y) = f_{Y|X}(y|x)f_X(x) = \begin{cases} \frac{1}{1-x}, & 0 < x < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Thus, for $0 < y < 1$, the marginal pdf of Y is

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx = \int_0^y \frac{1}{1-x} dx = -\ln(1-y).$$

- (2) The desired probability is

$$\begin{aligned} \mathbb{P}(1/3 < Y < 1/2) &= \int_{1/3}^{1/2} -\ln(1-y) dy \\ &= [(1-y)\ln(1-y) + y]_{1/3}^{1/2} \\ &= -\frac{7}{6}\ln 2 + \frac{2}{3}\ln 3 + \frac{1}{6} \approx 0.09. \end{aligned}$$

- (3) The conditional pdf of X given that $Y = y$ is

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \begin{cases} \frac{-1}{(1-x)\ln(1-y)}, & 0 < x < y, \\ 0, & \text{otherwise.} \end{cases}$$

□

2.3 Multivariate distributions

Now we extend the bivariate distribution to the multivariate distribution.

Definition 2.3.1 (Joint distribution function). The *joint distribution function* (joint cdf) of n random variables $X_1, X_2, \dots, X_n$ is the n -variate function F such that for any $(x_1, \dots, x_n) \in \mathbb{R}^n$,

$$F(x_1, \dots, x_n) = \mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n).$$

Similar as the bivariate case, F is monotone with each variable x_i . For each $i = 1, \dots, n$, the marginal distribution of X_i can be derived by

$$F_i(x_i) = \lim_{\substack{x_j \rightarrow \infty \\ j=1, \dots, n, j \neq i}} F(x_1, \dots, x_n).$$

Definition 2.3.2 (Joint pmf). We say that n random variables $X_1, \dots, X_n$ have a discrete joint distribution if the random vector $(X_1, \dots, X_n)$ can take only a finite number or a countably infinite sequence of distinct possible values. The *joint probability mass function (joint pmf)* is the function p defined on $\mathbb{R}^n$ such that for each $(x_1, \dots, x_n) \in \mathbb{R}^n$,

$$p(x_1, \dots, x_n) = \mathbb{P}(X_1 = x_1, \dots, X_n = x_n).$$

In vector notation, let $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{X} = (X_1, \dots, X_n)$; then the joint pmf satisfies, for all $\mathbf{x} \in \mathbb{R}^n$,

$$p(\mathbf{x}) = \mathbb{P}(\mathbf{X} = \mathbf{x}).$$

The joint pmf has the following properties:

- (Non-negativity) $p(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$;
- (Normalization) $\sum_{\mathbf{x} \in \mathbb{R}^n} p(\mathbf{x}) = 1$.
- For any subset $C \subseteq \mathbb{R}^n$, the probability that $\mathbf{X}$ lies in C is given by

$$\mathbb{P}(\mathbf{X} \in C) = \sum_{\mathbf{x} \in C} p(\mathbf{x}).$$

Example 2.3.1 (Rolling N Fair Dice). Consider the experiment of rolling N fair six-sided dice independently. For each $i = 1, 2, \dots, 6$, let X_i denote the number of dice that show the face value i (i.e., X_i counts the frequency of outcome i across the N rolls). The random vector $\mathbf{X} = (X_1, X_2, \dots, X_6)$ then characterizes

the full frequency distribution of outcomes from the N dice rolls, and its joint probability mass function (joint pmf) is given by:

$$p(n_1, n_2, \dots, n_6) = \binom{N}{n_1} \binom{N-n_1}{n_2} \cdots \binom{n_6}{n_6} \left(\frac{1}{6}\right)^N = \frac{N!}{n_1! n_2! \cdots n_6!} \left(\frac{1}{6}\right)^N,$$

where the non-negative integers $(n_1, n_2, \dots, n_6)$ satisfy $n_1 + n_2 + \cdots + n_6 = N$. Here, $\binom{k}{m} = \frac{m!}{k!(m-k)!}$ is the binomial coefficient (read as “ k choose m ”).

Definition 2.3.3 (Joint pdf). We say that n random variables $X_1, \dots, X_n$ have a *continuous joint distribution* if there is a nonnegative function f defined on $\mathbb{R}^n$ such that for every subset $C \subseteq \mathbb{R}^n$,

$$\mathbb{P}(\mathbf{X} \in C) = \int \cdots \int_{\mathbf{x} \in C} f(x_1, \dots, x_n) dx_1 \cdots dx_n$$

if the integral exists. The function f is called the *joint pdf* of $X_1, \dots, X_n$.

If f is a joint pdf, then $f \geq 0$ and $\int \cdots \int_{\mathbb{R}^n} f(x_1, \dots, x_n) dx_1 \cdots dx_n = 1$. As the univariate and bivariate cases the (joint) cdf and (joint) pdf have the following relation

$$F(x_1, \dots, x_n) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f(u_1, \dots, u_n) du_1 \cdots du_n,$$

$$f(x_1, \dots, x_n) = \frac{\partial^n}{\partial x_1 \cdots \partial x_n} F(x_1, \dots, x_n).$$

Example 2.3.2. Let $\mathbf{X} = (X_1, X_2, \dots, X_n)$ be an n -dimensional random vector with a joint continuous distribution over a bounded hyper-rectangle $R \subseteq \mathbb{R}^n$, where

$$R = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n \mid a_i \leq x_i \leq b_i \text{ for all } i = 1, 2, \dots, n\}.$$

The volume of R is given by

$$\text{Vol}(R) = \prod_{i=1}^n (b_i - a_i) = (b_1 - a_1)(b_2 - a_2) \cdots (b_n - a_n).$$

The joint probability density function of $\mathbf{X}$ is:

$$f(x_1, x_2, \dots, x_n) = \begin{cases} \frac{1}{\text{Vol}(R)}, & \text{if } (x_1, x_2, \dots, x_n) \in R, \\ 0, & \text{otherwise.} \end{cases}$$

Definition 2.3.4 (Independent random variables). n random variables $X_1, \dots, X_n$ are said to be *independent* (or mutually independent) if, for every non-empty index set $I \subseteq \{1, 2, \dots, n\}$ and for all sets $A_i \subseteq \mathbb{R}$ ($i \in I$),

$$\mathbb{P}\left(\bigcap_{i \in I} \{X_i \in A_i\}\right) = \prod_{i \in I} \mathbb{P}(X_i \in A_i). \quad (2.4)$$

Note that mutual independence is different from pairwise independence. The random variables $X_1, \dots, X_n$ are *pairwise independent* if any two of them are independent. To verify pairwise independence, since there are $\binom{n}{2}$ distinct pairs of random variables, we need to check Equation (2.2) $\binom{n}{2}$ times. However, there are $\binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = 2^n - 1$ non-empty index sets I . Thus, to verify *mutual independence*, we need to check Equation (2.4) $2^n - 1$ times.

Similar to the bivariate case, independence can be verified using the joint distribution function, joint pmf, or joint pdf. We state the following criterion without proof. Here, the marginal pmf and marginal pdf are defined in the same way as the marginal distribution function.

Theorem 2.3.1. *Let $X_1, \dots, X_n$ be n random variables with joint distribution function $F(x_1, \dots, x_n)$, and let $F_i(x_i)$ be the marginal distribution function of X_i for each i . Then $X_1, \dots, X_n$ are mutually independent if and only if*

$$F(x_1, \dots, x_n) = F_1(x_1)F_2(x_2) \cdots F_n(x_n).$$

Furthermore:

- If the random variables are jointly discrete with joint pmf $p(x_1, \dots, x_n)$ and marginal pmfs $p_i(x_i)$, then they are mutually independent if and only if

$$p(x_1, \dots, x_n) = p_1(x_1)p_2(x_2) \cdots p_n(x_n)$$

for all possible values $(x_1, \dots, x_n)$.

- If the random variables are jointly continuous with joint pdf $f(x_1, \dots, x_n)$ and marginal pdfs $f_i(x_i)$, then they are mutually independent if and only if

$$f(x_1, \dots, x_n) = f_1(x_1)f_2(x_2) \cdots f_n(x_n)$$

for almost all $(x_1, \dots, x_n) \in \mathbb{R}^n$.

2.4 Distribution of functions of random variables

Sometimes, random variables are not used directly in their original form; instead, we often work with functions of these random variables. For example, we might

be interested in the sum, the square, or some transformation of a given random variable, rather than the variable itself. In such cases, the original probability distribution is no longer sufficient, and we need to derive the distribution of the new random variable defined by the function. Therefore, it is essential to understand how to find the distribution of functions of random variables, as this is a fundamental technique in probability and statistics.

2.4.1 Distribution of functions of single variable

Functions of a discrete variable

Let X be a discrete random variable and let g be a real-valued function. Then $g(X)$ is also a discrete random variable. A direct method to find the probability mass function (pmf) of $g(X)$ is as follows: first identify all possible values of $g(X)$, then compute the probability that $g(X)$ takes each of these values.

Example 2.4.1 (Convenience store). Let X denote the daily number of bottles of mineral water sold at a neighborhood convenience store. The probability mass function (pmf) of X is provided as follows:

Possible Values of X	10	11	12	13
$P(X = x)$	0.25	0.30	0.30	0.15

The convenience store sells each bottle of mineral water at a fixed price of \$2, and it incurs a daily fixed cost of \$5 for storing and displaying the mineral water. Let Y represent the daily net revenue (in dollars) from selling mineral water; thus, $Y = 2X - 5$.

To find the pmf of Y , we first determine its possible values by substituting each possible value of X into the function $Y = 2X - 5$. These possible values of Y are 15, 17, 19, and 21. Next, we calculate the probability corresponding to each possible value of Y . For instance,

$$\mathbb{P}(Y = 15) = \mathbb{P}(2X - 5 = 15) = \mathbb{P}(X = 10) = 0.25.$$

The probabilities for the other possible values of Y can be derived in the same manner. The complete pmf of Y is shown below:

Possible Values of Y	15	17	19	21
$P(Y = y)$	0.25	0.30	0.30	0.15

Functions of a continuous variable

For the continuous case, let X be a continuous random variable and let g be a real-valued function. Note that $Y := g(X)$ is not necessarily a continuous random

variable. We turn to find the distribution function of Y at first. Denote by F_Y the df of Y . Then

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(g(X) \leq y) = \int_{\{x: g(x) \leq y\}} f(x) dx.$$

If Y is a continuous random variable, then its density $f_Y(y)$ satisfies

$$f_Y(y) = \frac{\partial F_Y(y)}{\partial y}$$

for all y such that F_Y is differentiable.

Example 2.4.2. Let X be a continuous random variable with pdf

$$f(x) = \begin{cases} \frac{1}{2}(1-x), & -1 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the pdf of X^2 .

Solution. Since $X \in (-1, 1)$, squaring X gives $Y = X^2 \in [0, 1)$. The cdf of Y is defined as $F_Y(y) = P(Y \leq y)$. For $0 \leq y < 1$:

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y}) \\ &= \int_{-\sqrt{y}}^{\sqrt{y}} f_X(x) dx = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2}(1-x) dx \\ &= \frac{1}{2} \left[x - \frac{x^2}{2} \right]_{-\sqrt{y}}^{\sqrt{y}} = \sqrt{y}. \end{aligned}$$

For $y < 0$, $F_Y(y) = 0$. For $y \geq 1$, $F_Y(y) = 1$. The pdf $f_Y(y)$ is

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \begin{cases} \frac{1}{2\sqrt{y}}, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

□

When g is one-to-one and differentiable, finding the pdf of $g(X)$ is easy.

Proposition 2.4.1. *Let X be a continuous random variable with pdf $f(x)$ supported on (a, b) . That is, $\mathbb{P}(a < X < b) = 1$ (where a and b may be infinite). Let $Y = g(X)$, where g is a one-to-one, differentiable real-valued function. Assume that the image of (a, b) under g is (α, β) . Then the pdf of Y is*

$$f_Y(y) = \begin{cases} f(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|, & \alpha < y < \beta, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. If g is increasing, then the cdf of Y is

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(g(X) \leq y) = \mathbb{P}(X \leq g^{-1}(y)) = F_X(g^{-1}(y)).$$

Differentiating $F_Y(y)$ to obtain the pdf $f_Y(y)$: for $y \in (\alpha, \beta)$,

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y)) = f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y).$$

If g is decreasing, then the cdf of Y is

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \leq y) = \mathbb{P}(g(X) \leq y) \\ &= \mathbb{P}(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y)). \end{aligned}$$

Differentiating $F_Y(y)$ to obtain the pdf $f_Y(y)$: for $y \in (\alpha, \beta)$,

$$f_Y(y) = -\frac{d}{dy} F_X(g^{-1}(y)) = -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y).$$

Combining both cases, the absolute value in the formula accounts for the sign difference, completing the proof. $\square$

Example 2.4.3. Let X be the continuous random variable given in Example 2.1.12 with the pdf: $f(x) = xe^{-x^2/2}$ for all $x > 0$. Find the pdf of $\ln(X)$.

Solution. Let $Y = \ln(X)$. Note that $\mathbb{P}(0 < X < +\infty) = 1$ and the image of $(0, +\infty)$ under $g(x) = \ln(x)$ is $(-\infty, +\infty)$. The inverse of g is $g^{-1}(y) = e^y$ for all $y \in (-\infty, +\infty)$ and $\frac{d}{dy} g^{-1}(y) = e^y$. Applying Proposition 2.4.1, we have

$$f_Y(y) = f(e^y)e^y = e^{2y}e^{-e^{2y}/2}, \quad -\infty < y < +\infty.$$

2.4.2 Distribution of functions of multivariable

We focus on bivariate functions of random variables. The methodology for finding the distribution of functions of multiple random variables is consistent across different cases. For discrete random variables, the process of deriving the distribution of their multivariate functions is analogous to that of the univariate case. We illustrate the idea through examples.

Discrete case

Let X_1 denote the daily number of small bottles of mineral water sold at a convenience store, and X_2 denote the daily number of large bottles of mineral water sold. The pmfs of X_1 and X_2 are independent and given below:

Possible Values of X_1	8	9	10
$P(X_1 = x_1)$	0.2	0.5	0.3
Possible Values of X_2	2	3	4
$P(X_2 = x_2)$	0.4	0.4	0.2

Example 2.4.4 (Sum of Random Variables). Let $Y_1 = X_1 + X_2$ represent the total number of mineral water bottles (small + large) sold daily.

Solution. We first find possible values of Y_1 . The possible values of Y_1 are $8 + 2 = 10$, ..., $10 + 4 = 14$, so $Y_1 \in \{10, 11, 12, 13, 14\}$.

We then calculate pmf of Y_1 . For independent X_1 and X_2 , the law of total probability yields

$$P(Y_1 = y) = \sum_{x_1} P(X_1 = x_1)P(X_2 = y - x_1).$$

For instance,

$$\begin{aligned} P(Y_1 = 11) &= P(X_1 = 8)P(X_2 = 3) + P(X_1 = 9)P(X_2 = 2) \\ &= 0.2 \times 0.4 + 0.5 \times 0.4 = 0.28. \end{aligned}$$

The complete pmf of Y_1 is:

Possible Values of Y_1	10	11	12	13	14
$P(Y_1 = y)$	0.08	0.28	0.36	0.22	0.06

□

Example 2.4.5 (Difference of Random Variables). Let $Y_2 = X_1 - X_2$ represent the difference between the number of small and large mineral water bottles sold daily.

Solution. The possible values of Y_2 are $\{4, 5, 6, 7, 8\}$. Now we Calculate pmf of Y_2 . It follows from the law of total probability and the independent assumption that

$$P(Y_2 = y) = \sum_{x_1} P(X_1 = x_1)P(X_2 = x_1 - y).$$

The complete pmf of Y_2 is:

Possible Values of Y_2	4	5	6	7	8
$P(Y_2 = y)$	0.04	0.18	0.34	0.32	0.12

□

Example 2.4.6 (Product of Random Variables). Let $Y_3 = X_1 X_2$.

Solution. The possible values of Y_3 are $\{16, 18, 20, 24, 27, 30, 32, 36, 40\}$. Using the law of total probability and the independent assumption, we have $P(Y_3 = y) = \sum_{x_1} P(X_1 = x_1)P(X_2 = y/x_1)$ (for integer y/x_1). For example, $P(Y_3 = 16) = P(X_1 = 8)P(X_2 = 2) = 0.08$. The complete pmf of Y_3 is:

Possible Values of Y_3	16	18	20	24	27	30	32	36	40
$P(Y_3 = y)$	0.08	0.2	0.12	0.08	0.2	0.12	0.04	0.1	0.06

□

Example 2.4.7 (Quotient of Random Variables). Let $Y_4 = \lfloor X_1/X_2 \rfloor$ where $\lfloor \cdot \rfloor$ is the floor function.

Solution. The possible values of Y_4 are $\{2, 3, 4, 5\}$. Note that

$$P(Y_4 = y) = \sum_{x_1, x_2: \lfloor x_1/x_2 \rfloor = y} P(X_1 = x_1)P(X_2 = x_2).$$

For example, $P(Y_4 = 2) = P(X_1 = 8, X_2 = 3) + P(X_1 = 8, X_2 = 4) + P(X_1 = 9, X_2 = 4) = 0.08 + 0.04 + 0.1 = 0.22$. The complete pmf of Y_4 is:

Possible Values of Y_4	2	3	4	5
$P(Y_4 = y)$	0.22	0.32	0.28	0.12

□

Example 2.4.8 (Maximum Function of Random Variables). Let $Y_5 = \max(X_1, X_2)$ represent the maximum daily sales volume between small and large mineral water bottles.

Solution. The possible values are $\max(8, 2) = 8$, ..., $\max(10, 4) = 10$; so $Y_5 \in \{8, 9, 10\}$. For independent X_1, X_2 ,

$$P(Y_5 = y) = P(X_1 = y, X_2 \leq y) + P(X_1 < y, X_2 = y).$$

For example, $P(Y_5 = 9) = P(X_1 = 9)P(X_2 \leq 9) + P(X_1 < 9)P(X_2 = 9) = 0.5 \times 1 + 0.2 \times 0 = 0.5$. The complete pmf of Y_5 is:

Possible Values of Y_5	8	9	10
$P(Y_5 = y)$	0.2	0.5	0.3

□

Example 2.4.9 (Minimum Function of Random Variables). Let $Y_6 = \min(X_1, X_2)$ represent the minimum daily sales volume between small and large mineral water bottles.

Solution. The possible values of Y_6 are $\{2, 3, 4\}$. The complete pmf of Y_6 is:

Possible Values of Y_6	2	3	4
$P(Y_6 = y)$	0.4	0.4	0.2

□

Continuous case

For the continuous case, let X and Y be jointly continuous random variables with joint probability density function (pdf) $f(x, y)$. Let $h(x, y)$ be a bivariate real-valued function, and denote $Z = h(X, Y)$. The distribution function (cdf) of Z is given by:

$$F_Z(z) = \mathbb{P}(Z \leq z) = \iint_{\{(x,y): h(x,y) \leq z\}} f(x, y) dx dy.$$

This problem reduces to evaluating the double integral above, with the key technique being the conversion of double integrals into iterated integrals (by specifying appropriate limits of integration). Once the cdf $F_Z(z)$ is obtained, if Z is a continuous random variable, its pdf can be derived by differentiating $F_Z(z)$ with respect to z . These methods are illustrated through the following example.

Example 2.4.10. Let X and Y be independent continuous random variables, each uniformly distributed on the interval $(0, 1)$. That is, their joint probability density function (pdf) is

$$f_{X,Y}(x, y) = \begin{cases} 1, & 0 < x < 1, 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

We will find the pdfs of the following functions of X and Y :

1. $S = X + Y$ (Sum)
2. $D = X - Y$ (Difference)
3. $P = XY$ (Product)
4. $Q = \frac{X}{Y}$ (Quotient)
5. $M = \max(X, Y)$ (Maximum)
6. $Z = \min(X, Y)$ (Minimum)

Solution. (1) The support of $S = X + Y$ is $(0, 2)$. We first find the cumulative distribution function (cdf)

$$F_S(s) = \mathbb{P}(S \leq s) = \iint_{x+y \leq s} f_{X,Y}(x, y) dx dy.$$

If $0 < s < 1$:

$$F_S(s) = \int_0^s \int_0^{s-x} 1 dy dx = \int_0^s (s-x) dx = \frac{s^2}{2}.$$

If $1 \leq s < 2$:

$$F_S(s) = 1 - \int_{s-1}^1 \int_{s-x}^1 1 dy dx = 1 - \int_{s-1}^1 (1 - (s-x)) dx = 1 - \frac{(2-s)^2}{2}.$$

Differentiating the cdf gives the pdf:

$$f_S(s) = \begin{cases} s, & 0 < s < 1, \\ 2-s, & 1 \leq s < 2, \\ 0, & \text{otherwise.} \end{cases}$$

(2) The support of $D = X - Y$ is $(-1, 1)$. The cdf is

$$F_D(d) = \mathbb{P}(D \leq d) = \mathbb{P}(X - Y \leq d) = \iint_{x-y \leq d} f_{X,Y}(x, y) dx dy.$$

If $-1 < d < 0$:

$$F_D(d) = \int_0^{1+d} \int_{x-d}^1 1 dy dx = \int_0^{1+d} (1-x+d) dx = \frac{(1+d)^2}{2}.$$

If $0 \leq d < 1$:

$$F_D(d) = 1 - \int_d^1 \int_0^{x-d} 1 dy dx = 1 - \int_d^1 (x-d) dx = 1 - \frac{(1-d)^2}{2}.$$

Differentiating the cdf gives the pdf:

$$f_D(d) = \begin{cases} 1+d, & -1 < d < 0, \\ 1-d, & 0 \leq d < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(3) The support of $P = XY$ is $(0, 1)$. The cdf is

$$F_P(p) = \mathbb{P}(XY \leq p) = \iint_{xy \leq p} f_{X,Y}(x, y) dx dy.$$

For $0 < p < 1$:

$$F_P(p) = \int_0^p \int_0^1 1 \, dy \, dx + \int_p^1 \int_0^{p/x} 1 \, dy \, dx = p + \int_p^1 \frac{p}{x} \, dx = p - p \ln p.$$

Differentiating the cdf gives the pdf:

$$f_P(p) = \begin{cases} -\ln p, & 0 < p < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(4) The support of $Q = \frac{X}{Y}$ is $(0, \infty)$. The cdf is

$$F_Q(q) = \mathbb{P}\left(\frac{X}{Y} \leq q\right) = \mathbb{P}(X \leq qY) = \iint_{x \leq qy} f_{X,Y}(x, y) \, dx \, dy.$$

If $0 < q < 1$:

$$F_Q(q) = \int_0^1 \int_0^{qy} 1 \, dx \, dy = \int_0^1 qy \, dy = \frac{q}{2}.$$

If $q \geq 1$:

$$F_Q(q) = 1 - \int_0^1 \int_{qy}^1 1 \, dx \, dy = 1 - \int_0^{1/q} (1 - qy) \, dy = 1 - \frac{1}{2q}.$$

Differentiating the cdf gives the pdf:

$$f_Q(q) = \begin{cases} \frac{1}{2}, & 0 < q < 1, \\ \frac{1}{2q^2}, & q \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

(5) The support of $M = \max(X, Y)$ is $(0, 1)$. For $0 < m < 1$, the cdf is:

$$\begin{aligned} F_M(m) &= \mathbb{P}(\max(X, Y) \leq m) = \mathbb{P}(X \leq m, Y \leq m) \\ &= \mathbb{P}(X \leq m) \mathbb{P}(Y \leq m) = m \cdot m = m^2. \end{aligned}$$

Differentiating the cdf gives the pdf:

$$f_M(m) = \begin{cases} 2m, & 0 < m < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(6) The support of $Z = \min(X, Y)$ is $(0, 1)$. For $0 < z < 1$, the cdf is:

$$F_Z(z) = \mathbb{P}(\min(X, Y) \leq z) = 1 - \mathbb{P}(\min(X, Y) > z)$$

$$= 1 - \mathbb{P}(X > z, Y > z) = 1 - (1 - z)(1 - z) = 1 - (1 - z)^2.$$

Differentiating the cdf gives the pdf:

$$f_Z(z) = \begin{cases} 2(1 - z), & 0 < z < 1, \\ 0, & \text{otherwise.} \end{cases} \quad \square$$

When X and Y are independent, then the distribution of $Z = X + Y$ can be obtained by the *convolution* formula. Let X and Y be independent continuous random variables with pdf f_1 and f_2 . Then the distribution function of Z is

$$\begin{aligned} F_Z(z) &= \mathbb{P}(Z \leq z) \\ &= \iint_{x+y \leq z} f_1(x)f_2(y) \, dx \, dy \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{z-y} f_1(x)f_2(y) \, dx \, dy \\ &= \int_{-\infty}^{+\infty} F_1(z - y)f_2(y) \, dy. \end{aligned}$$

Differentiate F_Z to obtain the pdf of Z

$$f_Z(z) = \int_{-\infty}^{+\infty} f_1(z - y)f_2(y) \, dy.$$

Example 2.4.11 (Convolution). Let X and Y be independent with pdfs

$$f_X(x) = \begin{cases} 1, & 0 < x < 1, \\ 0, & \text{otherwise,} \end{cases} \quad f_Y(y) = \begin{cases} 1, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the pdf of $Z = X + Y$.

Solution. Using the convolution formula:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(t)f_Y(z - t) \, dt$$

The integrand $f_X(t)f_Y(z - t)$ is non-zero only if:

$$0 < t < 1 \quad \text{and} \quad 0 < z - t < 1 \implies z - 1 < t < z$$

If $0 < z < 1$, then $0 < t < z$ and

$$f_Z(z) = \int_0^z 1 \cdot 1 \, dt = z$$

If $1 \leq z < 2$, then $z - 1 < t < 1$ and

$$f_Z(z) = \int_{z-1}^1 1 \cdot 1 \, dt = 1 - (z - 1) = 2 - z$$

If $z \leq 0$ or $z \geq 2$, then the integrand is zero and $f_Z(z) = 0$.

In summary, the pdf of $Z = X + Y$ is:

$$f_Z(z) = \begin{cases} z, & 0 < z < 1, \\ 2 - z, & 1 \leq z < 2, \\ 0, & \text{otherwise.} \end{cases} \quad \square$$

Chapter 3

Expectations

3.1 Expectations of Random Variables

3.1.1 Expectations of discrete random variables

The **expected value** (or mean) of a random variable is a measure of its central tendency, representing the “average” value the variable takes over many trials. We distinguish between discrete and continuous random variables in the definition.

Definition 3.1.1. Let X be a discrete random variable with probability mass function (pmf) $p(x) = P(X = x)$ and support $S = \{x_1, x_2, \dots\}$. The expected value of X , denoted $\mathbb{E}[X]$ or μ_X , is defined as:

$$\mathbb{E}[X] = \sum_{x \in S} x \cdot p(x)$$

provided the sum converges absolutely ($\sum_{x \in S} |x| \cdot p(x) < \infty$).

Example 3.1.1. Let $X = c$ be a constant random variable, i.e. $\mathbb{P}(X = c) = 1$. Then $\mathbb{E}[X] = c$. For instance, $\mathbb{E}[3] = 3$.

Example 3.1.2. Toss a fair coin. Let $X = 1$ if a head occurs and 0 if a tail occurs. Then $\mathbb{E}[X] = 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = \frac{1}{2}$.

Example 3.1.3 (Example 2.4.1 revisit). Let X denote the daily number of bottles of mineral water sold at a neighborhood convenience store. The pmf of X is given below:

Possible Values of X	10	11	12	13
$P(X = x)$	0.25	0.30	0.30	0.15

Find the expected daily sales of mineral water.

Solution. From Definition 3.1.1,

$$\mathbb{E}[X] = (10 \times 0.25) + (11 \times 0.30) + (12 \times 0.30) + (13 \times 0.15) = 11.35.$$

The expected daily sales of mineral water is 11.35 bottles. $\square$

Example 3.1.4 (St. Petersburg Paradox). Consider a game of chance where a fair coin is flipped repeatedly until the first head appears. Let Y denote the payout (in dollars) of the game, where the payout for getting the first head on the k -th flip is $Y = 2^k$. Show that the expected value does not exist.

Solution. The pmf of Y is:

$$P(Y = 2^k) = \frac{1}{2^k} \quad \text{for } k = 1, 2, 3, \dots$$

The support of Y is $S_Y = \{2, 4, 8, 16, \dots\}$ (powers of 2).

Now calculate the expected value of Y :

$$\mathbb{E}[Y] = \sum_{k=1}^{\infty} 2^k \cdot P(Y = 2^k) = \sum_{k=1}^{\infty} 2^k \cdot \frac{1}{2^k} = \sum_{k=1}^{\infty} 1 = \infty.$$

The sum diverges to infinity, so the expected value $E[Y]$ does not exist (is not finite). $\square$

3.1.2 Expectations of continuous random variables

The definition of expectations of discrete random variables are quite natural. Now we extend it to continuous random variables. Now consider a continuous random variable X with pdf $f(x)$. We partition the real line into small intervals of length Δx . For any x , the approximate probability that X lies in the interval $[x, x + \Delta x]$ is

$$\mathbb{P}(x \leq X \leq x + \Delta x) \approx f(x) \Delta x.$$

Then, treating each interval like a discrete outcome with value x and probability $f(x)\Delta x$, the expectation becomes a Riemann sum:

$$\mathbb{E}[X] \approx \sum x f(x) \Delta x.$$

Taking the limit as the partition becomes infinitely fine ($\Delta x \rightarrow 0$), the sum converges to an integral. This limit gives the formal definition:

Definition 3.1.2. Let X be a continuous random variable with probability density function (pdf) $f(x)$. The expected value of X is defined as:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

provided the integral converges absolutely ($\int_{-\infty}^{\infty} |x| \cdot f(x) dx < \infty$).

The expected value of a continuous random variable is the continuous weighted average of the possible values, where the pdf $f(x)$ serves as the weighting function, just as the pmf does in the discrete case.

Example 3.1.5. Let X be uniformly distributed on $(0, 1)$ with pdf:

$$f(x) = \begin{cases} 1, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

The expected value of X is:

$$E[X] = \int_0^1 x \cdot 1 dx = \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2}$$

Example 3.1.6. Let X be the continuous random variable given in Example 2.1.12 with the pdf:

$$f(x) = xe^{-x^2/2}, \quad \text{for all } x > 0.$$

The expected value of X is

$$\begin{aligned} \mathbb{E}[X] &= \int_0^{\infty} x \cdot (xe^{-x^2/2}) dx = \int_0^{\infty} x^2 e^{-x^2/2} dx \\ &= \int_0^{\infty} \sqrt{2u} \cdot e^{-u} \cdot \frac{du}{\sqrt{2u}} = \sqrt{2} \int_0^{\infty} u^{1/2} e^{-u} du = \sqrt{\frac{\pi}{2}}. \end{aligned}$$

Example 3.1.7 (Cauchy Distribution). Let X be a continuous random variable with the standard Cauchy distribution, whose pdf is given by:

$$f(x) = \frac{1}{\pi(1+x^2)}, \quad \text{for } -\infty < x < \infty.$$

This pdf is valid because it is non-negative for all x and integrates to 1:

$$\int_{-\infty}^{\infty} \frac{1}{\pi(1+x^2)} dx = \frac{1}{\pi} [\arctan(x)]_{-\infty}^{\infty} = \frac{1}{\pi} \left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right) = 1.$$

To check if the expected value exists, we evaluate the integral for absolute convergence:

$$\int_{-\infty}^{\infty} |x|f(x) dx = \int_{-\infty}^{\infty} \frac{|x|}{\pi(1+x^2)} dx.$$

Due to symmetry, we can compute twice the integral over the positive half-line:

$$\int_{-\infty}^{\infty} \frac{|x|}{\pi(1+x^2)} dx = \frac{2}{\pi} \int_0^{\infty} \frac{x}{1+x^2} dx.$$

Using the substitution $u = 1 + x^2$, $du = 2x dx$, we get:

$$\frac{2}{\pi} \int_0^{\infty} \frac{x}{1+x^2} dx = \frac{1}{\pi} \int_1^{\infty} \frac{1}{u} du = \frac{1}{\pi} [\ln |u|]_1^{\infty} = \infty.$$

The expected value $E[X]$ for a standard Cauchy random variable does not exist.

3.2 Properties of Expectations

3.2.1 Expectation of a Function

Let X be a discrete or continuous random variable and let $g(x)$ be a real-valued function. Assume that $Z = g(X)$ is a discrete or continuous random variable with pmf $p_Z(x)$ or pdf $f_Z(x)$. Then the expectation of Z is

$$\mathbb{E}(X) = \sum_x xp_Z(x) \quad \text{or} \quad \int_{-\infty}^{+\infty} xf_Z(x) dx.$$

Example 3.2.1. Let X be a continuous random variable with pdf

$$f(x) = \begin{cases} \frac{1}{2}(1-x), & -1 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the expected value of X^2 .

Solution. Let $Y = X^2$. The pdf $f_Y(y)$ of Y is given in Example 2.4.2:

$$f_Y(y) = \begin{cases} \frac{1}{2\sqrt{y}}, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then its expectation is

$$\mathbb{E}[Y] = \int_{-\infty}^{+\infty} yf_Y(y) dy = \int_0^1 y \frac{1}{2\sqrt{y}} dy = \frac{1}{3}. \quad \square$$

Below, we present an alternative method for computing the expectation of $g(X)$ that avoids the need to explicitly derive the probability mass function (pmf) or probability density function (pdf) of the transformed random variable $g(X)$.

Proposition 3.2.1. *Let g be a real-valued function.*

1. *If X is a discrete random variable supported on $S = \{x_1, x_2, \dots\}$ with pmf $p(x)$, then*

$$\mathbb{E}[g(x)] = \sum_{x \in S} g(x)p(x).$$

2. *If X is a continuous random variable with pdf $f(x)$, then*

$$\mathbb{E}[g(x)] = \int_{-\infty}^{+\infty} g(x)f(x) dx.$$

Proof. (1) Let T be the image of S under g . Note that T is finite or countable. For each $y \in T$, define the preimage set $S_y = \{x \in S : g(x) = y\}$. The original set S can be expressed as a disjoint union: $S = \cup_{y \in T} S_y$. Let $Y = g(X)$ and let $p_Y(y)$ denote the pmf of Y . Then,

$$\begin{aligned} \mathbb{E}[Y] &= \sum_{y \in T} y \cdot p_Y(y) = \sum_{y \in T} y \cdot \mathbb{P}(g(X) = y) \\ &= \sum_{y \in T} y \cdot \sum_{x \in S_y} \mathbb{P}(X = x) \\ &= \sum_{y \in T} \sum_{x \in S_y} g(x) \mathbb{P}(X = x) \\ &= \sum_{x \in S} g(x)p(x). \end{aligned}$$

(2) We first prove the result for the case where $g \geq 0$. Let $Y = g(X)$, so $Y \geq 0$. The expected value of Y is given by

$$\begin{aligned} \mathbb{E}[Y] &= \int_0^{+\infty} y f_Y(y) dy \\ &= \int_0^{+\infty} \int_0^y f_Y(y) dt dy = \int_0^{+\infty} \int_t^{+\infty} f_Y(y) dy dt \\ &= \int_0^{+\infty} \mathbb{P}(Y > t) dt = \int_0^{+\infty} \mathbb{P}(g(X) > t) dt \\ &= \int_0^{+\infty} \int_{x: g(x) > t} f(x) dx dt = \int_0^{+\infty} \int_0^{g(x)} f(x) dt dx \end{aligned}$$

$$= \int_0^{+\infty} g(x)f(x) dx,$$

where we have applied Fubini's theorem twice to interchange the order of integration.

In general, $g(x) = g^+(x) - g^-(x)$ where $g^+(x) = \max(0, g(x))$ and $g^-(x) = \max(0, -g(x))$. Note that both g^+ and g^- are non-negative. Applying the above argument, the result follows. We left the details to the readers. $\square$

Example 3.2.2 (Example 3.2.1 revisit). Let X be a continuous random variable with pdf

$$f(x) = \begin{cases} \frac{1}{2}(1-x), & -1 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the expected value of X^2 .

Solution. By Proposition 3.2.1 (2), we have

$$\mathbb{E}[X^2] = \int_{-1}^1 x^2 \cdot \frac{1}{2}(1-x) dx = \frac{1}{3}. \quad \square$$

For functions of several variables, we have an analog of Proposition 3.2.1.

Proposition 3.2.2. Let $g : \mathbb{R}^n \rightarrow \mathbb{R}$ be an n -variate real-valued function.

1. If $X_1, \dots, X_n$ are jointly discrete with joint pmf p , then

$$\mathbb{E}[g(X_1, \dots, X_n)] = \sum_{\text{All } x_1, \dots, x_n} g(x_1, \dots, x_n)p(x_1, \dots, x_n).$$

2. If $X_1, \dots, X_n$ are jointly continuous with joint pdf f , then

$$\mathbb{E}[g(X_1, \dots, X_n)] = \int \cdots \int_{\mathbb{R}^n} g(x_1, \dots, x_n)f(x_1, \dots, x_n) dx_1 \cdots dx_n.$$

Proof. The proof is similar as that of Proposition 3.2.1. One can see also Proposition 2.1 in [2, Section 7.2] for the proof of the two variable case. $\square$

3.2.2 Expectation is a linear function

Letting $g(x_1, \dots, x_n) = x_1 + \cdots + x_n$ and in general $g(x_1, \dots, x_n) = a_1x_1 + \cdots + a_nx_n + b$ where $a_1, \dots, a_n, b \in \mathbb{R}$ in Proposition 3.2.2, we obtain the linear property of the expectation.

Proposition 3.2.3 (Expectation of a linear combination). *Let $X_1, \dots, X_n$ be random variables and $a_1, \dots, a_n, b$ are constants. Assume that $\mathbb{E}[X_i]$ is finite for $i = 1, \dots, n$. Then*

$$\mathbb{E}[a_1X_1 + \dots + a_nX_n + b] = a_1\mathbb{E}[X_1] + \dots + a_n\mathbb{E}[X_n] + b.$$

In particular,

$$\mathbb{E}[X_1 + \dots + X_n] = \mathbb{E}[X_1] + \dots + \mathbb{E}[X_n].$$

Proof. We only prove for the jointly continuous case. The proof of the jointly discrete is left to the readers.

Assume that $X_1, \dots, X_n$ are jointly continuous with joint pdf f . Let

$$g(x_1, \dots, x_n) = a_1x_1 + \dots + a_nx_n + b$$

where $a_1, \dots, a_n, b \in \mathbb{R}$. Then by Proposition 3.2.2,

$$\begin{aligned} & \mathbb{E}[a_1X_1 + \dots + a_nX_n + b] \\ &= \int \dots \int_{\mathbb{R}^n} g(x_1, \dots, x_n) f(x_1, \dots, x_n) dx_1 \dots dx_n \\ &= \sum_{i=1}^n a_i \int \dots \int_{\mathbb{R}^n} x_i f(x_1, \dots, x_n) dx_1 \dots dx_n \\ & \quad + b \int \dots \int_{\mathbb{R}^n} f(x_1, \dots, x_n) dx_1 \dots dx_n \\ &= \sum_{i=1}^n a_i \int_{\mathbb{R}} x_i \left(\int \dots \int_{\mathbb{R}^{n-1}} f(x_1, \dots, x_n) dx_1 \dots dx_n \right) dx_i + b \\ &= \sum_{i=1}^n a_i \int_{\mathbb{R}} x_i f_i(x_i) dx_i + b \\ &= \sum_{i=1}^n a_i \mathbb{E}[X_i] + b \end{aligned}$$

where f_i is the marginal pdf of X_i . □

We remark that Proposition 3.2.3 does NOT require the independence of $X_1, \dots, X_n$.

Example 3.2.3 (Convenience store). Let X denote the daily number of bottles of mineral water sold at a neighborhood convenience store with pmf:

Possible Values of X	10	11	12	13
$P(X = x)$	0.25	0.30	0.30	0.15

The convenience store sells each bottle of mineral water at a fixed price of \$2, and it incurs a daily fixed cost of \$5 for storing and displaying the mineral water. Let $Y = 2X - 5$ represent the daily net revenue (in dollars) from selling mineral water. Find the expected revenue $\mathbb{E}[Y]$.

Solution. Method I. The pmf of Y is given in Example 2.4.1:

Possible Values of Y	15	17	19	21
$P(Y = y)$	0.25	0.30	0.30	0.15

By the definition of expectation, we have

$$\mathbb{E}[Y] = (15 \times 0.25) + (17 \times 0.30) + (19 \times 0.30) + (21 \times 0.15) = 17.7.$$

Method II. Using the linearity of expectation (Proposition 3.2.3), then the expected revenue is

$$\mathbb{E}[Y] = \mathbb{E}[2X - 5] = 2\mathbb{E}[X] - \mathbb{E}[5] = 2 \times 11.35 - 5 = 17.7,$$

where $\mathbb{E}[X] = 11.35$ is given in Example 3.1.3. □

Example 3.2.4 (Matching problem). Suppose that at a party there are n guests, each with a unique hat. All hats are mixed up in a box, and each guest randomly takes one hat. Let X be the number of guests who get their own hat. Find the expected value of X .

Solution. For $i = 1, \dots, n$, let $X_i = 1$ or 0 depending on the i th guest takes his/her own hat or not. There are $n!$ different arrangement of n hats. Among them, $(n-1)!$ of the arrangements give the i th guest takes his/her own hat. So

$$\mathbb{P}(X_i = 1) = \frac{(n-1)!}{n!} = \frac{1}{n} \quad \text{and} \quad \mathbb{E}[X_i] = \frac{1}{n}$$

for all i . By Proposition 3.2.3,

$$\mathbb{E}[X] = \mathbb{E}[X_1 + \dots + X_n] = \mathbb{E}[X_1] + \dots + \mathbb{E}[X_n] = n \cdot \frac{1}{n} = 1.$$

□

Example 3.2.5 (Sampling with/without replacement). Suppose that there are R red balls and W white balls in a urn. Suppose that n ($n \leq R$) balls are selected from the box at random (with or without replacement), and let X denote the number of red balls that are selected. Determine the value of $\mathbb{E}[X]$.

Solution. Of course, one can find the pmf of X and determine $\mathbb{E}[X]$ by the definition of the expectation. Below, we provide an alternative way to find $\mathbb{E}[X]$ using the linear property of the expectation.

For $i = 1, \dots, n$, let $X_i = 1$ or 0 if the i th selected ball is red or white, respectively. Then $X = X_1 + \dots + X_n$. By Proposition 3.2.3,

$$\mathbb{E}[X] = \mathbb{E}[X_1 + \dots + X_n] = \mathbb{E}[X_1] + \dots + \mathbb{E}[X_n].$$

We need to find $\mathbb{E}[X_i]$ for all i .

With replacement. Suppose the balls are selected with replacement. Then

$$\mathbb{P}(X_i = 1) = \mathbb{P}(i\text{th selected ball is red}) = \frac{R}{R+W}.$$

and $\mathbb{P}(X_i = 0) = 1 - \mathbb{P}(X_i = 1) = \frac{W}{R+W}$. The expected value of X_i is

$$\mathbb{E}[X_i] = 1 \cdot \frac{R}{R+W} + 0 \cdot \frac{W}{R+W} = \frac{R}{R+W}.$$

Therefore,

$$\mathbb{E}[X] = \mathbb{E}[X_1] + \dots + \mathbb{E}[X_n] = \frac{nR}{R+W}.$$

Without replacement. Suppose the balls are selected without replacement. The law of total probability yields

$$\begin{aligned} \mathbb{P}(X_i = 1) &= \sum_{k=0}^{i-1} \mathbb{P}(X_i = 1 \mid X_1 + \dots + X_{i-1} = k) \mathbb{P}(X_1 + \dots + X_{i-1} = k) \\ &= \sum_{k=0}^{i-1} \frac{\binom{R-k}{1}}{\binom{R+W-i+1}{1}} \cdot \frac{\binom{R}{k} \binom{W}{i-k}}{\binom{R+W}{i-1}} = \frac{R}{R+W}. \end{aligned}$$

Thus

$$\mathbb{E}[X] = \mathbb{E}[X_1] + \dots + \mathbb{E}[X_n] = \frac{nR}{R+W}.$$

□

3.2.3 Expectation of products of independent random variables

In Proposition 3.2.2, letting $g(x_1, \dots, x_n) = x_1 \dots x_n$, we are able to calculate the expected value $\mathbb{E}[X_1 \dots X_n]$ of product of random variables $X_1, \dots, X_n$. In addition, if $X_1, \dots, X_n$ are independent, then we have a easy way to calculate $\mathbb{E}[X_1 \dots X_n]$.

Proposition 3.2.4 (Expectation of products). *Let $X_1, \dots, X_n$ be independent random variables with finite expected values. Then*

$$\mathbb{E}[X_1 \cdots X_n] = \mathbb{E}[X_1] \cdots \mathbb{E}[X_n] \quad (3.1)$$

Proof. We proof for the jointly continuous case. Assume that $X_1, \dots, X_n$ are jointly continuous with joint pdf f . Then $f(x_1, \dots, x_n) = f_1(x_1) \cdots f_n(x_n)$ where $f_i(x_i)$ is the marginal pdf of X_i for each i . By Proposition 3.2.2,

$$\begin{aligned} \mathbb{E}[X_1 \cdots X_n] &= \int \cdots \int_{\mathbb{R}^n} x_1 \cdots x_n f(x_1, \dots, x_n) dx_1 \cdots dx_n \\ &= \int \cdots \int_{\mathbb{R}^n} x_1 \cdots x_n f_1(x_1) \cdots f_n(x_n) dx_1 \cdots dx_n \\ &= \prod_{i=1}^n \left(\int_{\mathbb{R}} x_i f_i(x_i) dx_i \right) \\ &= \prod_{i=1}^n \mathbb{E}[X_i]. \end{aligned}$$

□

For dependent random variables, Equation (3.1) may not be true.

Example 3.2.6. Toss a coin. Let $X = 1$ or 0 if the coin lands on head or tail, respectively. Let $Y = 1 - X$. Then X and Y are dependent and $XY = 0$. Now we see that $\mathbb{E}[X] = \mathbb{E}[Y] = 1/2$, but $\mathbb{E}[XY] = 0 \neq 1/2 = \mathbb{E}[X]\mathbb{E}[Y]$.

3.3 Variances

While the expected value of a random variable describes its central tendency, it does not capture how widely the possible values are dispersed around this center. To quantify this degree of spread or variability, we introduce the variance of a random variable. Variance measures the average squared deviation of the random variable from its expected value, providing a formal way to characterize uncertainty and dispersion in a distribution.

3.3.1 Definition of variances

Definition 3.3.1 (Variance). The *variance* of a random variable X with expected value $\mu = \mathbb{E}[X]$ is defined as

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2].$$

A useful shortcut formula for computing the variance is

$$\text{Var}(X) = \mathbb{E}[X^2] - \mu^2.$$

This follows directly from the definition:

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - 2\mu \mathbb{E}[X] + \mu^2 \\ &= \mathbb{E}[X^2] - \mu^2.\end{aligned}$$

Since both X^2 and $(X - \mu)^2$ are functions of X , we can use Proposition 3.2.2 to calculate $\mathbb{E}[X^2]$ or $\mathbb{E}[(X - \mu)^2]$.

Example 3.3.1. A convenience store tracks the number of bottles of a popular energy drink sold per day, denoted by X . The probability distribution of X (based on historical data) is given below:

x (number of bottles)	0	1	2	3
$P(X = x)$	0.1	0.4	0.3	0.2

Find the variance of X .

Solution. We first compute the expected value $\mu = \mathbb{E}[X]$:

$$\mu = (0 \cdot 0.1) + (1 \cdot 0.4) + (2 \cdot 0.3) + (3 \cdot 0.2) = 0 + 0.4 + 0.6 + 0.6 = 1.6.$$

Then compute $\mathbb{E}[X^2]$:

$$\mathbb{E}[X^2] = (0^2 \cdot 0.1) + (1^2 \cdot 0.4) + (2^2 \cdot 0.3) + (3^2 \cdot 0.2) = 0 + 0.4 + 1.2 + 1.8 = 3.4.$$

Finally, applying the variance shortcut formula, we have

$$\text{Var}(X) = \mathbb{E}[X^2] - \mu^2 = 3.4 - (1.6)^2 = 3.4 - 2.56 = 0.84.$$

□

Example 3.3.2. The wait time (in minutes) for a customer at the convenience store checkout, denoted by X , follows a *triangular distribution* over $[0, 4]$ with peak at $x = 2$. Its probability density function (pdf) is:

$$f(x) = \begin{cases} \frac{x}{2} & \text{if } 0 \leq x < 2, \\ \frac{4-x}{2} & \text{if } 2 \leq x \leq 4, \\ 0 & \text{otherwise.} \end{cases}$$

Calculate $\text{Var}(X)$.

Solution. We first calculate the expected value

$$\mu = \mathbb{E}[X] = \int_0^2 x \cdot \frac{x}{2} dx + \int_2^4 x \cdot \frac{4-x}{2} dx = 4.$$

Then compute $\mathbb{E}[X^2]$ by using Proposition 3.2.2 with $g(x) = x^2$:

$$\mathbb{E}[X^2] = \int_{\mathbb{R}} x^2 f(x) dx = \int_0^2 x^2 \cdot \frac{x}{2} dx + \int_2^4 x^2 \cdot \frac{4-x}{2} dx = \frac{26}{3}.$$

Using the shortcut formula, we have

$$\text{Var}(X) = \mathbb{E}[X^2] - \mu^2 = \frac{26}{3} - (2)^2 = \frac{26}{3} - 4 = \frac{26-12}{3} = \frac{14}{3}.$$

□

Standard deviation

While variance provides a quantitative measure of the spread of a random variable's values around its mean, it has a critical limitation: it is expressed in *squared units* of the original random variable (e.g., if X measures height in centimeters, $\text{Var}(X)$ is in cm^2). This makes the variance difficult to interpret in the context of the original data scale. The standard deviation addresses this issue by taking the square root of the variance, which returns the measure of spread to the *original units* of X .

Definition 3.3.2 (Standard Deviation). The *standard deviation* of X , denoted by σ_X (or simply σ when the random variable is clear from context), is the non-negative square root of the variance:

$$\sigma_X = \sqrt{\text{Var}(X)} = \sqrt{\mathbb{E}[(X - \mu)^2]}.$$

Standard deviation is more intuitive for practical interpretation: it directly quantifies the “typical” or “average” deviation of the random variable from its mean, whereas variance (being a squared quantity) exaggerates larger deviations and is less easily relatable to the actual values of X . For example, if a random variable X represents annual income in thousands of dollars with $\text{Var}(X) = 100$, the standard deviation $\sigma_X = 10$ immediately tells us the typical deviation of income from the mean is \$10,000—a far more interpretable result than the variance of 100 (thousands of dollars squared).

3.3.2 Properties of Variances

Variance is not a linear function.

Proposition 3.3.1. *Let X be a random variable, and let a and b be constants. Then*

$$\text{Var}(aX + b) = a^2 \text{Var}(X).$$

Proof. Let $\mu = \mathbb{E}[X]$. By the linearity of expectation, we have

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b = a\mu + b.$$

Using the definition of variance, we write

$$\begin{aligned} \text{Var}(aX + b) &= \mathbb{E} \left[((aX + b) - \mathbb{E}[aX + b])^2 \right] \\ &= \mathbb{E} \left[((aX + b) - (a\mu + b))^2 \right] \\ &= \mathbb{E} \left[(aX - a\mu)^2 \right] \\ &= \mathbb{E} \left[a^2 (X - \mu)^2 \right] \\ &= a^2 \mathbb{E} \left[(X - \mu)^2 \right] \\ &= a^2 \text{Var}(X). \end{aligned}$$

This completes the proof. □

Proposition 3.3.2. *Let $X_1, \dots, X_n$ be independent random variables, and let $a_1, \dots, a_n$ be constants. Then*

$$\text{Var}(a_1X_1 + \dots + a_nX_n) = a_1^2 \text{Var}(X_1) + \dots + a_n^2 \text{Var}(X_n).$$

In particular,

$$\text{Var}(X_1 + \dots + X_n) = \text{Var}(X_1) + \dots + \text{Var}(X_n).$$

Proof. By Proposition 3.3.1, we may assume without loss of generality that $\mathbb{E}[X_i] = 0$ for all i . If this is not the case, we replace each X_i with $X_i - \mathbb{E}[X_i]$ — a substitution that leaves both sides of the equation unchanged, as centering a random variable does not affect its variance. Under this assumption, $\text{Var}(X_i) = \mathbb{E}[X_i^2]$ for all i . We thus compute:

$$\begin{aligned} \text{Var}(a_1X_1 + \dots + a_nX_n) &= \mathbb{E} \left[(a_1X_1 + \dots + a_nX_n)^2 \right] \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i a_j \mathbb{E}[X_i X_j] \end{aligned}$$

$$= \sum_{i=1}^n a_i^2 \mathbb{E}[X_i^2] + \sum_{\substack{i,j=1 \\ i \neq j}}^n a_i a_j \mathbb{E}[X_i X_j].$$

Since $X_1, \dots, X_n$ are independent, $\mathbb{E}[X_i X_j] = \mathbb{E}[X_i] \mathbb{E}[X_j] = 0$ for all $i \neq j$. The cross terms therefore vanish, and we obtain:

$$\begin{aligned} \text{Var}(a_1 X_1 + \dots + a_n X_n) &= \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 0 \\ &= \sum_{i=1}^n a_i^2 \text{Var}(X_i). \end{aligned}$$

This completes the proof. $\square$

The variance do reflect the spread of the values of a random variable.

Proposition 3.3.3. *Let X be a random variable with mean μ . If $\text{Var}(X) = 0$, then $\mathbb{P}(X = \mu) = 1$.*

Proof. We start from the definition of variance:

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = 0.$$

Since $(X - \mu)^2$ is a non-negative random variable, we can apply Markov's inequality (Theorem 3.3.1). For any $\varepsilon > 0$, we have

$$\mathbb{P}((X - \mu)^2 \geq \varepsilon^2) \leq \frac{1}{\varepsilon^2} \mathbb{E}[(X - \mu)^2] = \frac{0}{\varepsilon^2} = 0.$$

This implies that $\mathbb{P}(|X - \mu| \geq \varepsilon) = 0$ for all $\varepsilon > 0$. Taking the limit as $\varepsilon \rightarrow 0^+$, we conclude that

$$\mathbb{P}(X \neq \mu) = \lim_{\varepsilon \rightarrow 0^+} \mathbb{P}(|X - \mu| \geq \varepsilon) = 0.$$

Therefore,

$$\mathbb{P}(X = \mu) = 1 - \mathbb{P}(X \neq \mu) = 1.$$

This completes the proof. $\square$

3.3.3 Markov's inequality

Theorem 3.3.1 (Markov's Inequality). *Let Y be a non-negative random variable (i.e., $\mathbb{P}(Y \geq 0) = 1$) with finite expected value $\mathbb{E}[Y]$. For any real number $a > 0$, the following inequality holds:*

$$\mathbb{P}(Y \geq a) \leq \frac{\mathbb{E}[Y]}{a}.$$

Proof. We split the proof into two cases: discrete and continuous random variables.

Case 1: Discrete case

Let Y be a discrete random variable with support $S \subseteq [0, \infty)$ and probability mass function $p_Y(y) = \mathbb{P}(Y = y)$. Then

$$\begin{aligned} \mathbb{E}[Y] &= \sum_{y \in S} y \cdot p_Y(y) \\ &= \sum_{y \in S, y \geq a} y \cdot p_Y(y) + \sum_{y \in S, y < a} y \cdot p_Y(y) \\ &\geq \sum_{y \in S, y \geq a} y \cdot p_Y(y) \quad \text{since } Y \text{ is non-negative} \\ &\geq a \cdot \sum_{y \in S, y \geq a} p_Y(y) = a \cdot \mathbb{P}(Y \geq a). \end{aligned}$$

Dividing both sides by $a > 0$ gives:

$$\mathbb{P}(Y \geq a) \leq \frac{\mathbb{E}[Y]}{a}.$$

Case 2: Continuous case

Let Y be a continuous random variable with probability density function $f_Y(y)$ (where $f_Y(y) = 0$ for $y < 0$). Then

$$\begin{aligned} \mathbb{E}[Y] &= \int_0^\infty y \cdot f_Y(y) dy \\ &= \int_0^a y \cdot f_Y(y) dy + \int_a^\infty y \cdot f_Y(y) dy \\ &\geq \int_a^\infty y \cdot f_Y(y) dy \quad \text{since } Y \text{ is non-negative} \\ &\geq a \cdot \int_a^\infty f_Y(y) dy = a \cdot \mathbb{P}(Y \geq a). \end{aligned}$$

Dividing both sides by $a > 0$ yields the result:

$$\mathbb{P}(Y \geq a) \leq \frac{\mathbb{E}[Y]}{a}.$$

In both cases, the inequality holds. This completes the proof of Markov's Inequality. $\square$

Theorem 3.3.2 (Chebyshev's Inequality). *Let X be a random variable with finite expected value $\mu = \mathbb{E}[X]$ and finite variance $\sigma^2 = \text{Var}(X)$. For any $\varepsilon > 0$,*

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}.$$

Proof. Let $Y = (X - \mu)^2$, which is a non-negative random variable (since it is a square of a real number). By the definition of variance:

$$\mathbb{E}[Y] = \mathbb{E}[(X - \mu)^2] = \sigma^2.$$

For any $\varepsilon > 0$, note that the event $|X - \mu| \geq \varepsilon$ is equivalent to the event $(X - \mu)^2 \geq \varepsilon^2$. Applying Markov's Inequality to $Y = (X - \mu)^2$ with $a = \varepsilon^2$:

$$\mathbb{P}((X - \mu)^2 \geq \varepsilon^2) \leq \frac{\mathbb{E}[(X - \mu)^2]}{\varepsilon^2}.$$

Substitute the equivalence of events and the definition of variance into the inequality:

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}.$$

This completes the proof of Chebyshev's Inequality. $\square$

3.4 Covariance and correlation

3.4.1 Covariance

While variance quantifies the spread of a single random variable around its mean, it provides no insight into the *relationship* between two random variables. To measure the degree and direction of linear association between two random variables X and Y , we introduce the concept of covariance.

Definition 3.4.1 (Covariance). Let X and Y be random variables with finite expected values $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$, respectively. The *covariance* between X and Y , denoted by $\text{Cov}(X, Y)$ (or σ_{XY}), is defined as

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)].$$

Example 3.4.1. Let X and Y be continuous random variables with joint probability density function (pdf)

$$f_{X,Y}(x, y) = \begin{cases} 2, & \text{if } 0 \leq x \leq 1, 0 \leq y \leq x, \\ 0, & \text{otherwise.} \end{cases}$$

Compute $\text{Cov}(X, Y)$.

Solution. We first find the expected values of X :

$$\mu_X = \int_0^1 x \left(\int_0^x 2 \, dy \right) dx = \int_0^1 2x^2 \, dx = \frac{2}{3}.$$

Similarly, we have $\mu_Y = \frac{1}{3}$. Then by the definition of the covariance,

$$\text{Cov}(X, Y) = \int_0^1 \int_0^x \left(x - \frac{2}{3}\right) \left(y - \frac{1}{3}\right) 2 \, dy \, dx = \frac{1}{36}. \quad \square$$

A shortcut formula for covariance is:

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

This follows directly from the definition:

$$\begin{aligned} \text{Cov}(X, Y) &= \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] \\ &= \mathbb{E}[XY] - \mu_X \mathbb{E}[Y] - \mu_Y \mathbb{E}[X] + \mu_X \mu_Y \\ &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]. \end{aligned}$$

Of course, from the shortcut formula, we see that if X and Y are independent, then $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ and $\text{Cov}(X, Y) = 0$. However, the inverse is not true.

Example 3.4.2 (Zero Covariance Does Not Imply Independence). Let X be a random variable taking values $-1, 0, 1$ each with probability $1/3$, and define $Y = X^2$. Show that $\text{Cov}(X, Y) = 0$ but X and Y are not independent.

Solution. We first calculate the expected values of X , Y , and XY :

$$\mathbb{E}[X] = (-1) \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} = 0,$$

$$Y = X^2 \implies Y \text{ takes values } 0, 1 \text{ with } \mathbb{P}(Y = 0) = \frac{1}{3}, \mathbb{P}(Y = 1) = \frac{2}{3},$$

$$\mathbb{E}[Y] = 0 \cdot \frac{1}{3} + 1 \cdot \frac{2}{3} = \frac{2}{3},$$

$$XY = X \cdot X^2 = X^3 = X \implies \mathbb{E}[XY] = \mathbb{E}[X] = 0.$$

Using the shortcut formula for covariance:

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0 - (0) \cdot \frac{2}{3} = 0.$$

Now we show these two random variables are dependent. We test a specific case:

$$\mathbb{P}(X = 1, Y = 0) = 0 \quad (\text{since } X = 1 \implies Y = 1),$$

but

$$\mathbb{P}(X = 1)\mathbb{P}(Y = 0) = \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9} \neq 0.$$

Since the joint probability does not equal the product of marginal probabilities, X and Y are *not* independent. $\square$

The covariance is a bilinear function.

Proposition 3.4.1. *Let X , Y and Z be random variables, and let a and b be constants.*

1. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$;
2. $\text{Cov}(aX, Y) = a \text{Cov}(X, Y)$, $\text{Cov}(X, bY) = b \text{Cov}(X, Y)$;
3. $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$.

Proof. The proof follows directly from the definition of covariance and the linearity of expectation. The details are left to the reader. $\square$

Using covariance, we are able to compute the variance of linear combinations without assuming independent.

Proposition 3.4.2. *Let $X_1, \dots, X_n$ be random variables, and let $a_1, \dots, a_n$ be constants. Then*

$$\text{Var}(a_1X_1 + \dots + a_nX_n) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j).$$

In particular,

$$\text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2) + 2 \text{Cov}(X_1, X_2).$$

Proof. Note that $\text{Var}(X) = \text{Cov}(X, X)$. Then since the covariance is bilinear, we have

$$\begin{aligned} \text{Var}(a_1X_1 + \dots + a_nX_n) &= \text{Cov}(a_1X_1 + \dots + a_nX_n, a_1X_1 + \dots + a_nX_n) \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j) \\ &= \sum_{i=1}^n a_i^2 \text{Var}(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j). \end{aligned}$$

We are done. $\square$

3.4.2 Correlation

While covariance provides a measure of how two random variables X and Y tend to deviate from their respective means in a coordinated fashion, its value is not standardized and is heavily dependent on the units and scales of X and Y .

This makes it difficult to meaningfully compare the strength of association across different pairs of variables, as the units of covariance are simply the product of the units of X and Y . To address this limitation, we introduce the *correlation coefficient*.

Definition 3.4.2 (Correlation Coefficient). Let X and Y be random variables with finite positive variances $\text{Var}(X)$ and $\text{Var}(Y)$, respectively. The *correlation coefficient* between X and Y , denoted by ρ_{XY} (or simply ρ), is defined as

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}}.$$

Proposition 3.4.3 (Properties of the Correlation). *Let X and Y be random variables with finite positive variances, and let ρ_{XY} be their correlation coefficient. Then the following properties hold:*

1. $-1 \leq \rho_{XY} \leq 1$.
2. (Invariance under linear transformations) For any constants a, b, c, d with $a, c \neq 0$,

$$\rho_{aX+b, cY+d} = \text{sign}(ac) \cdot \rho_{XY}.$$

In particular, if a and c have the same sign, the correlation coefficient is unchanged.

3. (Perfect linear relationship) $\rho_{XY} = \pm 1$ if and only if there exist constants $a > 0$ and b such that $Y = \pm aX + b$ almost surely.
4. If X and Y are independent, then $\rho_{XY} = 0$.

Proof. (1) Let $\mu_X = \mathbb{E}[X]$, $\mu_Y = \mathbb{E}[Y]$, $\sigma_X^2 = \text{Var}(X)$, and $\sigma_Y^2 = \text{Var}(Y)$. Consider the variance of the random variable $Z = \frac{X - \mu_X}{\sigma_X} \pm \frac{Y - \mu_Y}{\sigma_Y}$. Since variance is non-negative, we have:

$$\text{Var}\left(\frac{X - \mu_X}{\sigma_X} \pm \frac{Y - \mu_Y}{\sigma_Y}\right) \geq 0.$$

Expanding the variance using the properties of covariance:

$$\text{Var}\left(\frac{X - \mu_X}{\sigma_X}\right) + \text{Var}\left(\frac{Y - \mu_Y}{\sigma_Y}\right) \pm 2 \text{Cov}\left(\frac{X - \mu_X}{\sigma_X}, \frac{Y - \mu_Y}{\sigma_Y}\right) \geq 0.$$

Since $\text{Var}\left(\frac{X - \mu_X}{\sigma_X}\right) = \frac{\text{Var}(X)}{\sigma_X^2} = 1$ and $\text{Cov}\left(\frac{X - \mu_X}{\sigma_X}, \frac{Y - \mu_Y}{\sigma_Y}\right) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \rho_{XY}$, the inequality becomes:

$$1 + 1 \pm 2\rho_{XY} \geq 0 \quad \implies \quad 2 \pm 2\rho_{XY} \geq 0.$$

This gives $-1 \leq \rho_{XY} \leq 1$.

(2) Let $U = aX + b$ and $V = cY + d$, where $a, c \neq 0$. First, we compute the covariance and variances of U and V :

$$\text{Cov}(U, V) = \text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y),$$

$$\text{Var}(U) = \text{Var}(aX + b) = a^2 \text{Var}(X) = a^2 \sigma_X^2 \implies \sigma_U = |a| \sigma_X,$$

$$\text{Var}(V) = \text{Var}(cY + d) = c^2 \text{Var}(Y) = c^2 \sigma_Y^2 \implies \sigma_V = |c| \sigma_Y.$$

By definition of correlation:

$$\rho_{UV} = \frac{\text{Cov}(U, V)}{\sigma_U \sigma_V} = \frac{ac \text{Cov}(X, Y)}{|a| \sigma_X \cdot |c| \sigma_Y} = \frac{ac}{|a||c|} \cdot \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \text{sign}(ac) \cdot \rho_{XY}.$$

If a and c have the same sign, then $ac > 0$, so $\text{sign}(ac) = 1$, and $\rho_{UV} = \rho_{XY}$.

(3) ‘ $\Leftarrow$ ’ Suppose $Y = aX + b$ almost surely, with $a \neq 0$. Then

$$\text{Cov}(X, Y) = \text{Cov}(X, aX + b) = a \text{Var}(X),$$

and $\text{Var}(Y) = a^2 \text{Var}(X)$. Thus,

$$\rho_{XY} = \frac{a \text{Var}(X)}{\sqrt{\text{Var}(X)} \sqrt{a^2 \text{Var}(X)}} = \frac{a \text{Var}(X)}{|a| \text{Var}(X)} = \text{sign}(a).$$

If $a > 0$, then $\rho_{XY} = 1$; if $a < 0$, then $\rho_{XY} = -1$.

‘ $\implies$ ’ Suppose $\rho_{XY} = 1$. From the proof of (1), this implies:

$$\text{Var}\left(\frac{X - \mu_X}{\sigma_X} - \frac{Y - \mu_Y}{\sigma_Y}\right) = 0.$$

By Proposition 3.3.3,

$$\mathbb{P}\left(\frac{X - \mu_X}{\sigma_X} - \frac{Y - \mu_Y}{\sigma_Y} = C\right) = 1$$

for some constant C . Rearranging gives $Y = \frac{\sigma_Y}{\sigma_X} X + \left(\mu_Y - \frac{\sigma_Y}{\sigma_X} \mu_X - C \sigma_Y\right)$ and the slope $a = \frac{\sigma_Y}{\sigma_X}$ is positive.

The case for $\rho_{XY} = -1$ is similar, starting from $\text{Var}\left(\frac{X - \mu_X}{\sigma_X} + \frac{Y - \mu_Y}{\sigma_Y}\right) = 0$, which yields a negative slope $a = -\frac{\sigma_Y}{\sigma_X}$.

(4) If X and Y are independent, then $\text{Cov}(X, Y) = 0$ and $\rho_{XY} = 0$. $\square$

Remark 3.4.1. 1) As illustrated in Example 3.4.2, $\rho_{XY} = 0$ does *not* imply that X and Y are independent.

2) A value of $\rho_{XY} = 0$ indicates the absence of a linear relationship between X and Y , but it does *not* mean that no relationship exists between them whatsoever. For instance, non-linear relationships such as $Y = X^2$ (see Example 3.4.2) or $Y = \sin X$ can still hold even when $\rho_{XY} = 0$.

Chapter 4

Special distributions

4.1 Discrete Distributions

In this section, we will study the most important examples of discrete distributions.

4.1.1 Bernoulli Distribution

The **Bernoulli distribution** (commonly referred to as the two-point distribution) is the most fundamental discrete probability distribution in statistics. It describes the outcome of a single independent random experiment (called a *Bernoulli trial*) with exactly two mutually exclusive and exhaustive outcomes: “success” and “failure”. Notably, it is a special case of the binomial distribution where the number of trials $n = 1$.

Denote the probability of “success” in a single trial to be p , where $0 \leq p \leq 1$, and the probability of “failure” is $q = 1 - p$. Let a random variable X follow a Bernoulli distribution (denoted as $X \sim \text{Bern}(p)$). Its key characteristics are:

1. **Binary & discrete support:** X can only take two values:
 - $X = 1$ (success, probability p);
 - $X = 0$ (failure, probability $q = 1 - p$).
2. **Single trial:** It only models the result of one trial (no repetition or multiple attempts).
3. **Fixed probability:** The success probability p remains constant for the single trial (no variation).

4. **Exhaustive outcomes:** Exactly one of the two outcomes (success/failure) must occur (no other possibilities).

Probability Mass Function (PMF)

The probability mass function (PMF) of a Bernoulli random variable $X \sim \text{Bern}(p)$ is formally defined as:

$$P(X = x) = \begin{cases} p & \text{if } x = 1, \\ 1 - p & \text{if } x = 0, \\ 0 & \text{otherwise.} \end{cases}$$

A compact, unified form of the PMF (easier for derivation and calculation) is:

$$P(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}.$$

Expected Value (Mean)

The expected value (denoted as $E(X)$ or μ) measures the long-run average outcome of repeated Bernoulli trials. It is calculated using the definition of expected value for discrete random variables:

$$E(X) = \sum_{x \in \{0,1\}} x \cdot P(X = x)$$

Substitute the PMF values into the formula:

$$E(X) = (0 \cdot (1 - p)) + (1 \cdot p) = p$$

Intuitively, this means the expected outcome of a single Bernoulli trial equals the probability of success.

Variance and Standard Deviation

The variance (denoted as $\text{Var}(X)$ or σ^2) quantifies the spread of the random variable around its mean. It is calculated via the formula:

$$\text{Var}(X) = E[(X - E(X))^2] = E(X^2) - [E(X)]^2$$

Step 1: Calculate $E(X^2)$

Since X only takes values 0 and 1, $X^2 = X$ (because $0^2 = 0$ and $1^2 = 1$). Thus:

$$E(X^2) = \sum_{x \in \{0,1\}} x^2 \cdot P(X = x) = (0^2 \cdot (1 - p)) + (1^2 \cdot p) = p$$

Step 2: Compute the Variance

Substitute $E(X^2) = p$ and $E(X) = p$ into the variance formula:

$$\text{Var}(X) = p - p^2 = p(1 - p) = pq$$

Step 3: Standard Deviation

The standard deviation (denoted as σ) is the square root of the variance, representing the average deviation from the mean:

$$\sigma = \sqrt{\text{Var}(X)} = \sqrt{p(1 - p)}$$

Example 4.1.1. Suppose you flip a fair coin once (a classic Bernoulli trial), where “head” = success ($p = 0.5$) and “tail” = failure ($q = 0.5$). Let $X = 1$ for head and $X = 0$ for tail ($X \sim \text{Bern}(0.5)$):

- Expected value: $E(X) = 0.5$ (the long-run average of flipping heads is 0.5 per trial);
- Variance: $\text{Var}(X) = 0.5 \times (1 - 0.5) = 0.25$;
- Standard deviation: $\sigma = \sqrt{0.25} = 0.5$.

Key Takeaways

1. The Bernoulli (two-point) distribution models a single binary trial, defined by one parameter p (success probability).
2. Its expected value is $E(X) = p$, variance is $\text{Var}(X) = p(1 - p)$, and standard deviation is $\sigma = \sqrt{p(1 - p)}$.
3. The PMF can be compactly written as $P(X = x) = p^x(1 - p)^{1-x}$ ($x \in \{0, 1\}$), and it is a special case of the binomial distribution with $n = 1$.

4.1.2 Geometric Distribution

The geometric distribution models the **number of independent Bernoulli trials needed to get the first successful outcome**, and it has **two standard, widely used definitions** (a critical distinction to avoid confusion). It is defined by a single parameter: the constant probability of success p in a single trial ($0 < p \leq 1$), with the probability of failure denoted $q = 1 - p$.

Two Standard Definitions

Definition Type	Random Variable	Discrete Support
Type 1		
No. of trials until the first success (includes the successful trial)	$X \sim \text{Ge}(p)$	$X \in \{1, 2, 3, \dots\}$
Type 2		
No. of failures before the first success (excludes the successful trial)	$Y \sim \text{Ge}^*(p)$	$Y \in \{0, 1, 2, \dots\}$

Note: The two definitions are directly related: $X = Y + 1$ and $Y = X - 1$. This relationship is used to derive the expected value and variance of one from the other.

Probability Mass Function (PMF)

The PMF gives the probability of observing the first success on a specific trial (Type 1) or a specific number of failures before the first success (Type 2):

1. Type 1 ($X \sim \text{Ge}(p)$):

$$P(X = k) = q^{k-1}p, \quad k = 1, 2, 3, \dots$$

Interpretation: $k - 1$ consecutive failures (probability q^{k-1}) followed by 1 success (probability p).

2. Type 2 ($Y \sim \text{Ge}^*(p)$):

$$P(Y = k) = q^k p, \quad k = 0, 1, 2, \dots$$

Interpretation: k consecutive failures (probability q^k) followed by 1 success (probability p).

Expected Value (Mean)

Proposition 4.1.1. *Let $X \sim \text{Ge}(p)$ and $Y \sim \text{Ge}^*(p)$. Then*

$$E(X) = \frac{1}{p} \quad \text{and} \quad E(Y) = \frac{1}{p} - 1.$$

Intuition: If the success probability is $p = 0.25$ ($1/4$), the expected number of trials to get the first success is 4 and the expected number of failures before the first success is 3.

Proof. By definition of expected value for a discrete random variable:

$$E[X] = \sum_{k=1}^{\infty} k \cdot P(X = k) = \sum_{k=1}^{\infty} kq^{k-1}p.$$

Start with the infinite geometric series (valid for $|q| < 1$):

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q}.$$

Differentiate both sides with respect to q :

$$\frac{d}{dq} \left(\sum_{k=0}^{\infty} q^k \right) = \frac{d}{dq} \left(\frac{1}{1-q} \right).$$

Left-hand side (LHS):

$$\sum_{k=1}^{\infty} kq^{k-1}.$$

Right-hand side (RHS) (using chain rule):

$$\frac{1}{(1-q)^2}.$$

Thus:

$$\sum_{k=1}^{\infty} kq^{k-1} = \frac{1}{(1-q)^2}.$$

Substitute the differentiated series into the expected value formula:

$$E[X] = p \cdot \sum_{k=1}^{\infty} kq^{k-1} = p \cdot \frac{1}{(1-q)^2}.$$

Since $1 - q = p$:

$$E[X] = p \cdot \frac{1}{p^2} = \frac{1}{p}.$$

Using the relationship $X = Y + 1$ and linearity of expectation, we immediately obtain

$$E(Y) = E(X - 1) = \frac{1}{p} - 1.$$

□

Variance and Standard Deviation

Variance is invariant under constant subtraction/addition, namely, $\text{Var}(aX+b) = a^2\text{Var}(X)$, so **both definitions have the same variance**:

Proposition 4.1.2. *Let $X \sim \text{Ge}(p)$ and $Y \sim \text{Ge}^*(p)$. Then*

1. *Variance (for both X and Y):*

$$\text{Var}(X) = \text{Var}(Y) = \frac{q}{p^2} = \frac{1-p}{p^2}.$$

2. *Standard Deviation:*

$$\sigma_X = \sigma_Y = \sqrt{\frac{1-p}{p^2}} = \frac{\sqrt{1-p}}{p}.$$

Proof. The variance of a random variable is defined as:

$$\text{Var}(X) = E[X^2] - (E[X])^2.$$

The expected value (mean) of X is known to be:

$$E[X] = \frac{1}{p},$$

so we only need to compute $E[X^2]$. By definition of expected value for a discrete random variable:

$$E[X^2] = \sum_{k=1}^{\infty} k^2 P(X=k) = \sum_{k=1}^{\infty} k^2 q^{k-1} p.$$

Using the infinite series result (valid for $|q| < 1$):

$$\sum_{k=1}^{\infty} k^2 q^{k-1} = \frac{1+q}{(1-q)^3} = \frac{1+q}{p^3}.$$

Substitute back into $E[X^2]$:

$$E[X^2] = p \cdot \frac{1+q}{p^3} = \frac{1+q}{p^2}$$

Substitute $E[X]$ and $E[X^2]$ into the variance formula:

$$\text{Var}(X) = \frac{1+q}{p^2} - \left(\frac{1}{p}\right)^2.$$

Simplify the expression:

$$\text{Var}(X) = \frac{1 + q - 1}{p^2} = \frac{q}{p^2}.$$

Since $q = 1 - p$, we can rewrite the variance as:

$$\text{Var}(X) = \frac{1 - p}{p^2}.$$

Variance is invariant under constant subtraction/addition, so $\text{Var}(Y) = \text{Var}(X)$. Taking square root, we obtain the standard deviation σ_X and σ_Y . $\square$

Example 4.1.2. For a fair coin ($p = 0.5, q = 0.5$):

- $E(X) = 2$ (2 flips for the first head), $E(Y) = 1$ (1 tail before the first head);
- $\text{Var}(X) = \text{Var}(Y) = 2$, $\sigma = \sqrt{2} \approx 1.414$.

4.1.3 Binomial Distribution

The **binomial distribution** is a foundational discrete probability distribution in statistics and probability theory. It quantifies the number of successful outcomes in a fixed set of **independent, identical** Bernoulli trials (each trial has only two mutually exclusive results: “success” or “failure”).

Definition 4.1.1. A random variable X follows a **binomial distribution** (denoted $X \sim \text{Bin}(n, p)$) if it counts the number of “successes” in n trials that satisfy four core conditions:

1. **Fixed trials:** Exactly n independent trials (sampling attempts) are performed.
2. **Independence:** The outcome of one trial does not influence the outcome of any other trial.
3. **Constant success probability:** Each trial has the same probability p of success (and $1 - p$ of failure).
4. **Binary outcomes:** Each trial results in either “success” or “failure” (mutually exclusive, exhaustive).

Example 4.1.3. Consider flipping the same fair coin n times **independently**:

- Let X = total number of Heads (successes) in the n flips.

Here, X is a Binomial random variable with parameters n (number of trials) and $p = 0.5$ (success probability per trial). We denote this as:

$$X \sim \text{Binomial}(n, 0.5).$$

Example 4.1.4. In **with-replacement sampling**, after each draw from a population, the selected element is *returned to the population* before the next draw. This ensures:

1. The population size/composition is unchanged for every trial, so the success probability $p = \frac{K}{N}$ (where K = number of success elements, N = total population size) is **constant**.
2. Trials are **independent**: Drawing a success/failure in one trial does not alter the odds for future trials (the population is reset).

All binomial conditions are satisfied *exactly*, so the number of successes in n with-replacement draws is a binomial random variable.

Probability Mass Function (PMF)

The Probability Mass Function (PMF) of a binomial distribution gives the probability of observing exactly k successes in n trials:

$$P(X = k) = \binom{n}{k} p^k q^{n-k}, \quad k = 0, 1, 2, \dots, n$$

Where:

- $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ (read as “n choose k”) is the **binomial coefficient**, representing the number of ways to select k successful trials out of n total trials.
- p^k : Probability of k successes.
- q^{n-k} : Probability of $n - k$ failures in the remaining trials.

Example 4.1.5. If you roll a fair die ($p = 1/6$ for rolling a 6, “success”) 3 times ($n = 3$), the probability of rolling exactly 2 sixes is:

$$P(X = 2) = \binom{3}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^1 = 3 \times \frac{1}{36} \times \frac{5}{6} = \frac{15}{216} \approx 0.0694$$

Binomial as Sum of Independent Bernoulli Variables

The binomial random variable X is **exactly equal to the sum of n independent Bernoulli random variables**. Here's the intuition with coin flips:

- Each coin flip is a Bernoulli trial: Let X_i (for $i = 1, 2, \dots, n$) represent the i -th flip, where $X_i = 1$ (Heads/success) and $X_i = 0$ (Tails/failure), with $X_i \sim \text{Bernoulli}(p)$.
- The flips are independent: The outcome of flip i does not affect flip j (for $i \neq j$).
- The total number of heads in n flips is the sum of these individual Bernoulli variables:

$$X = X_1 + X_2 + \dots + X_n.$$

- This sum satisfies the binomial PMF.

Expected Value (Mean)

The expected value (denoted $E(X)$ or μ) of a binomial distribution represents the **long-run average number of successes** in repeated sets of n trials.

Proposition 4.1.3. *Let $X \sim B(n, p)$. Then*

$$E(X) = np.$$

Proof. A binomial random variable X can be decomposed into the sum of n independent Bernoulli random variables $X_1, X_2, \dots, X_n$, i.e.,

$$X = X_1 + X_2 + \dots + X_n.$$

For a single Bernoulli trial X_i , $E(X_i) = p$ (the expected value of a Bernoulli distribution). By the **linearity of expectation** (valid even for dependent variables, but here trials are independent), the expected value of the sum is the sum of expected values:

$$E(X) = E(X_1) + E(X_2) + \dots + E(X_n) = np.$$

□

Example 4.1.6. If you flip a fair coin ($p = 0.5$) 100 times ($n = 100$), the expected number of heads is $E(X) = 100 \times 0.5 = 50$.

Variance and Standard Deviation

Variance (denoted $\text{Var}(X)$ or σ^2) measures the spread of the binomial distribution around its mean; standard deviation (σ) is the square root of variance (in the same units as the random variable).

Proposition 4.1.4. *Let $X \sim B(n, p)$. Then*

$$\text{Var}(X) = np(1 - p) \text{ and } \sigma = \sqrt{np(1 - p)}.$$

Proof. For independent random variables, the variance of the sum equals the sum of variances. For a single Bernoulli trial X_i , $\text{Var}(X_i) = p(1 - p)$. Thus:

$$\text{Var}(X) = \text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n) = np(1 - p) = np(1 - p).$$

The standard deviation is:

$$\sigma = \sqrt{\text{Var}(X)} = \sqrt{np(1 - p)}.$$

□

Example 4.1.7. Using the coin flip example ($n = 100$, $p = 0.5$):

- Variance: $\text{Var}(X) = 100 \times 0.5 \times 0.5 = 25$
- Standard deviation: $\sigma = \sqrt{25} = 5$

This means the number of heads typically varies by about 5 from the mean (50) in 100 flips.

Key Takeaways

1. The binomial distribution ($X \sim B(n, p)$) models successes in n independent Bernoulli trials with fixed success probability p , and has discrete outcomes from 0 to n .
2. Its expected value (mean) is $E(X) = np$, representing the average number of successes over repeated trials.
3. Variance is $\text{Var}(X) = np(1 - p)$ (standard deviation $\sigma = \sqrt{np(1 - p)}$), quantifying the spread of outcomes around the mean.

4.1.4 Hypergeometric Distribution

A discrete random variable X follows a binomial distribution (denoted as $X \sim \text{Binomial}(n, p)$) if it models the number of successes in n **independent, with-replacement** sampling. In contrast, the hypergeometric distribution describes the number of successes in a fixed number of **without-replacement** draws from a finite population with a known number of success and failure elements.

Definition 4.1.2. A discrete random variable X follows a hypergeometric distribution is denoted as $X \sim \text{Hypergeometric}(N, K, n)$, where

- N : Total size of the population
- K : Number of “success” elements in the population
- n : Number of draws (sample size), where $0 \leq n \leq N$
- X : Discrete random variable = number of successes in n draws

Probability Mass Function (PMF)

For a hypergeometric random variable X , the probability of observing exactly k successes in n draws is:

$$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

where: $\binom{a}{b} = \frac{a!}{b!(a-b)!}$ (binomial coefficient, “a choose b”) and the valid range for k is $\max(0, n - (N - K)) \leq k \leq \min(K, n)$ (ensures binomial coefficients are non-zero).

Example 4.1.8. An urn contains 50 balls: 15 red balls and 35 blue balls.

- (a) Suppose we sample 10 balls **with replacement** (each ball is returned to the urn before the next draw). Calculate the probability of drawing exactly 4 red balls in the 10 draws.
- (b) Suppose we sample 10 balls **without replacement** (no ball is returned to the urn after drawing). Calculate the probability of drawing exactly 4 red balls in the 10 draws.

Solution. Let X be the number of red balls in the 10 draws.

- (a) **With Replacement Sampling:** $X \sim \text{Binomial}(n, p)$.

$$P(X = 4) = \binom{10}{4} (0.3)^4 (0.7)^6 = 210 \times 0.0009529569 \approx 0.2001.$$

(b) **Without Replacement Sampling:** $X \sim \text{Hypergeometric}(N, K, n)$.

$$P(X = k) = \frac{\binom{15}{4} \binom{35}{6}}{\binom{50}{10}} \frac{1365 \times 1623160}{10272278170} = \frac{2215613400}{10272278170} \approx 0.2157.$$

□

Expected Value and Variance

Proposition 4.1.5. *Let $X \sim \text{Hypergeometric}(N, K, n)$. Then*

$$E[X] = n \cdot \frac{K}{N}.$$

Proof. The expected value of X is:

$$E[X] = \sum_{k=0}^n k \cdot P(X = k) = \sum_{k=0}^n k \cdot \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}.$$

The $k = 0$ term is 0, so we start summing from $k = 1$:

$$E[X] = \frac{1}{\binom{N}{n}} \sum_{k=1}^n k \binom{K}{k} \binom{N-K}{n-k}.$$

Use the identity $k \binom{K}{k} = K \binom{K-1}{k-1}$ (valid for $k \geq 1$):

$$E[X] = \frac{K}{\binom{N}{n}} \sum_{k=1}^n \binom{K-1}{k-1} \binom{N-K}{n-k}.$$

Let $m = k - 1$ (so $k = m + 1$, sum from $m = 0$ to $m = n - 1$):

$$E[X] = \frac{K}{\binom{N}{n}} \sum_{m=0}^{n-1} \binom{K-1}{m} \binom{N-K}{(n-1)-m}.$$

Vandermonde's Identity: $\sum_{m=0}^t \binom{A}{m} \binom{B}{t-m} = \binom{A+B}{t}$. Here, $A = K - 1$, $B = N - K$, $t = n - 1$:

$$\sum_{m=0}^{n-1} \binom{K-1}{m} \binom{N-K}{(n-1)-m} = \binom{(K-1) + (N-K)}{n-1} = \binom{N-1}{n-1}.$$

Use $\binom{N}{n} = \frac{N}{n} \binom{N-1}{n-1}$:

$$E[X] = \frac{K \cdot \binom{N-1}{n-1}}{\frac{N}{n} \binom{N-1}{n-1}} = n \cdot \frac{K}{N}.$$

□

Proposition 4.1.6. *Let $X \sim \text{Hypergeometric}(N, K, n)$. Then*

$$\text{Var}(X) = n \cdot \frac{K(N-K)(N-n)}{N^2(N-1)}.$$

Proof. Variance is defined as:

$$\text{Var}(X) = E[X^2] - (E[X])^2.$$

We first compute $E[X(X-1)]$ (a shortcut for discrete distributions):

$$E[X(X-1)] = \sum_{k=0}^n k(k-1) \cdot P(X=k).$$

Use $k(k-1)\binom{K}{k} = K(K-1)\binom{K-2}{k-2}$ (valid for $k \geq 2$; $k=0,1$ terms are 0):

$$E[X(X-1)] = \frac{K(K-1)}{\binom{N}{n}} \sum_{k=2}^n \binom{K-2}{k-2} \binom{N-K}{n-k}.$$

Let $m = k-2$ (sum from $m=0$ to $m=n-2$) and apply Vandermonde's Identity:

$$\sum_{m=0}^{n-2} \binom{K-2}{m} \binom{N-K}{(n-2)-m} = \binom{N-2}{n-2}.$$

Thus:

$$E[X(X-1)] = \frac{K(K-1)\binom{N-2}{n-2}}{\binom{N}{n}}.$$

Simplify with $\binom{N}{n} = \frac{N(N-1)}{n(n-1)}\binom{N-2}{n-2}$:

$$E[X(X-1)] = n(n-1) \cdot \frac{K(K-1)}{N(N-1)}.$$

Since $E[X^2] = E[X(X-1)] + E[X]$,

$$E[X^2] = n(n-1) \frac{K(K-1)}{N(N-1)} + n \frac{K}{N}.$$

Substitute $E[X^2]$ and $(E[X])^2 = n^2 \frac{K^2}{N^2}$:

$$\text{Var}(X) = \left[n(n-1) \frac{K(K-1)}{N(N-1)} + n \frac{K}{N} \right] - n^2 \frac{K^2}{N^2}.$$

Factor and simplify (let $p = \frac{K}{N}$, population success probability):

$$\text{Var}(X) = np(1-p) \cdot \frac{N-n}{N-1}$$

Or in full form (without p):

$$\text{Var}(X) = n \cdot \frac{K(N-K)(N-n)}{N^2(N-1)}.$$

□

Binomial as the limit of hypergeometric

Despite their differences, binomial and hypergeometric distributions are closely related, with the binomial distribution being a limiting case of the hypergeometric distribution. When the population size N becomes very large (approaching infinity) relative to the sample size n , the hypergeometric distribution approximates the binomial distribution. This is because removing a small sample from a large population has a negligible effect on the population composition—effectively, the probability of success $p = \frac{K}{N}$ remains nearly constant, and the draws become approximately independent. Formally, as $N \rightarrow \infty$ and $\frac{K}{N} \rightarrow p$ (while n and p remain fixed), the hypergeometric PMF converges to the binomial PMF:

$$\lim_{N \rightarrow \infty, \frac{K}{N} \rightarrow p} \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}} = \binom{n}{k} p^k (1-p)^{n-k}$$

If we define $p = \frac{K}{N}$ (the proportion of successes in the hypergeometric population), the expected value of both distributions takes the same form: $E[X] = np$. This reflects that the expected number of successes is the product of the number of trials/draws and the success probability/proportion.

4.1.5 Negative Binomial Distribution

The negative binomial distribution (often referred to as the inverse binomial distribution in some contexts) is another discrete probability distribution closely related to the binomial distribution. Unlike the binomial distribution (which fixes the number of trials and counts successes), the negative binomial distribution fixes the number of successes and counts the number of trials required to achieve those successes, or the number of failures before the r -th success.

There are two common formulations of the negative binomial distribution. We focus on the most widely used one: the number of **independent, with-replacement trials** X required to achieve exactly r successes (where r is a positive integer, $r \geq 1$), which is denoted as $X \sim \text{NegBin}(r, p)$, where p ($0 < p < 1$) is the probability of success in a single independent trial.

Negative Binomial and Binomial Distribution

The negative binomial distribution is the “inverse” of the binomial distribution:

- Binomial: Fix number of trials (n), count successes (k).
- Negative Binomial: Fix number of successes (r), count trials (k) needed.

Probability Mass Function (PMF)

For a negative binomial random variable X , the probability that exactly k trials are needed to achieve r successes is:

$$P(X = k) = \binom{k-1}{r-1} p^r q^{k-r}$$

where $k \geq r$ (i.e., $k = r, r+1, r+2, \dots$). Explanation:

- $\binom{k-1}{r-1}$: Number of ways to arrange $r-1$ successes in the first $k-1$ trials (the k -th trial must be a success).
- p^r : Probability of r successes (one on the k -th trial, $r-1$ in the first $k-1$ trials).
- q^{k-r} : Probability of $k-r$ failures in the first $k-1$ trials.

Example 4.1.9. Consider flipping a fair coin (probability of Heads = 0.5, defined as “success”; Tails = 0.5, defined as “failure”). All flips are independent.

- Calculate the probability of getting exactly 5 Heads in 8 flips.
- Calculate the probability that you need exactly 8 total flips to get your 5th Head.

Solution. Consider flipping a fair coin.

- Let X be the number of Heads in 8 flips. $X \sim \text{Binomial}(n, p)$.

$$P(X = 5) = \binom{8}{5} (0.5)^5 (0.5)^3 = 56 \times 0.00390625 = 0.21875.$$

- Let Y be the random total number of flips to achieve 5 successes. $Y \sim \text{NegativeBinomial}(r, p)$.

$$P(Y = 8) = \binom{8-1}{5-1} (0.5)^5 (0.5)^{8-5} = 35 \times 0.00390625 = 0.13671875.$$

□

Negative Binomial as Sum of Independent Geometric Variables

A Negative Binomial (NB) random variable $Y \sim \text{NegativeBinomial}(r, p)$ is **exactly equal to the sum of r independent Geometric random variables**. Here's the intuition with coin flips:

- A Geometric random variable Y_i models the number of flips needed to get the i -th Head after the $(i-1)$ -th Head (i.e., the “gap” between consecutive successes).
- The total number of flips to get r Heads is the sum of these gaps:

$$Y = Y_1 + Y_2 + \cdots + Y_r.$$

- Each $Y_i \sim \text{Geometric}(p)$:

$$P(Y_i = k) = (1 - p)^{k-1} p \quad \text{for } k = 1, 2, 3, \dots$$

- $Y = n$ means that the r -th Head must occur on the n -th flip (success on flip n) and there are exactly $r - 1$ Heads in the first $n - 1$ flips (failures on the remaining $(n - 1) - (r - 1) = n - r$ flips). By independence:

$$P(Y = n) = \binom{n-1}{r-1} p^{r-1} (1-p)^{n-r} \cdot p = \binom{n-1}{r-1} p^r (1-p)^{n-r},$$

which is exactly the PMF of $Y \sim \text{NegativeBinomial}(r, p)$.

Expected Value and Variance

Proposition 4.1.7. *For $X \sim \text{NegBin}(r, p)$, the expected number and the variance are:*

$$E[X] = \frac{r}{p} \quad \text{and} \quad \text{Var}(X) = \frac{rq}{p^2}.$$

Proof. The negative binomial distribution is the sum of r independent geometric random variables ($X = X_1 + X_2 + \cdots + X_r$, where $X_i \sim \text{Geometric}(p)$). Since $E[X_i] = \frac{1}{p}$, linearity of expectation gives:

$$E[X] = \sum_{i=1}^r E[X_i] = \frac{r}{p}.$$

On the other hand, each geometric random variable X_i has variance $\frac{1-p}{p^2}$. For independent variables, variance of the sum is the sum of variances:

$$\text{Var}(X) = \sum_{i=1}^r \text{Var}(X_i) = r \frac{1-p}{p^2}.$$

□

4.1.6 Poisson distribution

The **Poisson distribution** is a fundamental discrete probability distribution in statistics, primarily used to model the **number of rare, independent events occurring in a fixed interval of time, space, or volume** (e.g., the number of customers arriving at a store in an hour, the number of defective items in a batch, or the number of earthquakes in a region per year). It is named after the French mathematician Siméon Denis Poisson and is also a limiting case of the binomial distribution under specific conditions.

Definition 4.1.3. A random variable X is said to have a Poisson distribution (denoted as $X \sim \text{Poisson}(\lambda)$) if the Probability Mass Function (PMF) of X is

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

Where:

- $e \approx 2.71828$: The base of the natural logarithm;
- λ : The average number of events in the interval (the distribution's only parameter);
- $k!$: The factorial of k (e.g., $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$; by definition, $0! = 1$);
- $e^{-\lambda}$: A normalization constant that ensures the sum of all probabilities over $k = 0$ to ∞ equals 1 (total probability = 1).

Example 4.1.10. Suppose a coffee shop receives an average of 4 customers per 10-minute interval ($\lambda = 4$). The number of customers arriving in a fixed interval of time is generally regarded as following the Poisson distribution. The probability of exactly 3 customers arriving in 10 minutes is:

$$P(X = 3) = \frac{e^{-4} \times 4^3}{3!} = \frac{0.0183 \times 64}{6} \approx \frac{1.1712}{6} \approx 0.1952$$

This means there is a 19.5% chance of exactly 3 customers arriving in 10 minutes.

Expected Value and Variance

The expected value (denoted $E(X)$ or μ) of a Poisson distribution is the long-run average number of events occurring in the fixed interval and the variance (denoted $\text{Var}(X)$ or σ^2) measures the spread of the number of events around the mean.

Proposition 4.1.8. For $X \sim \text{Poisson}(\lambda)$, the expected value and the variance are:

$$E(X) = \lambda \text{ and } \text{Var}(X) = \lambda.$$

Proof. Using the definition of expected value for discrete random variables:

$$E(X) = \sum_{k=0}^{\infty} k \cdot P(X = k) = \sum_{k=0}^{\infty} k \cdot \frac{e^{-\lambda} \lambda^k}{k!}.$$

Since the term for $k = 0$ is 0, we start from $k = 1$ and simplify (using $k! = k \times (k-1)!$):

$$E(X) = e^{-\lambda} \lambda \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = e^{-\lambda} \lambda e^{\lambda} = \lambda.$$

(The sum $\sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}$ is the Taylor series expansion of e^{λ} .)

On the other hand, observe that

$$E(X^2) = \lambda + \lambda^2.$$

Substituting $E(X) = \lambda$ into the variance formula $\text{Var}(X) = E(X^2) - [E(X)]^2$:

$$\text{Var}(X) = (\lambda + \lambda^2) - \lambda^2 = \lambda.$$

□

The standard deviation (denoted σ)—the average deviation from the mean—is the square root of the variance:

$$\sigma = \sqrt{\text{Var}(X)} = \sqrt{\lambda}.$$

Example 4.1.11. Using the coffee shop example ($\lambda = 4$):

- Expected number of customers in 10 minutes: $E(X) = 4$;
- Variance: $\text{Var}(X) = 4$;
- Standard deviation: $\sigma = \sqrt{4} = 2$.

This means the number of customers typically varies by about 2 from the mean (4) in a 10-minute interval.

The Poisson Distribution as a Limit

The Poisson distribution is the **limit of the binomial distribution** when:

- The number of trials $n \rightarrow \infty$ (very large number of trials);
- The success probability $p \rightarrow 0$ (very low probability of success per trial);
- The product $np = \lambda$ (constant average number of successes).

Formally:

$$\lim_{n \rightarrow \infty, p \rightarrow 0, np = \lambda} \binom{n}{k} p^k (1-p)^{n-k} = \frac{e^{-\lambda} \lambda^k}{k!}$$

Example 4.1.12. If a factory produces 1000 items ($n = 1000$) with a 0.005 defect rate ($p = 0.005$) per item, the average number of defects is $\lambda = np = 5$. The probability of exactly 3 defects can be calculated via binomial (complex) or Poisson (simpler) distribution—both yield nearly identical results (0.1404).

Key Takeaways

1. The Poisson distribution ($X \sim \text{Poisson}(\lambda)$) models the number of rare, independent events in a fixed interval, defined solely by λ (the average event rate, $\lambda > 0$).
2. Its key property is that **expected value** = **variance** = λ , with standard deviation $\sigma = \sqrt{\lambda}$.
3. The PMF is $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$ (for $k \geq 0$), and it is the limiting case of the binomial distribution for large n , small p , and constant $np = \lambda$.

At the end of this section, we illustrate the connections among the six discrete distributions through the example of flipping coins.

Distribution	Core Scenario (Coin Flip Context)	Key Relationships to Other Distributions
Bernoulli	Outcome of 1 single coin flip (1 = Heads, 0 = Tails)	- Building block for Binomial distribution - Special case of Binomial with $n = 1$
Binomial	Number of Heads in n independent flips	- Sum of n independent Bernoulli variables - Limit of Hypergeometric as $N \rightarrow \infty$ - Approximated by Poisson ($n \rightarrow \infty, p \rightarrow 0, np = \lambda$)
Hypergeometric	Number of Heads in n without-replacement draws	- Exact for finite-population sampling - Converges to Binomial when $N \gg n$
Poisson	Number of rare Heads in a very large number of trials	- Approximation of Binomial for rare events (n large, p small, $\lambda = np$)
Geometric	Number of flips until the first Head appears	- Special case of Negative Binomial with $r = 1$ - Building block for Negative Binomial
Negative Binomial	Number of flips until the r -th Head appears	- Sum of r independent Geometric variables - Generalization of Geometric ($r > 1$)

Table 4.1: Relationships Between Key Discrete Distributions (Coin Flip Examples)

4.2 Continuous Distributions

Unlike discrete distributions (which model random variables taking discrete and at most countable values), continuous probability distributions describe random variables that can take any value within a continuous interval (e.g., time, height,

weight, temperature). For continuous random variables, we use Probability Density Function (PDF) instead of PMF, and the probability of the variable taking any single value is 0 (we calculate probabilities over intervals). Below are the most commonly used continuous distributions.

4.2.1 Uniform Distribution

A continuous random variable X is said to follow a uniform distribution over the interval $[a, b]$ (denoted as $X \sim U(a, b)$) if its PDF is constant within the interval $[a, b]$ and zero outside this interval. This characteristic implies that every value in the interval $[a, b]$ has the same probability density, reflecting equal likelihood of occurrence.

Key Use Cases

- Generating random numbers between two values (e.g., $X \sim U(0, 1)$ for random numbers in $[0, 1]$).
- Modeling waiting time for events with fixed intervals (e.g., bus waiting time).
- Random sampling from a continuous range with equal likelihood for all values.

Probability Density Function (PDF)

The PDF of $X \sim U(a, b)$ is:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

The constant $\frac{1}{b-a}$ ensures the total area under the PDF curve equals 1 (total probability).

Expected Value and Variance

Proposition 4.2.1. *Let $X \sim U(a, b)$. Then the expected value of X is the midpoint of the interval $[a, b]$:*

$$E[X] = \frac{a + b}{2}.$$

And the variance is

$$\text{Var}(X) = \frac{(b - a)^2}{12}.$$

Proof. We use the definitional formulas for continuous distributions for the derivations:

1. Expected Value: $E[X] = \int_{-\infty}^{+\infty} x \cdot f(x) dx$;

2. Variance: $\text{Var}(X) = E[X^2] - (E[X])^2$, where $E[X^2] = \int_{-\infty}^{+\infty} x^2 \cdot f(x) dx$.

Since $f(x) = 0$ outside $[a, b]$, the integral reduces to the interval $[a, b]$:

$$E[X] = \int_a^b x \cdot \frac{1}{b-a} dx.$$

Factor out the constant $\frac{1}{b-a}$ (independent of x):

$$E[X] = \frac{1}{b-a} \int_a^b x dx = \frac{1}{b-a} \cdot \frac{1}{2} (b^2 - a^2) = \frac{a+b}{2}.$$

To compute the variance, we first calculate $E[X^2]$. Restrict the integral to $[a, b]$ and factor out the constant:

$$E[X^2] = \frac{1}{b-a} \int_a^b x^2 dx = \frac{1}{b-a} \cdot \frac{1}{3} (b^3 - a^3).$$

Simplify using the difference of cubes ($b^3 - a^3 = (b-a)(b^2 + ab + a^2)$), we obtain

$$E[X^2] = \frac{1}{b-a} \cdot \frac{1}{3} (b-a) (a^2 + ab + b^2) = \frac{a^2 + ab + b^2}{3}.$$

Substitute $E[X] = \frac{a+b}{2}$ and $E[X^2] = \frac{a^2+ab+b^2}{3}$ into the variance formula:

$$\text{Var}(X) = \frac{a^2 + ab + b^2}{3} - \left(\frac{a+b}{2} \right)^2 = \frac{(b-a)^2}{12}.$$

□

4.2.2 Exponential Distribution

A continuous random variable X follows an exponential distribution with rate parameter λ (denoted $X \sim \text{Exp}(\lambda)$) if it models the time between consecutive independent events in a Poisson process, where $X \geq 0$ (time cannot be negative).

Probability Density Function (PDF)

The PDF of $X \sim \text{Exp}(\lambda)$ is:

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Cumulative Distribution Function (CDF)

The CDF $F(x) = P(X \leq x)$ for $X \sim \text{Exp}(\lambda)$ is:

$$F(x) = \begin{cases} 1 - e^{-\lambda x} & \text{if } x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Expected Value and Variance

Proposition 4.2.2. *The expected value of $X \sim \text{Exp}(\lambda)$ (average time between events) is:*

$$E[X] = \frac{1}{\lambda}$$

and the variance is

$$\text{Var}(X) = \frac{1}{\lambda^2}.$$

The proof is trivial by using the definitional formulas for continuous distributions:

1. Expected Value: $E[X] = \int_{-\infty}^{+\infty} x \cdot f(x) dx$;
2. Variance: $\text{Var}(X) = E[X^2] - (E[X])^2$, where $E[X^2] = \int_{-\infty}^{+\infty} x^2 \cdot f(x) dx$.

Memoryless Property

For a continuous random variable X , the memoryless property is formally defined as:

$$P(X > s + t \mid X > s) = P(X > t)$$

for all non-negative real numbers $s \geq 0$ and $t \geq 0$.

Proposition 4.2.3. *A continuous random variable X following an exponential distribution with rate parameter $\lambda > 0$ (denoted $X \sim \text{Exp}(\lambda)$) possesses the memoryless property.*

The memoryless property implies that the future probability of an event is independent of past elapsed time. For example, if X models the time until a machine fails, the probability that the machine runs for an additional t hours (given it has run for s hours) equals the probability that a new machine runs for t hours. In short, the machine “forgets” how long it has already been running; its future reliability is independent of its past operating time.

Proof. To derive the memoryless property, we only need to prove that

$$P(X > s + t \mid X > s) = P(X > t).$$

Let $A = \{X > s + t\}$ and $B = \{X > s\}$. Since $X > s + t \implies X > s$, we have $A \cap B = A$, so:

$$P(X > s + t \mid X > s) = \frac{P(X > s + t \cap X > s)}{P(X > s)} = \frac{P(X > s + t)}{P(X > s)}.$$

Using the CDF of the exponential distribution:

$$P(X > s + t) = e^{-\lambda(s+t)}, \quad P(X > s) = e^{-\lambda s},$$

we obtain

$$P(X > s + t \mid X > s) = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda(s+t)+\lambda s} = e^{-\lambda t}.$$

Since $P(X > t) = e^{-\lambda t}$, we have proved

$$P(X > s + t \mid X > s) = P(X > t).$$

□

4.2.3 Normal Distribution (Gaussian Distribution)

A continuous random variable X follows a normal distribution with mean μ and variance σ^2 (denoted $X \sim N(\mu, \sigma^2)$) if its Probability Density Function (PDF) has a symmetric, bell-shaped form centered at μ , with spread determined by $\sigma > 0$.

Key Parameters

- μ : Mean (expected value), median, and mode of the distribution; shifts the curve horizontally.
- σ^2 : Variance; measures the spread of the distribution.
- σ : Standard deviation ($\sigma > 0$), square root of variance; interpretable measure of spread.

Probability Density Function (PDF)

The PDF of $X \sim N(\mu, \sigma^2)$ is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}},$$

where $\pi \approx 3.1416$ and $e \approx 2.7183$ are constants, and the normalization term ensures total probability equals 1.

Cumulative Distribution Function (CDF)

The CDF of $X \sim N(\mu, \sigma^2)$ is:

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt.$$

Standard Normal Distribution

The standard normal distribution has $\mu = 0$ and $\sigma = 1$ (denoted $Z \sim N(0, 1)$), with PDF:

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}.$$

The standard normal distribution has a well-documented CDF, denoted as $\Phi(z) = P(Z \leq z)$, with symmetry property:

$$\Phi(-z) = 1 - \Phi(z).$$

Any normal random variable $X \sim N(\mu, \sigma^2)$ can be converted to a standard normal variable Z using the z-score transformation:

$$Z = \frac{X - \mu}{\sigma}.$$

This transformation is widely used to calculate probabilities for any normal distribution using standard normal tables or statistical software.

Proposition 4.2.4. *For any normal variable $X \sim N(\mu, \sigma^2)$, let*

$$Z = \frac{X - \mu}{\sigma}.$$

Then $Z \sim N(0, 1)$.

This transformation consists of two key steps:

- Subtract the mean μ from X : $X - \mu$. This shifts the distribution's center to 0 (eliminating the effect of μ).
- Divide by the standard deviation σ : $\frac{X - \mu}{\sigma}$. This scales the distribution to have a standard deviation of 1 (eliminating the effect of σ).

Proof. First of all, let's derive the CDF of Z . By definition of CDF:

$$F_Z(z) = P(Z \leq z)$$

Substitute $Z = \frac{X-\mu}{\sigma}$ and rearrange the inequality ($\sigma > 0$, inequality direction unchanged):

$$F_Z(z) = P\left(\frac{X-\mu}{\sigma} \leq z\right) = P(X \leq \sigma z + \mu).$$

From the CDF of X , $P(X \leq \sigma z + \mu) = F_X(\sigma z + \mu)$, so:

$$F_Z(z) = \int_{-\infty}^{\sigma z + \mu} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt.$$

Applying the Leibniz Rule: for $F(y) = \int_a^{b(y)} h(t)dt$

$$F'(y) = h(b(y)) \cdot b'(y),$$

we find that

$$f_Z(z) = \frac{d}{dz} F_Z(z) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(\sigma z + \mu - \mu)^2}{2\sigma^2}} \cdot \sigma = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}},$$

which is the PDF of the standard normal distribution. Thus, $Z \sim N(0, 1)$. $\square$

Practical Use

To calculate $P(X \leq x)$ for $X \sim N(\mu, \sigma^2)$, one can

- Compute the z-score: $z = \frac{x-\mu}{\sigma}$
- Find $P(Z \leq z)$ using the standard normal CDF.

Example 4.2.1. For $X \sim N(1, 2)$, calculate $P(0.5 < X < 1.5)$.

Solution. Standardize both bounds:

$$z_1 = \frac{0.5 - 1}{\sqrt{2}} \approx -0.35, \quad z_2 = \frac{1.5 - 1}{\sqrt{2}} \approx 0.35.$$

Then

$$P(0.5 < X < 1.5) = P(-0.35 < Z < 0.35) = \Phi(0.35) - \Phi(-0.35).$$

Using symmetry $\Phi(-z) = 1 - \Phi(z)$:

$$\Phi(-0.35) = 1 - \Phi(0.35).$$

From the table:

$$\Phi(0.35) = 0.6368.$$

Thus

$$P(-0.35 < Z < 0.35) = 0.6368 - (1 - 0.6368) = 0.2736.$$

Standard Normal CDF Table $\Phi(z) = P(Z \leq z)$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621

□

Expected Value and Variance

The normal distribution is defined by two parameters: μ and σ^2 , which are actually the mean and the variance of the distribution.

Proposition 4.2.5. *For $X \sim N(\mu, \sigma^2)$, the expected value and variance are*

$$E[X] = \mu \text{ and } \text{Var}(X) = \sigma^2.$$

Proof. Lets firstly derive the mean and variance of the Standard Normal Distribution ($Z \sim N(0, 1)$). By definition:

$$E[Z] = \int_{-\infty}^{+\infty} z \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz.$$

The integrand $z \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$ is an odd function ($g(-z) = -g(z)$). The integral of an odd function over $(-\infty, +\infty)$ is 0. Using substitution $u = \frac{z^2}{2}$ (so $du = z dz$):

$$E[Z] = \frac{1}{\sqrt{2\pi}} \int_{+\infty}^{+\infty} e^{-u} du = 0.$$

Next lets use $\text{Var}(Y) = E[Y^2] - (E[Y])^2$ to calculate $\text{Var}(Z)$. Since $E[Z] = 0$, $\text{Var}(Z) = E[Z^2]$:

$$E[Z^2] = \int_{-\infty}^{+\infty} z^2 \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz.$$

Applying integration by parts:

$$E[Z^2] = -\frac{z}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \Big|_{-\infty}^{+\infty} + \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{z^2}{2}} dz.$$

The boundary term is 0 (exponential decay dominates polynomial growth). The Gaussian integral $\int_{-\infty}^{+\infty} e^{-\frac{z^2}{2}} dz = \sqrt{2\pi}$, so:

$$E[Z^2] = \frac{1}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = 1.$$

Thus:

$$\text{Var}(Z) = 1 - 0^2 = 1.$$

Now we are ready to derive the mean and variance of the General Normal Distribution. The general normal variable is $X = \sigma Z + \mu$, where $Z \sim N(0, 1)$. And thus

$$E[X] = E[\sigma Z + \mu] = \sigma E[Z] + \mu = \sigma \cdot 0 + \mu = \mu$$

and

$$\text{Var}(X) = \text{Var}(\sigma Z + \mu) = \text{Var}(\sigma Z) = \sigma^2 \text{Var}(Z) = \sigma^2 \cdot 1 = \sigma^2.$$

□

Critical Values

For a normal distribution (either $X \sim N(\mu, \sigma^2)$ or standard normal $Z \sim N(0, 1)$), a **critical value** is a threshold that divides the distribution into a critical region (rejection region) and a non-critical region (acceptance region). It is used in hypothesis testing and confidence intervals.

Definition 4.2.1. For a significance level α , the critical value z_α (standard normal) is defined to be:

$$P(Z > z_\alpha) = \alpha.$$

For $X \sim N(\mu, \sigma^2)$, the critical value x_α is derived by reversing standardization:

$$x_\alpha = \mu + z_\alpha \cdot \sigma$$

where z_α is the standard normal critical value.

Example 4.2.2. Find $z_{0.05}$ and $x_{0.05}$ for $X \sim N(\mu = 10, \sigma = 4)$.

Solution. From the Standard Normal CDF Table, $z_{0.05} \approx 1.645$. On the other hand,

$$x_{0.05} = 10 + 1.645 \times 4 = 16.58$$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952

□

4.2.4 t-Distribution

The t-distribution is a continuous probability distribution used in statistical inference for small sample sizes ($n < 30$) when the population standard deviation σ is unknown. It is defined by the **degrees of freedom (df)**, denoted v , typically $v = n - 1$ for a sample of size n .

PDF of the t-Distribution

For a t-distributed random variable T with v degrees of freedom, the PDF is:

$$f_T(t) = \frac{\Gamma\left(\frac{v+1}{2}\right)}{\sqrt{v\pi} \cdot \Gamma\left(\frac{v}{2}\right)} \left(1 + \frac{t^2}{v}\right)^{-\frac{v+1}{2}}, \quad -\infty < t < +\infty.$$

where $\Gamma(\cdot)$ is the gamma function.

Comparison with Standard Normal Distribution

The t-distribution and standard normal distribution ($Z \sim N(0, 1)$) are symmetric and centered at 0. However, their shapes differ significantly, especially for small degrees of freedom. The differences and connections can be clearly illustrated by comparing their PDF graphs:

1. **Thicker tails:** The t-distribution has heavier tails, meaning higher probability of extreme values.
2. **Lower peak:** The t-distribution's PDF peak is lower than the standard normal.
3. **Dependence on df:** The t-distribution's shape changes with v ; smaller v = thicker tails, larger v = more like standard normal.

When $v \geq 30$, the t-distribution is nearly identical to the standard normal distribution, so the standard normal can be used as an approximation.

Critical Values

A **critical value** $t_{\alpha,v}$ (for df v and significance level α) divides the t-distribution into critical and non-critical regions.

Definition 4.2.2. The critical value $t_{\alpha,v}$ is defined to be:

$$P(T > t_{\alpha,v}) = \alpha,$$

where T is the t-distributed random variable with v degrees of freedom.

Critical values depend on both α and v , and are found using a t-table:

$v \setminus \alpha$	0.10	0.05	0.025	0.01	0.005
1	3.078	6.314	12.706	31.821	63.657
5	1.476	2.015	2.571	3.365	4.032
10	1.372	1.812	2.228	2.764	3.169
15	1.341	1.753	2.131	2.602	2.947
20	1.325	1.725	2.086	2.528	2.845
30	1.310	1.697	2.042	2.457	2.750
100	1.290	1.660	1.984	2.364	2.626
∞	1.282	1.645	1.960	2.326	2.576

Note: The last row ($v = \infty$) is the standard normal critical value.

Example 4.2.3. Find $t_{0.05,15}$ and $t_{0.05,100}$.

Solution. From the t-table, $t_{0.05,15} = 1.753$.

For $v = 100$ (large degrees of freedom), the t-distribution is nearly identical to the standard normal distribution. $t_{0.05,100} = 1.660$, close to $z_{0.05} = 1.645$. $\square$

4.2.5 Chi-Square Distribution

The chi-square distribution (χ^2 -distribution) is a continuous, right-skewed probability distribution used in statistical inference, independence tests, and confidence intervals for variances. It is defined by the **degrees of freedom (df)**, denoted v .

PDF of the Chi-Square Distribution

For $X \sim \chi^2(v)$ (chi-square random variable with v df), the PDF is:

$$f_X(x) = \frac{1}{2^{v/2}\Gamma\left(\frac{v}{2}\right)} x^{\frac{v}{2}-1} e^{-\frac{x}{2}}, \quad x \geq 0$$

where $\Gamma(\cdot)$ is the gamma function.

Key Properties

- Mean: $E[X] = v$
- Variance: $\text{Var}(X) = 2v$
- Additivity: If $X_1 \sim \chi^2(v_1)$ and $X_2 \sim \chi^2(v_2)$ are independent, then $X_1 + X_2 \sim \chi^2(v_1 + v_2)$.

Relationship with the Standard Normal Distribution

The chi-square distribution has a direct and fundamental relationship with the standard normal distribution ($Z \sim N(0, 1)$)—it is defined as the sum of the squares of independent standard normal random variables. This relationship is the foundation of many statistical applications of the chi-square distribution.

Proposition 4.2.6. *Let $Z_1, Z_2, \dots, Z_v$ be v independent standard normal random variables ($Z_i \sim N(0, 1)$). Then:*

$$X = Z_1^2 + Z_2^2 + \dots + Z_v^2 \sim \chi^2(v)$$

In other words, the chi-square distribution with v df is the distribution of the sum of squares of v independent standard normal variables.

Critical Values of the Chi-Square Distribution

A **critical value** $\chi_{\alpha, v}^2$ (for df v and significance level α) divides the chi-square distribution into critical and non-critical regions.

Definition 4.2.3. The critical value $\chi_{\alpha, v}^2$ is defined to be:

$$P(X > \chi_{\alpha, v}^2) = \alpha,$$

where $X \sim \chi^2(v)$.

Critical values depend on α and v , and are found using a chi-square table:

$v \setminus \alpha$	0.995	0.95	0.10	0.05	0.025	0.01
1	0.000	0.004	2.706	3.841	5.024	6.635
5	0.412	1.145	9.236	11.070	12.833	15.086
10	2.156	3.940	15.987	18.307	20.483	23.209
15	4.601	7.261	22.307	24.996	27.488	30.578
20	7.434	10.851	28.412	31.410	34.170	37.566
30	13.787	18.493	40.256	43.773	46.979	50.892

Example 4.2.4. Find $\chi_{0.95, 15}^2$ and $\chi_{0.05, 15}^2$.

Solution. From the chi-square table, $\chi_{0.95,15}^2 = 7.261$ and $\chi_{0.05,15}^2 = 24.996$. $\square$

Note: unlike the t-distribution and standard normal distribution which are symmetric about 0, the chi-square distribution only takes non-negative values (since it is derived from the sum of squared standard normal variables). For this reason, the critical value corresponding to $\alpha = 0.05$ ($\chi_{0.05,v}^2$) and the critical value corresponding to $\alpha = 0.95$ ($\chi_{0.95,v}^2$) are not opposites of each other. In contrast, for symmetric distributions like the t-distribution or standard normal distribution, the critical value for $\alpha = 0.05$ (right-tailed) and $\alpha = 0.95$ (left-tailed) are negatives of one another, which is not the case for the right-skewed chi-square distribution.

Chapter 5

Statistics

In previous chapters, we developed probability models for a single random variable. However, many problems in probability involve several random variables simultaneously. For example, a professor must first get on a bus near her house and then transfer to a second bus. If the waiting time (in minutes) at each stop has a uniform distribution with $A = 0$ and $B = 5$, what's the probability that the total waiting time Y is less than 5 minutes?

Let X_1 be the waiting time at the first stop and X_2 be the waiting time at the second stop. Then the desired probability is

$$P(X_1 + X_2 < 5)$$

To find the solution, we need the joint probability distribution of X_1 and X_2 .

5.1 Statistics and their distributions

In application, we are usually interested to draw conclusion about a population distribution based on the information of some sample. For example, we want to understand an important question: the “average daily study time” of all university students in the country. We call this unknown, true average the population mean μ . Clearly, it's impossible to ask every single student about their study time.

Our solution is to select a small group of university students (e.g., randomly choosing $n = 500$ students) and measure their daily study time to obtain a set of data $x_1, x_2, \dots, x_{500}$. This group is our **sample**.

Our question is: How can we use the sample data from these 500 students to make a reasonable estimate and inference about the average daily study time

μ for all university students in the country? The most natural approach is to calculate the mean of the study times for these 500 students:

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_{500}}{500}$$

This $\bar{x}$ is a specific value we compute based on the sample data.

Any value calculated from sample data, like the sample mean $\bar{x}$, that is used to describe sample characteristics is called a **statistic**.

In fact, the longest study time in the sample (maximum value), the median study time in the sample, and the variance of study times in the sample, are also statistics. Typically, a statistic is used to estimate an unknown parameter of the population (e.g., using the sample mean $\bar{x}$ to estimate the population mean μ , or the sample variance to estimate the population variance σ^2).

5.1.1 Random Samples

To ensure that our sample statistics can effectively represent the population and be analyzed using probability theory tools, we typically require the sample to be a random sample.

Definition 5.1.1. The rv's $X_1, X_2, \dots, X_n$ are said to form a random sample of size n if

1. The X_i 's are independent rv's.
2. Every X_i has the same probability distribution.

Remark 5.1.1. If conditions 1 and 2 are satisfied, $X_1, X_2, \dots, X_n$ are said to be independent and identically distributed (iid). If sampling is either with replacement or from an infinite population, conditions 1 and 2 are satisfied exactly.

Since a statistic (such as the sample mean $\bar{X}$) is a random variable, how does it take on values? We introduce the following concept.

Definition 5.1.2. The distribution of all possible values of a statistic and their corresponding probabilities is called the sampling distribution of that statistic.

5.1.2 Sampling Distribution

How do we obtain and study the sampling distribution?

Method 1: Theoretical Derivation

In some ideal cases, such as when the population distribution is known (e.g., normal distribution, exponential distribution) or the population is very small, we can use probability theory to mathematically derive the exact sampling distribution of a statistic.

Proposition 5.1.1. *Since a statistic is a function of $X_1, X_2, \dots, X_n$, the probability distribution of a statistic is the joint distribution of $X_1, X_2, \dots, X_n$. If $X_1, X_2, \dots, X_n$ are independent, the joint pmf or pdf is the product of marginal pdf of X_i .*

Example 5.1.1. Service time for a certain type of bank transaction is a random variable having an exponential distribution with parameter λ . Suppose X_1 and X_2 are service times for two different customers, assumed independent of each other. Consider the total service time $T_0 = X_1 + X_2$ for the two customers, also a statistic.

Solution. Both X_1 and X_2 have an exponential distribution with parameter λ , and they are independent of each other. So their joint pdf is

$$f(x_1, x_2) = \lambda^2 e^{-\lambda x_1} e^{-\lambda x_2}.$$

Thus the cdf of T_0 is for any $t \geq 0$

$$\begin{aligned} P(T_0 \leq t) &= P(X_1 + X_2 \leq t) = \iint_{x_1 + x_2 \leq t} f(x_1, x_2) dx_1 dx_2 \\ &= \lambda^2 \int_0^t e^{-\lambda x_1} \left(\int_0^{t-x_1} e^{-\lambda x_2} dx_2 \right) dx_1 \\ &= 1 - e^{-\lambda t} - \lambda t e^{-\lambda t} \end{aligned}$$

and the pdf of T_0 is

$$f_{T_0}(t) = \begin{cases} \lambda^2 t e^{-\lambda t}, & t \geq 0, \\ 0, & t < 0. \end{cases}$$

□

Method 2: Simulation Experiment

We can use a computer to simulate the sampling process multiple times.

1. **Set the Population:** We can assume a population distribution (e.g., we can assume an average study time of μ with a standard deviation of σ).

2. **Repeated Sampling:** The computer repeatedly draw a large number of independent samples from this assumed population (e.g., $k = 10,000$ times, each time drawing $n = 500$ students' data).
3. **Calculate the Statistic:** For each sample drawn, compute its sample mean $\bar{x}$. This gives us k values of $\bar{x}$.
4. **Observe the Distribution:** Plot these k $\bar{x}$ values as a histogram. This histogram approximates the shape of the sampling distribution of the sample mean $\bar{X}$.

Example 5.1.2. The population distribution for our simulation study is normal with $\mu = 8.25$ and $\sigma = .75$, as pictured in Figure below. We actually performed

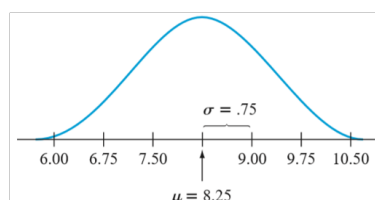


Figure 5.1: Normal population distribution with $\mu = 8.25$ and $\sigma = 0.75$.

four different experiments, with 500 replications for each one. In the first experiment, 500 samples of $n = 5$ observations each were generated using Minitab, and the sample sizes for the other three were $n = 10$, $n = 20$, and $n = 30$, respectively. The sample mean was calculated for each sample, and the resulting histograms of $\bar{x}$ values appear in the Figure (a): $n = 5$; (b): $n = 10$; (c): $n = 20$; (d): $n = 30$.

- The first thing to notice about the histograms is their shape. To a reasonable approximation, each of the four looks like a normal curve. Second, each histogram is centered approximately at 8.25, the mean of the population being sampled.
- For the larger sample sizes, most of the $\bar{x}$ values are quite close to 8.25.
- $\bar{X}$ based on a large n tends to be closer to μ than does $\bar{X}$ based on a small n .

5.1.3 The distribution of the Sample

In last section, we learned that to estimate the average daily study time μ (population mean) of university students, we take a sample and calculate the sample mean $\bar{X}$.

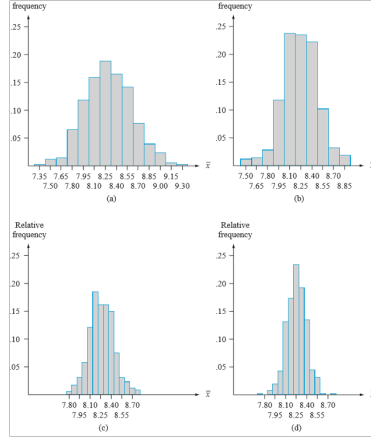


Figure 5.2: Histograms of the sample mean $\bar{X}$ for different sample sizes ($n = 5, 10, 20, 30$).

This $\bar{X}$ is a random variable with its own probability distribution, called the sampling distribution. Through theoretical analysis or computer simulation, we can estimate the sampling distribution of the statistic. Naturally, this leads to the following questions:

1. Is it reasonable to use the sample mean $\bar{X}$ to estimate the population mean μ ?
2. If $\bar{X}$ varies around μ , how much does it vary? Is our estimate “stable”?

Proposition 5.1.2. *Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with mean value μ and standard deviation σ . Then*

1. $E(\bar{X}) = \mu$ ($\mu_{\bar{X}} = \mu$),
2. $\text{Var}(\bar{X}) = \sigma^2/n$ ($\sigma_{\bar{X}} = \sigma/\sqrt{n}$).

Proof. Let $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$. By linearity of expectation,

$$E(\bar{X}) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \cdot n\mu = \mu.$$

Since the sample is independent,

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}.$$

Thus $\sigma_{\bar{X}} = \sigma/\sqrt{n}$. □

- This tells us $\bar{X}$ is an **unbiased estimator** of μ .
- While any single sample mean might not be exactly μ , the distribution of all possible $\bar{X}$ values is **centered precisely at μ** .
- The variance of $\bar{X}$ depends on the population's variance (σ) and the sample size (n).
- As sample size (n) increases, the standard error ($\sigma_{\bar{X}}$) decreases. This means larger samples lead to more precise estimates of μ .

Proposition 5.1.3. *Let $T_0 = X_1 + X_2 + \cdots + X_n$ (the sample total), then $T_0 = n\bar{X}$, and*

$$E(T_0) = n\mu \quad \text{and} \quad \text{Var}(T_0) = n\sigma^2.$$

Proof. By definition, $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, so $T_0 = \sum_{i=1}^n X_i = n\bar{X}$. By linearity of expectation,

$$E(T_0) = \sum_{i=1}^n E(X_i) = n\mu.$$

Since the sample is independent,

$$\text{Var}(T_0) = \sum_{i=1}^n \text{Var}(X_i) = n\sigma^2.$$

□

Now that we know where the sample mean $\bar{X}$ is centered ($E(\bar{X}) = \mu$) and how spread out it is ($\text{Var}(\bar{X}) = \sigma^2/n$), can we describe the actual shape of the sampling distribution?

Proposition 5.1.4. *If our random sample $X_1, X_2, \dots, X_n$ is drawn from a normal population with mean μ and standard deviation σ , then for any sample size n , the distribution of the sample mean $\bar{X}$ is also exactly normal,*

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

Similarly, the sample total T_0 is also normally distributed:

$$T_0 \sim N(n\mu, n\sigma^2).$$

Proof. Since $X_1, \dots, X_n$ are independent and each $X_i \sim N(\mu, \sigma^2)$, their sum

$$T_0 = \sum_{i=1}^n X_i$$

is normal with mean $n\mu$ and variance $n\sigma^2$, so $T_0 \sim N(n\mu, n\sigma^2)$. Because $\bar{X} = T_0/n$ is a linear transformation of a normal random variable, it is also normal:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

□

Theorem 5.1.1. Let $X_1, X_2, \dots, X_n$ be n independent random variables, where each X_i is normally distributed with mean μ_i and variance σ_i^2 . Let $a_1, a_2, \dots, a_n$ be real constants.

Consider the linear combination Y defined as:

$$Y = a_1X_1 + a_2X_2 + \dots + a_nX_n.$$

Then Y is also normally distributed. The mean and variance of Y is:

$$E(Y) = \sum_{i=1}^n a_i\mu_i \quad \text{and} \quad \text{Var}(Y) = \sum_{i=1}^n a_i^2\sigma_i^2.$$

Therefore,

$$Y \sim N\left(\sum_{i=1}^n a_i\mu_i, \sum_{i=1}^n a_i^2\sigma_i^2\right).$$

Example 5.1.3. For our sample of 16 students (from a Normal population with $\mu = 18, \sigma = 4$), what is the probability that their average study time $\bar{X}$ falls between 17 and 19 hours?

Solution. Note that

$$\bar{X} \sim N(18, 1).$$

To use standard normal tables, we convert $\bar{X}$ to a Z-score:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

then

$$P(17 \leq \bar{X} \leq 19) = P(-1 \leq Z \leq 1) = \Phi(1) - \Phi(-1) = 0.6826.$$

□

5.1.4 The Central Limit Theorem (CLT)

In many practical situations, the population distribution of X_i (e.g., individual study times) might not be normal. If the X_i 's are not from a normal distribution, can we still say anything concrete about the distribution of their sample mean $\bar{X}$, especially when our sample size n gets larger?

When dealing with the distribution of $\bar{X}$ from non-normal populations, a powerful tool comes to our rescue: **the Central Limit Theorem**. It provides an astonishing answer to our question about the distribution of $\bar{X}$ from non-normal populations.

Theorem 5.1.2 (CLT). *Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with mean μ and variance σ^2 . Then if the sample size n is sufficiently large, the sampling distribution of the sample mean $\bar{X}$ will be approximately Normal:*

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right).$$

Similarly, the distribution of the sample total T_0 will also be approximately Normal:

$$T_0 \approx N(n\mu, n\sigma^2).$$

Example 5.1.4. Suppose the true national average weekly study time for college students is μ (unknown), and the true population standard deviation is $\sigma = 7$ hours. The shape of the population distribution of study times is unknown. We decide to take a large random sample of $n = 100$ students. What is the approximate probability that the sample mean study time ($\bar{X}$) we calculate from these 100 students will be within 1 hour of the true, unknown national average μ ? That is, what is $P(\mu - 1 \leq \bar{X} \leq \mu + 1)$?

Solution. By the CLT, $\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$, so

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{7}{10} = 0.7.$$

Thus

$$P(\mu - 1 \leq \bar{X} \leq \mu + 1) = P\left(|Z| \leq \frac{1}{0.7}\right) = P(|Z| \leq 1.4286).$$

Therefore,

$$P(|Z| \leq 1.4286) = 2\Phi(1.4286) - 1 \approx 0.847.$$

□

5.1.5 The distribution of a Linear Combination

The sample mean $\bar{X}$ and sample total T_0 are two statistics that are widely used in application. To understand more about the distribution of $\bar{X}$ and T_0 , we will learn the distribution of a linear combination of rv's.

Given a collection of n random variables $X_1, X_2, \dots, X_n$ and n numerical constants $a_1, a_2, \dots, a_n$, the rv

$$Y = a_1X_1 + \dots + a_nX_n$$

is called a linear combination of the X_i 's.

Suppose $X_1, X_2, \dots, X_n$ have the expected values $\mu_1, \mu_2, \dots, \mu_n$ respectively, and variances $\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2$, respectively.

Proposition 5.1.5. *Whether or not the X_i 's are independent,*

$$E(a_1X_1 + \dots + a_nX_n) = a_1E(X_1) + \dots + a_nE(X_n) = a_1\mu_1 + \dots + a_n\mu_n.$$

Proof. Let $Y = \sum_{i=1}^n a_iX_i$. By linearity of expectation,

$$E(Y) = E\left(\sum_{i=1}^n a_iX_i\right) = \sum_{i=1}^n E(a_iX_i) = \sum_{i=1}^n a_iE(X_i) = \sum_{i=1}^n a_i\mu_i.$$

Independence is not required for this result. □

Proposition 5.1.6. *If the X_i 's are independent (or just uncorrelated), then*

$$\text{Var}(a_1X_1 + \dots + a_nX_n) = a_1^2 \text{Var}(X_1) + \dots + a_n^2 \text{Var}(X_n) = a_1^2\sigma_1^2 + \dots + a_n^2\sigma_n^2.$$

Proof. Let $Y = \sum_{i=1}^n a_iX_i$. Then

$$\text{Var}(Y) = \text{Var}\left(\sum_{i=1}^n a_iX_i\right) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{1 \leq i < j \leq n} a_i a_j \text{Cov}(X_i, X_j).$$

If the X_i 's are independent (or uncorrelated), then $\text{Cov}(X_i, X_j) = 0$ for $i \neq j$, so

$$\text{Var}(Y) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) = \sum_{i=1}^n a_i^2 \sigma_i^2.$$

□

Proposition 5.1.7. *For any $X_1, X_2, \dots, X_n$,*

$$\text{Var}(a_1X_1 + \dots + a_nX_n) = \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j).$$

A special case: the Difference Between Two Random Variables. Let $Y = X_1 - X_2$. We then have the following result.

$$E(Y) = E(X_1) - E(X_2) \quad \text{for any two rv's } X_1 \text{ and } X_2.$$

$$\text{Var}(Y) = \text{Var}(X_1) + \text{Var}(X_2) \quad \text{if } X_1 \text{ and } X_2 \text{ are two independent rv's.}$$

5.2 Point Estimation

5.2.1 Point Estimation

What is Point Estimation? How can we use a sample to make our best guess about an unknown characteristic of a population?

Definition 5.2.1. A point estimate is a single numerical value used to estimate an unknown population parameter (denoted as θ).

If we select a suitable statistic (a function of the sample data) and compute its value from the observed sample, this statistic is called the point estimator (denoted as $\hat{\theta}$). Our main goal is to compute a number from the sample that is a sensible value for the parameter.

A natural way to measure the closeness of an estimate $\hat{\theta}$ to the true parameter θ is the squared difference:

$$(\hat{\theta} - \theta)^2.$$

Definition 5.2.2 (Mean Squared Error (MSE)). An overall measure of an estimator's accuracy is its expected squared error:

$$\text{MSE} = E[(\hat{\theta} - \theta)^2].$$

Example 5.2.1. Let $X_1, X_2, \dots, X_n$ be a random sample from a Bernoulli distribution with parameter p . We want to estimate the unknown parameter p .

Consider the following two estimators for p :

1. Estimator 1 ($\hat{p}_1$): The Sample Mean $\hat{p}_1 = \frac{1}{n} \sum_{i=1}^n X_i$.
2. Estimator 2 ($\hat{p}_2$): The Constant Estimator $\hat{p}_2 = \frac{1}{2}$.

What is $\text{MSE}(\hat{p}_1)$, $\text{MSE}(\hat{p}_2)$?

Solution. It follows that

$$\text{MSE}(\hat{p}_1) = \frac{p(1-p)}{n} \quad \text{and} \quad \text{MSE}(\hat{p}_2) = \left(\frac{1}{2} - p\right)^2.$$

If the true value of p is exactly 0.5:

$$\text{MSE}(\hat{p}_1) = \frac{1}{4n} \quad \text{and} \quad \text{MSE}(\hat{p}_2) = 0.$$

So, if we knew $p = 0.5$, $\hat{p}_2$ would be the preferred estimator. If the true value of p is far from 0.5 (e.g., $p = 0.1$):

$$\text{MSE}(\hat{p}_1) = \frac{0.09}{n} \quad \text{and} \quad \text{MSE}(\hat{p}_2) = 0.6.$$

So, the sample mean $\hat{p}_1$ is a much better estimator if $n = 10$. $\square$

Definition 5.2.3. A point estimator $\hat{\theta}$ is unbiased if its expected value is equal to the true value of the parameter θ , for every possible value of θ , i.e.,

$$E(\hat{\theta}) = \theta$$

If $\hat{\theta}$ is not unbiased, the difference $E(\hat{\theta}) - \theta$ is called the bias of $\hat{\theta}$. Note that

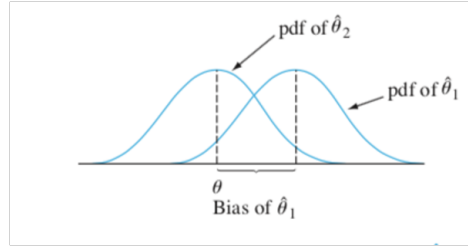


Figure 5.3: Illustration of the bias of an estimator.

$\hat{\theta}$ is unbiased if its sampling distribution is always “centered” at the true value of the parameter. Note that “centered” here means that the expected value, not the median, of the distribution of $\hat{\theta}$ is equal to θ .

Proposition 5.2.1. When X is a binomial rv with parameters n and p , the sample proportion $\hat{p} = X/n$ is an unbiased estimator of p .

Proposition 5.2.2. If $X_1, X_2, \dots, X_n$ is a random sample from a distribution with mean μ , then $\bar{X}$ is an unbiased estimator of μ .

Proof. By definition, $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$. By linearity of expectation,

$$E(\bar{X}) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \cdot n\mu = \mu.$$

Thus $\bar{X}$ is unbiased for μ . $\square$

Proposition 5.2.3. *If $X_1, X_2, \dots, X_n$ is a random sample from a distribution with mean μ and variance σ^2 , then the estimator*

$$\hat{\sigma}^2 = S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$$

is unbiased for estimating σ^2 .

Proof. Let $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$. Use the identity

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2.$$

Taking expectations and using $E[(X_i - \mu)^2] = \sigma^2$ and $E[(\bar{X} - \mu)^2] = \text{Var}(\bar{X}) = \sigma^2/n$, we get

$$E\left[\sum_{i=1}^n (X_i - \bar{X})^2\right] = n\sigma^2 - n \cdot \frac{\sigma^2}{n} = (n-1)\sigma^2.$$

Therefore,

$$E(S^2) = \frac{1}{n-1} E\left[\sum_{i=1}^n (X_i - \bar{X})^2\right] = \sigma^2.$$

So S^2 is an unbiased estimator of σ^2 . □

If I have a biased estimator, can we change it into an unbiased one? The example below shows how to do this.

Example 5.2.2. Suppose that X , the reaction time to a certain stimulus, has a uniform distribution $X \sim U[0, \theta]$. Since θ is the largest possible time in the entire population of reaction times, consider as a first estimator the largest sample reaction time:

$$\hat{\theta}_1 = \max(X_1, X_2, \dots, X_n).$$

Note that $\hat{\theta}_1$ is a biased estimator, since we always have $\hat{\theta}_1 \leq \theta$. Can we modify $\hat{\theta}_1$ to obtain an unbiased estimator of θ ?

Solution. Since X has a uniform distribution on the interval from 0 to θ ,

$$E(\hat{\theta}_1) = \int_0^\theta \frac{y n y^{n-1}}{\theta^n} dy = \frac{n}{n+1} \cdot \frac{y^{n+1}}{\theta^n} \Big|_0^\theta = \frac{n}{n+1} \cdot \theta < \theta.$$

The bias of $\hat{\theta}_1$ is given by $\frac{n\theta}{n+1} - \theta = -\frac{\theta}{n+1}$, which approaches 0 as n gets large.

Consider the estimator

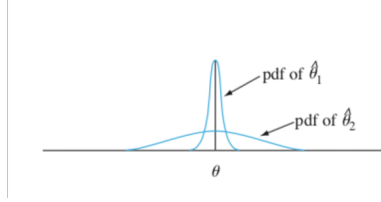
$$\hat{\theta}_2 = \frac{n+1}{n} \cdot \max(X_1, \dots, X_n).$$

The mean value of this estimator is

$$\begin{aligned} E(\hat{\theta}_2) &= E\left[\frac{n+1}{n} \cdot \max(X_1, \dots, X_n)\right] \\ &= \frac{n+1}{n} \cdot E[\max(X_1, \dots, X_n)] \\ &= \frac{n+1}{n} \cdot \frac{n}{n+1} \theta = \theta. \end{aligned}$$

□

Definition 5.2.4. The resulting $\hat{\theta}$ is called the minimum variance unbiased estimator (MVUE) of θ .



An estimator with smaller variance is more likely to produce an estimate close to the true value of θ .

Theorem 5.2.1. Let $X_1, X_2, \dots, X_n$ be a random sample from a normal distribution with parameters μ and σ . Then the estimator $\hat{\mu} = \bar{X}$ is the MVUE of μ .

Proof. Since $X_1, \dots, X_n$ are i.i.d. normal, the sample mean $\bar{X}$ is unbiased for μ . The statistic $\bar{X}$ is sufficient for μ (with σ known), and among unbiased estimators of μ it has the minimum variance σ^2/n . Therefore, $\bar{X}$ is the MVUE of μ . □

Example 5.2.3. Suppose that X , the reaction time to a certain stimulus, has a uniform distribution $X \sim U[0, \theta]$. The estimator

$$\hat{\theta}_1 = \frac{n+1}{n} \max(X_1, X_2, \dots, X_n)$$

is unbiased for θ . We can easily prove that the estimator

$$\hat{\theta}_2 = 2\bar{X}$$

is unbiased for θ .

Proof. Let $M = \max(X_1, \dots, X_n)$. Then $F_M(m) = (m/\theta)^n$ for $0 \leq m \leq \theta$, and

$$E[M] = \int_0^\theta m \cdot \frac{n}{\theta^n} m^{n-1} dm = \frac{n}{n+1} \theta.$$

Therefore,

$$E[\hat{\theta}_1] = \frac{n+1}{n} E[M] = \theta.$$

Also, $E[X_i] = \theta/2$, so $E[\bar{X}] = \theta/2$ and thus

$$E[\hat{\theta}_2] = 2E[\bar{X}] = \theta.$$

□

Definition 5.2.5. The standard error of an estimator $\hat{\theta}$ is its standard deviation $\sigma_{\hat{\theta}} = \sqrt{V(\hat{\theta})}$.

- It is the magnitude of a typical deviation between an estimate and the value of θ .
- If the standard error itself involves unknown parameters whose values can be estimated, substitution of these estimates into $\sigma_{\hat{\theta}}$ yields the estimated standard error of the estimator.

5.2.2 Methods of Point Estimation

In this section, we will discuss two “constructive” methods for obtaining point estimators: the method of moments and the method of maximum likelihood.

Let $X_1, X_2, \dots, X_n$ be a random sample from a pmf or pdf $f(x)$.

Definition 5.2.6. For $k = 1, 2, 3, \dots$, the k th population moment, or k th moment of the distribution $f(x)$ is $E(X^k)$. Then, the k th sample moment is

$$A_k = \frac{1}{n} \sum_{i=1}^n X_i^k.$$

Theorem 5.2.2 (Law of Large Numbers). *Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with pmf or pdf $f(x)$. For all $k \geq 1$, when $n \rightarrow \infty$,*

$$\frac{1}{n} \sum_{i=1}^n X_i^k \xrightarrow{P} E[X^k].$$

The Law of Large Numbers states that as the sample size (n) increases, the sample moments converge in probability to the true population moments.

Method of Moments

1. Express the first m population moments in terms of $\theta_1, \dots, \theta_m$:

$$E(X^k) = \mu_k(\theta_1, \theta_2, \dots, \theta_m), \quad k = 1, \dots, m.$$

2. Set sample moments equal to population moments:

$$A_k = E(X^k)$$

i.e.,

$$\frac{1}{n} \sum_{i=1}^n X_i^k = \mu_k(\theta_1, \theta_2, \dots, \theta_m), \quad k = 1, \dots, m.$$

3. Solve the system of m equations for $\theta_1, \dots, \theta_m$ to get the moment estimators.
4. Substitute sample data to get estimates.

Example 5.2.4. Let $X_1, X_2, \dots, X_n$ be a random sample from a generalized negative binomial distribution with parameters r and p . Now we aim to estimate the parameters r and p . ($E(X) = r(1-p)/p$ and $\text{Var}(X) = r(1-p)/p^2$)

Solution. Let $\bar{X}$ be the sample mean and S^2 be the sample variance. From the moment equations

$$\bar{X} = \frac{r(1-p)}{p}, \quad S^2 = \frac{r(1-p)}{p^2},$$

we get

$$p = \frac{\bar{X}}{S^2}.$$

Substitute into the mean equation:

$$r = \frac{\bar{X} p}{1-p} = \frac{\bar{X}^2}{S^2 - \bar{X}}.$$

Thus the moment estimators are

$$\hat{p} = \frac{\bar{X}}{S^2}, \quad \hat{r} = \frac{\bar{X}^2}{S^2 - \bar{X}}.$$

□

Maximum Likelihood Estimation

If $X_1, X_2, \dots, X_n$ have joint pmf or pdf $f(x_1, \dots, x_n; \theta_1, \dots, \theta_m)$, when $x_1, \dots, x_n$ are the observed sample values, this joint pmf/pdf, viewed as a function of the parameters $\theta_1, \dots, \theta_m$, is called the likelihood function $L(x_1, \dots, x_n; \theta_1, \dots, \theta_m)$.

Since $\ln(\cdot)$ is a monotonic function, maximizing $L(\theta)$ is equivalent to maximizing $\ln(L(\theta))$.

Steps (Calculus-based):

1. Write down the likelihood function $L(\theta_1, \dots, \theta_m)$.
2. Take the natural logarithm: $\ln(L(\theta_1, \dots, \theta_m))$.
3. Take partial derivatives with respect to each parameter θ_i and set them to zero:

$$\frac{\partial}{\partial \theta_i} \ln(L(\cdot)) = 0.$$

4. Solve the resulting system of equations for $\theta_1, \dots, \theta_m$.
5. Replace observed values x_i with random variables X_i to get the estimators.

Example 5.2.5. Suppose $X_1, X_2, \dots, X_n$ is a random sample from an exponential distribution with parameter λ . Here, we want to estimate the parameter λ .

Solution. The pdf is $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. The likelihood function is

$$L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n \exp\left(-\lambda \sum_{i=1}^n x_i\right).$$

The log-likelihood is

$$\ell(\lambda) = n \ln \lambda - \lambda \sum_{i=1}^n x_i.$$

Differentiate and set to zero:

$$\ell'(\lambda) = \frac{n}{\lambda} - \sum_{i=1}^n x_i = 0.$$

Thus the MLE is

$$\hat{\lambda}_{\text{MLE}} = \frac{n}{\sum_{i=1}^n x_i} = \frac{1}{\bar{X}}.$$

□

5.3 Interval Estimation

5.3.1 Confidence Interval

Instead of a single guess, let's calculate a range or interval that likely contains the true population parameter. This range is called an **interval estimate** or **confidence interval**.

We calculate a lower limit ($\hat{\theta}_L$) and an upper limit ($\hat{\theta}_U$) based on our sample data. We then state that we believe the true parameter θ lies within this interval:

$$\hat{\theta}_L \leq \theta \leq \hat{\theta}_U$$

Confidence Level ($1 - \alpha$): If we can find two statistics Θ_L and Θ_U such that

$$P(\Theta_L \leq \theta \leq \Theta_U) = 1 - \alpha$$

for some $0 < \alpha < 1$.

- The interval $(\hat{\theta}_L, \hat{\theta}_U)$, computed from selected sample, is called a $100(1 - \alpha)\%$ -confidence interval.
- The fraction $1 - \alpha$ is called the degree of confidence or confidence level.
- Common levels are 90%, 95%, or 99% (i.e., $\alpha = 0.1, 0.05, 0.01$).
- A 95% confidence level means that if we were to repeat our experiment many, many times, we expect 95% of the CIs to contain the true unknown parameter.

Precision (Width of the Interval): The precision of an interval estimate is represented by the width of the interval, i.e., $\hat{\theta}_U - \hat{\theta}_L$.

- A narrow interval means our estimate is more precise. A wide interval means we're less certain about the exact value.
- If you want higher confidence (say, 99% instead of 95%), you'll generally need a wider, less precise interval.

Estimate the Mean

Let's focus on one of the most common tasks: estimating the population average μ . Imagine we're in a perfect situation:

1. The population distribution is normal.
2. The value of the population standard deviation σ is known.

Let $X_1, X_2, \dots, X_n$ be a random sample, i.e., $X_i \sim N(\mu, \sigma^2)$ for all i and they are independent. Then for any n ,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right),$$

i.e., $\bar{X}$ is normally distributed with mean μ and variance σ^2/n .

Z-distribution: Let $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$. Then

$$Z \sim N(0, 1).$$

Proposition 5.3.1. *If after observing $X_1 = x_1, X_2 = x_2, \dots, X_n = x_n$, we compute the observed sample mean $\bar{x}$ and substitute $\bar{x}$ into*

$$\left(\bar{X} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right)$$

in place of $\bar{X}$, the resulting fixed interval

$$\left(\bar{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right)$$

is called a $100(1 - \alpha)\%$ confidence interval.

Proof. Since $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$, we have

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

Let $z_{\alpha/2}$ satisfy $P(Z \leq z_{\alpha/2}) = 1 - \alpha/2$. Then

$$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha,$$

which is equivalent to

$$P\left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha.$$

Therefore, after observing $\bar{x}$, the interval

$$\left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right)$$

is a $100(1 - \alpha)\%$ confidence interval for μ . □

95% confidence level: From tables, we know that 95% of the Z -values fall between -1.96 and $+1.96$.

Proposition 5.3.2. *A 95% confidence interval for μ for the mean μ of a normal population when the value of σ is known is given by*

$$\left(\bar{x} - 1.96 \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \cdot \frac{\sigma}{\sqrt{n}} \right)$$

or equivalently, by

$$\bar{x} \pm 1.96 \cdot \frac{\sigma}{\sqrt{n}}.$$

Example 5.3.1. We want to find the true average preferred keyboard height (μ) for typists. Here, we measure 31 typists ($n = 31$) and find their average preferred height is $\bar{x} = 80$ cm. We assume height is normally distributed and we know the population standard deviation is $\sigma = 2$ cm. Let's calculate a 95% CI for μ .

Solution. Since σ is known, the 95% CI is

$$\bar{x} \pm 1.96 \cdot \frac{\sigma}{\sqrt{n}}.$$

Compute the margin:

$$1.96 \cdot \frac{2}{\sqrt{31}} \approx 0.704.$$

Thus the 95% CI is

$$(80 - 0.704, 80 + 0.704) \approx (79.3, 80.7) \text{ cm.}$$

□

How Many Samples Do We Need?

Often, before we even start collecting data, we have an idea of how precise we need our estimate to be. We might want our CI to be no wider than a certain amount (w). How many samples (n) must we collect to achieve this desired width?

For a normal population with known σ , a $100(1 - \alpha)\%$ CI for μ has width

$$w = 2z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}.$$

Solving for n gives

$$n = \left(2z_{\alpha/2} \cdot \frac{\sigma}{w} \right)^2.$$

Here $z_{\alpha/2}$ is the standard normal critical value, σ is the population standard deviation, and w is the desired total width of the CI.

5.3.2 Large-sample confidence intervals for a population mean and proportion

In the last section, we're in a perfect situation:

1. The population distribution is normal.
2. The value of the population standard deviation σ is known.

Unfortunately, the ideal case is rare. What if our population isn't exactly normal, or (more commonly) we don't know σ ?

The Central Limit Theorem (CLT): The CLT is a cornerstone of statistics. It says that if our sample size (n) is large enough, the distribution of the sample mean ($\bar{X}$) will be approximately normal, regardless of the original population's shape.

A Large-Sample Interval for μ with known σ :

Let $X_1, X_2, \dots, X_n$ be a random sample from a (not necessarily normal) population having a mean μ and standard deviation σ .

Provided that n is large, the CLT implies that

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

approximately. Then

$$P\left(-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right) \approx 1 - \alpha.$$

Proposition 5.3.3. *A large-sample confidence interval for μ with the confidence level approximately $100(1 - \alpha)\%$ is*

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}.$$

A Large-Sample Interval for μ with σ unknown:

In practical, the population standard deviation σ is hardly to be known. If n is large, the sample standard deviation S such that

$$S^2 = \frac{\sum (X_i - \bar{X})^2}{n - 1}$$

becomes a very good estimate of σ . Now, we consider the standardized variable

$$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}}.$$

Proposition 5.3.4. *A large-sample confidence interval for μ with the confidence level approximately $100(1 - \alpha)\%$ is*

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{S}{\sqrt{n}}.$$

5.3.3 Intervals based on a normal population distribution

We know our population is normal, but we don't know σ , and our sample size (n) is small. With small n , S might not be a very good estimate for σ .

The Student's t-Distribution:

Developed by William Sealy Gosset (who worked at the Guinness Brewery), the Student's t -distribution was created to handle small-sample inference when the population standard deviation is unknown, giving more reliable uncertainty than the normal approximation in this setting.

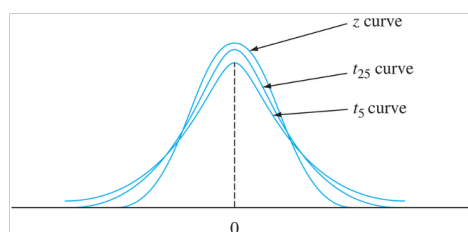
Theorem 5.3.1. *Let $X_1, \dots, X_n$ be a random sample from a normal distribution with both μ and σ unknown. Then the random variable*

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a probability distribution called a t -distribution with $n - 1$ degree of freedom (df), i.e., $T \sim t(n - 1)$.

Let t_ν denote the density function curve for ν df:

- Like the Z -distribution, it's bell-shaped and centered at 0.
- The t -distribution has “fatter” or heavier tails than the Z -distribution. This means it gives more probability to values far from the mean.



Degrees of Freedom (df): This ($\nu = n - 1$) determines the shape of the t -curve.

- The smaller the df (smaller n), the more spread out the curve.

- As df gets larger (larger n), the t -curve gets closer and closer to the Z -curve. In fact, the Z -curve is often called the t -curve with infinite df.

We look these up in a t -table, using our desired α and the correct df.

ν	α					
	.10	.05	.025	.01	.005	.0005
1	3.078	6.314	12.706	31.821	63.657	318.31
2	1.886	2.920	4.303	6.965	9.925	22.326
3	1.638	2.353	3.182	4.541	5.841	10.213
4	1.533	2.132	2.776	3.747	4.604	7.173
5	1.476	2.015	2.571	3.365	4.032	5.893
6	1.440	1.943	2.447	3.143	3.707	5.208
7	1.415	1.895	2.365	2.998	3.499	4.785
8	1.397	1.860	2.306	2.896	3.355	4.501
9	1.383	1.833	2.262	2.821	3.250	4.297
40	1.303	1.684	2.021	2.423	2.704	3.307
50	1.299	1.676	2.009	2.403	2.678	3.262
60	1.296	1.671	2.000	2.390	2.660	3.232
120	1.289	1.658	1.980	2.358	2.617	3.160
∞	1.282	1.645	1.960	2.326	2.576	3.090

Proposition 5.3.5. Let $X_1, \dots, X_n$ be a random sample from a normal population with mean μ . Denote by $\bar{x}$ and s the sample mean and sample standard deviation computed from the results of a random sample. Then a $100(1 - \alpha)\%$ confidence interval for μ is

$$\bar{x} - t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}}.$$

or, more compactly, $\bar{x} \pm t_{\alpha/2, n-1} \cdot s/\sqrt{n}$.

Summary

Estimate the mean μ		
Population	Normal	Not normal
σ known	for all n , $\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$	for large n , $\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$ approx.
100(1 - α)% CI: $\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$		with width control $n = \left(2z_{\alpha/2} \cdot \frac{\sigma}{w}\right)^2$
σ unknown	for all n , $\frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t(n-1)$	for large n , $\frac{\bar{x} - \mu}{s/\sqrt{n}} \sim N(0, 1)$ approx
100(1 - α)% CI: $\bar{x} \pm t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}}$		100(1 - α)% CI: $\bar{x} \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$ without width control

5.4 Hypotheses

5.4.1 Hypotheses and Test Procedures

In many practical problems, we are not just satisfied with providing an estimate or an interval for an unknown parameter. We are more concerned about whether a specific assertion about the parameter is true.

Question: When we have two contradictory claims about a parameter of a population, how do we use sample data to determine which claim is more credible?

Tests of Hypotheses: Tests a specific claim about a parameter to decide between two contradictory statements. For example, determining if a new drug is more effective than an existing one.

Hypotheses: A claim about the value of a single parameter, the values of several parameters, or the form of an entire probability distribution.

Null Hypotheses (H_0): The claim that is initially assumed to be true (the “prior belief” claim). It is often the statement that a researcher wants to disprove.

- Frequently stated as an equality claim (e.g., $\mu = \mu_0$).
- Represents the “no change,” or “no effect,” viewpoint.

Alternative Hypotheses (H_a): A claim that is contradictory to H_0 . Often the claim that the researcher would really like to validate. It can take several forms:

- $\mu > \mu_0$ (right-tailed test)
- $\mu < \mu_0$ (left-tailed test)
- $\mu \neq \mu_0$ (two-tailed test)

5.4.2 Test Procedures

Test Procedures: A rule, based on sample data, for deciding whether to reject H_0 .

Test Statistic: A function of the sample data on which the decision (reject H_0 or do not reject H_0) is to be based.

Rejection Region : The set of all test statistic values for which H_0 will be rejected.

- If the calculated test statistic falls outside this region, H_0 is not rejected.

Decision Rule: The null hypothesis will be rejected if and only if the observed or computed test statistic value falls in the rejection region.

Example 5.4.1. A manufacturer’s past production of circuit boards had a 10% defect rate. An engineer suggests a change in the process and believes it will reduce the defect rate.

Let p be the true proportion of defective boards from the changed process. We test

$$H_0 : p = 0.1 \quad \text{versus} \quad H_a : p < 0.1.$$

Suppose we take a random sample of $n = 200$ boards and let X be the number of defectives observed; the test statistic is X .

If H_0 is true, we expect X to be around $np = 20$, so a value of X that is sufficiently smaller than 20 provides evidence against H_0 . For instance, we could reject H_0 when $X \leq k$ for some small integer (e.g., $k = 12$); otherwise, we do not reject H_0 .

5.4.3 Risks in Decision Making

Since decisions are based on sample information, we always risk making mistakes. What types of errors are these?

Type I Error (Rejecting a true H_0 , α error): Occurs if we reject H_0 when H_0 is actually true. (rejecting a true null)

- Its probability is denoted by $\alpha = P(\text{reject } H_0 \mid H_0 \text{ true})$.
- α is also known as the significance level.

Type II Error (Failing to reject a false H_0 , β error): Occurs if we do not reject H_0 when H_0 is actually false. (accepting a false null)

- Its probability is denoted by $\beta = P(\text{accept } H_0 \mid H_0 \text{ false})$.

For a fixed sample size n and a chosen test statistic, decreasing the size of the rejection region to obtain a smaller value of α results in a larger value of β . Conversely, enlarging the rejection region increases α but decreases β . Both α and β are small. This usually requires a larger sample size.

1. Specify the probability of Type I error, α (significance level).

- Common values for α are 0.05, 0.01, 0.10.
- The choice of α depends on the seriousness of making a Type I error.

2. Under this α level, choose a rejection region that makes β as small as possible.

Example 5.4.2. Under specified test conditions, the drying time of a certain paint is known to be normally distributed with mean $\mu = 75$ min and standard deviation $\sigma = 9$ min. Chemists propose a new additive intended to reduce the average drying time, and it is believed that drying times with this additive remain normal with $\sigma = 9$. We take a random sample of size $n = 25$ for testing.

Parameter: μ = the true mean drying time with the additive.

Hypotheses: $H_0 : \mu = 75, \quad H_a : \mu < 75$

Test Statistic: $\bar{X}$, the sample mean.

Rejection Region: assume we reject H_0 when $\bar{x} \leq 70.8$.

Let $X_1, X_2, \dots, X_{25}$ denote the 25 drying times. Then

$$\bar{X} \sim N\left(\mu, \frac{81}{25}\right),$$

so the Type I error probability is

$$\begin{aligned}\alpha &= P(\text{reject } H_0 \mid H_0 \text{ true}) = P(\bar{X} \leq 70.8 \mid \mu = 75) \\ &= \Phi\left(\frac{70.8 - 75}{1.8}\right) \approx 0.01.\end{aligned}$$

If the true mean is $\mu = 72$, then the Type II error probability is

$$\begin{aligned}\beta &= P(\text{fail to reject } H_0 \mid \mu = 72) = P(\bar{X} > 70.8 \mid \mu = 72) \\ &= 1 - \Phi\left(\frac{70.8 - 72}{1.8}\right) \approx 0.7486.\end{aligned}$$

Remark. If we want a significance level $\alpha = 0.05$, then the rejection region should be chosen so that

$$P(\bar{X} \leq c \mid \mu = 75) = 0.05.$$

Since $\bar{X} \sim N(75, 81/25)$ under H_0 , we obtain

$$c = 75 + z_{0.05} \cdot \frac{9}{\sqrt{25}} = 75 + (-1.645) \cdot 1.8 \approx 72.0,$$

so a suitable rejection region is $\bar{x} \leq 72.0$.

5.4.4 Tests About a Population Mean

We now consider three common cases for tests about a population mean:

1. **Normal population, σ known.** This is the most idealized case. The Z -test is used.
2. **Large-sample tests (any population distribution, σ known or unknown).** When n is sufficiently large (e.g., $n > 30$ or $n > 40$), the CLT implies that a Z -test can be used as an approximation. If σ is unknown, it is replaced by the sample standard deviation S .
3. **Normal population, σ unknown.** When n may be small and σ is unknown, we assume the population is normal and use the t -test.

Case 1: Normal population, σ known

The rejection region is determined by the form of the alternative hypothesis H_a at significance level α . The cutoff value is chosen so that the probability of a Type I error is controlled at the desired level α .

Upper-tailed test ($H_a : \mu > \mu_0$). Evidence in favor of H_a arises only when the sample mean $\bar{x}$ is sufficiently larger than μ_0 (equivalently, when the test statistic z is sufficiently positive). The rejection region is

$$z \geq z_\alpha.$$

To justify this cutoff, note that under H_0 the test statistic satisfies $Z \sim N(0, 1)$, so the Type I error probability is

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ true}) = P(Z \geq c) = 1 - \Phi(c).$$

Setting $1 - \Phi(c) = \alpha$ gives $P(Z \geq c) = \alpha$, so $c = z_\alpha$. Hence the rejection region $z \geq z_\alpha$ has Type I error probability α . This is called an *upper-tailed* test because the rejection region consists only of large values of the test statistic.

Lower-tailed test ($H_a : \mu < \mu_0$). Evidence in favor of H_a arises only when the sample mean $\bar{x}$ is sufficiently smaller than μ_0 (equivalently, when the test statistic z is sufficiently negative). The rejection region is

$$z \leq -z_\alpha.$$

Two-tailed test ($H_a : \mu \neq \mu_0$). Evidence in favor of H_a arises when the discrepancy between $\bar{x}$ and μ_0 is sufficiently large (equivalently, when $|z|$ is sufficiently large). The rejection region is

$$z \geq z_{\alpha/2} \quad \text{or} \quad z \leq -z_{\alpha/2}.$$

If the null hypothesis H_0 is false, the Type II error probability β depends on the true mean μ (denoted μ') and on the form of the alternative hypothesis H_a . For the case $H_a : \mu > \mu_0$, the Type II error is the probability of failing to reject H_0 when the true mean is μ' , that is,

$$\beta(\mu') = P\left(\bar{X} < \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}} \mid \mu'\right) = \Phi\left(z_\alpha + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right).$$

For a level- α test with known σ ,

$$\beta(\mu') = \begin{cases} \Phi\left(z_\alpha + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right), & H_a : \mu > \mu_0, \\ 1 - \Phi\left(-z_\alpha + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right), & H_a : \mu < \mu_0, \\ \Phi\left(z_{\alpha/2} + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right) - \Phi\left(-z_{\alpha/2} + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right), & H_a : \mu \neq \mu_0, \end{cases}$$

where $\Phi(z)$ denotes the standard normal cdf.

To achieve prescribed α and β for a specific alternative mean μ' , the sample size can be obtained by a simple algebraic argument. For a one-tailed test, use the first line of $\beta(\mu')$ and set it equal to the desired β :

$$\beta = \Phi\left(z_\alpha + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right).$$

Then

$$z_\beta = z_\alpha + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}, \quad \text{and} \quad n = \left[\frac{\sigma(z_\alpha + z_\beta)}{\mu_0 - \mu'} \right]^2,$$

which is the stated sample-size formula for a one-tailed test.

For a two-tailed test, use the two-tailed expression for $\beta(\mu')$. Solving exactly for n is cumbersome, and a standard approximation replaces z_α by $z_{\alpha/2}$, yielding

$$n \approx \left[\frac{\sigma(z_{\alpha/2} + z_\beta)}{\mu_0 - \mu'} \right]^2.$$

Example 5.4.3. Let μ denote the true average tread life (in miles) of a certain type of tire. Consider testing $H_0 : \mu = 30000$ versus $H_a : \mu > 30000$ based on a sample of size $n = 16$ from a normal population with $\sigma = 1500$. A level $\alpha = 0.01$ test has $z_\alpha = z_{0.01} = 2.33$. The Type II error probability when $\mu = 31000$ is

$$\beta(31000) = \Phi\left(2.33 + \frac{30000 - 31000}{1500/\sqrt{16}}\right) = \Phi(-0.34) \approx 0.3669.$$

Since $z_{0.10} = 1.28$, requiring the level-0.01 test to also have $\beta(31000) = 0.10$ yields

$$n = \left[\frac{1500(2.33 + 1.28)}{30000 - 31000} \right]^2 \approx 29.32.$$

Thus the sample size should be rounded up to $n = 30$ tires.

Case 2: Large-Sample Tests (approximate Z-test)

When the sample size is large, we can test the mean without assuming a normal population. Let $X_1, X_2, \dots, X_n$ be a random sample from a population with mean μ and standard deviation σ , and consider

$$H_0 : \mu = \mu_0.$$

By the CLT, if n is large then the test statistic

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

is approximately standard normal when H_0 is true.

When σ is unknown, we use the sample standard deviation

$$S = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$$

and the approximate Z -statistic

$$Z = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}.$$

If H_0 is true and n is large, this Z is approximately $N(0, 1)$.

Rejection region. The rejection regions are the same as in the Z -test with known σ , using z_α or $z_{\alpha/2}$ as critical values:

- $H_a : \mu > \mu_0$ (upper-tailed): reject H_0 if $z \geq z_\alpha$.
- $H_a : \mu < \mu_0$ (lower-tailed): reject H_0 if $z \leq -z_\alpha$.
- $H_a : \mu \neq \mu_0$ (two-tailed): reject H_0 if $z \geq z_{\alpha/2}$ or $z \leq -z_{\alpha/2}$.

Example 5.4.4. A dynamic cone penetrometer (DCP) is used to measure material resistance to penetration. Suppose that for a particular application it is required that the true average DCP value be less than 30. Using the following $n = 52$ observations, carry out a large-sample test at significance level $\alpha = 0.05$:

14.1	14.5	15.5	16.0	16.0	16.7	16.9	17.1	17.5	17.8
17.8	18.1	18.2	18.3	18.3	19.0	19.2	19.4	20.0	20.0
20.8	20.8	21.0	21.5	23.5	27.5	27.5	28.0	28.3	30.0
30.0	31.6	31.7	31.7	32.5	33.5	33.9	35.0	35.0	35.0
36.7	40.0	40.0	41.3	41.7	47.5	50.0	51.0	51.8	54.4
55.0	57.0								

Parameter: μ = average DCP value.

Hypotheses: $H_0 : \mu = 30$, $H_a : \mu < 30$.

Test statistic:

$$z = \frac{\bar{x} - 30}{s/\sqrt{n}}.$$

For a level-0.05 lower-tailed test, reject H_0 when $z \leq -z_{0.05} = -1.645$.

With $n = 52$, $\bar{x} = 28.76$, and $s = 12.2647$,

$$z = \frac{28.76 - 30}{12.2647/\sqrt{52}} = -0.73.$$

Since $-0.73 > -1.645$, the test statistic does not fall in the rejection region, so H_0 is not rejected at the 0.05 level.

Case 3: Normal Population, σ Unknown (t-test)

When the population is normally distributed, the sample size is small, and the population standard deviation σ is unknown, we test the mean using the t -procedure. Let $X_1, X_2, \dots, X_n$ be a random sample from a normal population and consider

$$H_0 : \mu = \mu_0.$$

The test statistic is

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}},$$

where

$$S = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}.$$

When H_0 is true, T follows a t -distribution with $n-1$ degrees of freedom.

Rejection region. For a level- α test, use $t_{\alpha, n-1}$ or $t_{\alpha/2, n-1}$ as critical values:

- $H_a : \mu > \mu_0$ (upper-tailed): reject H_0 if $t \geq t_{\alpha, n-1}$.
- $H_a : \mu < \mu_0$ (lower-tailed): reject H_0 if $t \leq -t_{\alpha, n-1}$.
- $H_a : \mu \neq \mu_0$ (two-tailed): reject H_0 if $t \geq t_{\alpha/2, n-1}$ or $t \leq -t_{\alpha/2, n-1}$.

Example 5.4.5. An analyst measures the mean thickness (in mm) of a coating that is assumed to be normally distributed. The target is 12 mm. Using the following $n = 10$ measurements, test $H_0 : \mu = 12$ versus $H_a : \mu < 12$ at $\alpha = 0.05$:

11.6, 11.7, 11.8, 11.9, 12.0, 12.1, 12.2, 12.3, 12.4, 11.5.

Here $\bar{x} = 11.95$ and $s = 0.3028$, so

$$t = \frac{\bar{x} - 12}{s/\sqrt{n}} = \frac{11.95 - 12}{0.3028/\sqrt{10}} \approx -0.52.$$

With $n-1 = 9$ degrees of freedom, the 0.05 lower-tailed critical value is $-t_{0.05, 9} = -1.833$. Since $-0.52 > -1.833$, we do not reject H_0 at the 0.05 level.

Chapter 6

The modern theory of probability

6.1 Markov chains

6.1.1 Terminology and Basic Properties of Markov chains

Markov chains provide a model for sequential information that allows future outcomes to depend on previous ones, albeit in a very specific way (the defining *Markov property*). Researchers in numerous fields employ Markov chains to model the phenomena they study. Recent examples include predicting changes in electricity demand; modeling the motion of sperm whales off the Galapagos Islands; Chinese citizens changing their cellphone service; keeping track of inpatient bed usage at hospitals; monitoring patterns in Web browser histories to deploy better-targeted advertising; the evolution of drought conditions over time; and the dynamics of capital assets.

In this subsection, we introduce the basic vocabulary and notation of Markov chains. We begin with the following classic (if slightly artificial) example, which will serve as a thread throughout the chapter.

Example 6.1.1 (Taxi Zone Markov Chain). A city has three different taxi zones, numbered 1, 2, and 3. A taxi driver operates his cab in all three zones. The probability that his next passenger has a destination in a particular one of these zones depends on where the passenger is picked up. Specifically, whenever the driver is in zone 1, the probability his next passenger is going to zone 1 is .3, to zone 2 is .2, and to zone 3 is .5. Starting in zone 2, the probability his next passenger is going to zone 1 is .1, to zone 2 is .8, and to zone 3 is .1. Finally,

whenever he is in zone 3, the probability his next passenger is going to zone 1 is .4, to zone 2 is .4, and to zone 3 is .2. These probabilities are encapsulated in the state diagram in Fig. 6.1.

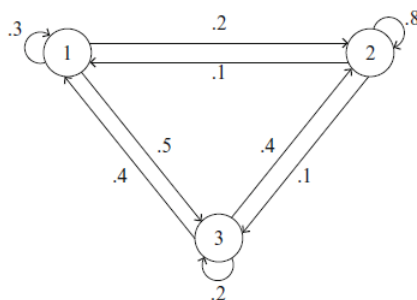


Figure 6.1: State diagram for Example 6.1.1

In every such state diagram, the sum of the probabilities on branches exiting any state must equal 1. For example, in Fig. 6.1 the probabilities exiting state 2 (i.e., zone 2) are .1, .8, and .1. We include in this calculation the probability .8 indicated by a “loop” in the state diagram, which simply means that the taxi driver has .8 probability of staying in zone 2 once he has dropped off a fare in zone 2.

Define X_0 to be the zone in which the taxi driver starts and X_n ($n \geq 1$) to be the zone where he drops off his n -th fare. Since $X_0, X_1, X_2, \dots$ “occur” in sequence, they are often referred to as a chain. More precisely, this particular sequence is a *finite-state, discrete-time, time-homogeneous Markov chain*. Each of these terms will be explained shortly.

In Example 6.1.1, each of the X_n for $n \geq 0$ assumes the value 1, 2, or 3 according to the destination zone. The zones collectively constitute the *states* of our chain, and so the *state space* is {zone 1, zone 2, zone 3}, although we will often drop the state names and just use the integers $\{1, 2, 3\}$. States can be identified with physical locations, levels (such as high/medium/low), dollar amounts, or just about anything else. We’ll sometimes refer to the X_n as random variables, even though they are not necessarily numerical (which goes against the definition from previous Chapter). The random variable X_0 is called the *initial state* of the chain. A *discrete-space chain* is one for which the number of possible states is finite or countably infinite. If there are finitely many possible states, we have a *finite-state chain*.

Since time was indexed by the discrete listing $n = 0, 1, 2, \dots$, the sequence of zones the taxi driver visited in Example 6.1.1 is called a *discrete-time chain*. The *continuous-time chains* is often indexed as $\{X_t : t \in [0, \infty)\}$, which are useful for modeling behavior continuously in time rather than just at discrete time points (e.g., tracking over time the number of people looking at a particular Web site). The taxi driver chain is also *time-homogeneous*, in that the specified probabilities do not change over time. One could imagine a different, more complicated model where the probabilities specified in Example 6.1.1 apply during morning hours but not in the evening, so that the probability of taking a fare from zone 1 to zone 3 is .5 for $n = 1$ (beginning of the work day) but is 1, say, for $n = 20$ (end of his shift).

Example 6.1.2 (Random Walk Markov Chain). A random walk. Imagine a marker initially placed at 0 on the number line. A fair coin is flipped repeatedly; each head moves the marker one integer to the right, while each tail moves it one integer to the left. Let $X_0 = 0$, the initial state, and X_n = the marker's position after n coin flips for $n \geq 1$. Each member of the chain can only take on a finite set of values: X_1 is either +1 or -1, X_2 is one of -2, 0, or 2, and so on. However, the collection of all possible states across all time indices comprises the entire set of integers: $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$. Thus, this so-called "random walk" is an infinite-state (though still discrete-state) chain; it is partially illustrated in Fig. 6.2.

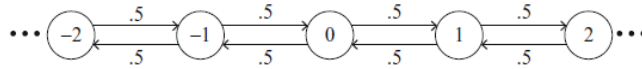


Figure 6.2: State diagram for Example 6.1.2

The Markov Property

All of the preceding examples have an important feature known as the *Markov property*. Loosely speaking, it says that in order to know where the chain will go next (say, X_{n+1}), it suffices to know where the chain is now (the value of X_n). In particular, once the current state is specified, the path that brought the chain to that state is irrelevant. Consider, for example, the random walk of Example 6.1.2: if for any particular n we have $X_n = 4$, then we know $X_{n+1} = 3$ or 5 with probability .5 each. It does not matter whether the chain arrived at 4 quickly ($0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$) or by a more circuitous route; the probability distribution of the next state in the chain is the same. This notion is formalized in the following definition.

Definition 6.1.1. Let $X_0, X_1, X_2, \dots$ be a sequence of random variables (a chain) on some discrete state space. The sequence is said to have the *Markov property* if, for any time index n and any set of (not necessarily distinct) states $s_0, s_1, \dots, s_n, s_{n+1}$,

$$P(X_{n+1} = s_{n+1} \mid X_0 = s_0, X_1 = s_1, \dots, X_n = s_n) = P(X_{n+1} = s_{n+1} \mid X_n = s_n) \quad (1)$$

Such a sequence $\{X_n; n = 0, 1, 2, \dots\}$ is called a *Markov chain*.

The conditional probabilities specified in Eq. (1) are called the *one-step transition probabilities* of the chain, or sometimes just *transition probabilities*. These are precisely the probabilities specified in Examples 6.1.1-6.1.2. It's critical to recognize that these are *conditional* probabilities: they specify the likelihood of the next member of the chain X_{n+1} being in any particular state, given the current state of the chain X_n .

Example 6.1.3 (Example 6.1.1 continued). The sequence of successive zones visited by our taxi driver is characterized by nine one-step transition probabilities. For example, it is stated that the driver “transitions” from zone 1 to zone 3 with probability .5, which means that for any time index n ,

$$P(X_{n+1} = 3 \mid X_n = 1) = .5$$

This probability does not depend on the value of n , because the chain is time-homogeneous. Instead of writing $P(X_{n+1} = 3 \mid X_n = 1) = .5$, we will sometimes abbreviate with $P(1 \rightarrow 3) = .5$ to emphasize the idea of transitioning from one state to another. Thus, the complete set of one-step transition probabilities for the taxi driver is

$$\begin{aligned} P(1 \rightarrow 1) &= .3 & P(1 \rightarrow 2) &= .2 & P(1 \rightarrow 3) &= .5 \\ P(2 \rightarrow 1) &= .1 & P(2 \rightarrow 2) &= .8 & P(2 \rightarrow 3) &= .1 \\ P(3 \rightarrow 1) &= .4 & P(3 \rightarrow 2) &= .4 & P(3 \rightarrow 3) &= .2 \end{aligned}$$

6.1.2 The Transition Matrix and the C-K Equations

Subsection 6.1.1 introduced the notion of a Markov chain and its characteristic one-step transition probabilities. In this subsection, we will develop a systematic way to determine the probability that a chain moves from one state to another in two steps (or three or four ...) by considering all the intermediate paths the chain may have taken. Such calculations are facilitated by aggregating the transition probabilities into a matrix.

The Transition Matrix

The one-step transition probabilities for the taxi driver example were displayed in Example 6.1.3 as a 3×3 array. It would be more efficient to simply specify the probabilities themselves in that same format, with the understanding that the probability in the i -th row and j -th column indicates the transition probability $P(i \rightarrow j)$, the chance the taxi driver takes his next fare to zone j given that he picks up the fare in zone i . Such a representation will be critical to understanding how various multi-step transition probabilities are calculated.

Definition 6.1.2. Let $(X_0, X_1, X_2, \dots)$ be a finite-state, time-homogeneous Markov chain, and index the states of the chain by the positive integers $1, 2, \dots, s$. The *(one-step) transition matrix* of the Markov chain is the $s \times s$ matrix $\mathbf{P}$ whose (i, j) -th entry is given by

$$p_{ij} = P(i \rightarrow j) = P(X_{n+1} = j \mid X_n = i)$$

for $i = 1, \dots, s$ and $j = 1, \dots, s$.

Example 6.1.4 (Example 6.1.3 continued). The one-step transition matrix for our taxi driver example is

$$\mathbf{P} = \begin{bmatrix} .3 & .2 & .5 \\ .1 & .8 & .1 \\ .4 & .4 & .2 \end{bmatrix}$$

which is identical in format to the display in Example 6.1.3. The entries are interpreted as the preceding definition suggests, e.g., the upper left entry (first row, first column) of the matrix is

$$p_{11} = P(1 \rightarrow 1) = P(X_{n+1} = 1 \mid X_n = 1) = .3,$$

i.e., the conditional probability that his next fare is dropped off somewhere in zone 1 given that the taxi is currently in zone 1.

Computation of Multistep Transition Probabilities

We now turn to the determination of multistep transition probabilities. Given that a Markov chain is currently in state i , what is the probability it will be in state j two steps later (i.e., after two transitions)? Three steps later? We begin with the following definition.

Definition 6.1.3. Let $X_0, X_1, X_2, \dots$ be a time-homogeneous Markov chain. For any positive integer k , the *k -step transition probabilities* are defined by

$$p^{(k)}(i \rightarrow j) = P(X_{n+k} = j \mid X_n = i) \quad (2)$$

where i and j range across the states of the chain (typically $1, \dots, s$). For $k = 1$, we will typically revert to the previous notation: $P^{(1)}(i \rightarrow j) = P(i \rightarrow j)$.

The superscript (k) in Expression (2) does not indicate taking the k -th power; it is simply notation representing “in k steps.” The next example illustrates how these k -step transition probabilities can be calculated, and how they can be represented compactly in terms of powers of the one-step transition matrix.

Example 6.1.5 (Example 6.1.4 continued). Suppose our taxi driver just dropped off a fare in zone 3, so that that is his current state. What is the probability that his *second* fare, counting from now, takes him to zone 1? That is, we wish to determine $P(X_{n+2} = 1 \mid X_n = 3) = P^{(2)}(3 \rightarrow 1)$. The calculation method is suggested by Fig. 6.3. Consider all the possible destinations of the $n + 1$ -st fare, i.e., all the intermediate steps the taxi driver could take from zone 3 to zone 1, and then employ the Law of Total Probability (applied here to conditional probabilities).

The partitioning events in the Law of Total Probability are the possible states at time $n + 1$:

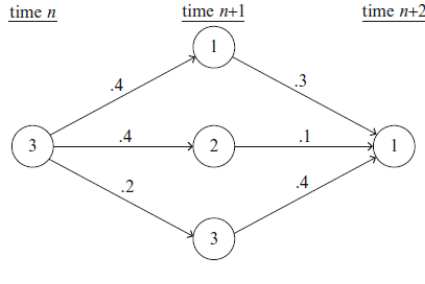


Figure 6.3: Transitioning from state 3 to state 1 in two time steps

$$\begin{aligned}
 P^{(2)}(3 \rightarrow 1) &= P(X_{n+2} = 1 \mid X_n = 3) \\
 &= P(X_{n+1} = 1 \mid X_n = 3)P(X_{n+2} = 1 \mid X_n = 3, X_{n+1} = 1) \\
 &\quad + P(X_{n+1} = 2 \mid X_n = 3)P(X_{n+2} = 1 \mid X_n = 3, X_{n+1} = 2) \\
 &\quad + P(X_{n+1} = 3 \mid X_n = 3)P(X_{n+2} = 1 \mid X_n = 3, X_{n+1} = 3)
 \end{aligned}$$

By the Markov property, $P(X_{n+2} = 1 \mid X_n = 3, X_{n+1} = 1) = P(X_{n+2} = 1 \mid X_{n+1} = 1)$, and the other two probabilities involving conditioning on X_n and

X_{n+1} simplify analogously. Thus,

$$\begin{aligned}
 P^{(2)}(3 \rightarrow 1) &= P(X_{n+1} = 1 \mid X_n = 3)P(X_{n+2} = 1 \mid X_{n+1} = 1) \\
 &\quad + P(X_{n+1} = 2 \mid X_n = 3)P(X_{n+2} = 1 \mid X_{n+1} = 2) \\
 &\quad + P(X_{n+1} = 3 \mid X_n = 3)P(X_{n+2} = 1 \mid X_{n+1} = 3) \\
 &= P(3 \rightarrow 1)P(1 \rightarrow 1) + P(3 \rightarrow 2)P(2 \rightarrow 1) + P(3 \rightarrow 3)P(3 \rightarrow 1) \\
 &= (.4)(.3) + (.4)(.1) + (.2)(.4) = .24
 \end{aligned}$$

For later reference, the last expression could be written in terms of the elements of the transition matrix $\mathbf{P}$; specifically, it's $p_{31}p_{11} + p_{32}p_{21} + p_{33}p_{31}$.

Similarly, the conditional probability that his second fare wants to be dropped off in zone 2 is computed by

$$\begin{aligned}
 P^{(2)}(3 \rightarrow 2) &= P(3 \rightarrow 1)P(1 \rightarrow 2) + P(3 \rightarrow 2)P(2 \rightarrow 2) + P(3 \rightarrow 3)P(3 \rightarrow 2) \\
 &= \sum_{j=1}^3 P(3 \rightarrow j)P(j \rightarrow 2) = \sum_{j=1}^3 p_{3j}p_{j2} \\
 &= (.4)(.2) + (.4)(.8) + (.2)(.4) = .48
 \end{aligned}$$

Finally, the probability the taxi driver finds himself back in zone 3 after two fares is

$$P^{(2)}(3 \rightarrow 3) = \sum_{j=1}^3 p_{3j}p_{j3} = (.4)(.5) + (.4)(.1) + (.2)(.2) = .28$$

This makes sense, since the taxi driver must arrive in one of the three zones at time $n + 2$, and $1 - (.24 + .48) = .28$.

The sums of products of matrix entries that appear repeatedly in the preceding example should look familiar: they are the same manner of computation that arises when one matrix is multiplied by another (or, here, a matrix is multiplied by itself). Indeed, consider what happens if we multiply the one-step transition matrix $\mathbf{P}$ from Example 6.1.4 by itself:

$$\mathbf{P}^2 = \mathbf{P}\mathbf{P} = \begin{bmatrix} .3 & .2 & .5 \\ .1 & .8 & .1 \\ .4 & .4 & .2 \end{bmatrix} \begin{bmatrix} .3 & .2 & .5 \\ .1 & .8 & .1 \\ .4 & .4 & .2 \end{bmatrix} = \begin{bmatrix} .31 & .42 & .27 \\ .15 & .70 & .15 \\ .24 & .48 & .28 \end{bmatrix}$$

The entries in the bottom row—.24, .48, .28—are precisely the two-step transition probabilities computed in Example 6.1.5. Specifically, the (3, 1) entry of $\mathbf{P}^2$ is $P^{(2)}(3 \rightarrow 1) = .24$, the (3, 2) entry of $\mathbf{P}^2$ is $P^{(2)}(3 \rightarrow 2) = .48$, and the (3, 3) entry of $\mathbf{P}^2$ is $P^{(2)}(3 \rightarrow 3) = .28$. It should come as no surprise that the

other six entries of $\mathbf{P}^2$ follow the same pattern: the (i, j) entry of $\mathbf{P}^2$ is equal to $P^{(2)}(i \rightarrow j)$. Hence, we can obtain all nine two-step transition probabilities with a single matrix computation (which, of course, can be facilitated by Matlab or other matrix-capable software).

The foregoing result can be generalized to an arbitrary fixed number of steps: to find the three-step transition probabilities, for example, one only needs to compute the matrix $\mathbf{P}^3$. It is not necessary to consider explicitly the many different paths by which the Markov chain could transition from state i to state j in three steps and add up all the corresponding probabilities (this is, secretly, what the threefold matrix multiplication does). The most general result is often referred to as the set of *Chapman-Kolmogorov Equations*.

If a Markov chain has one-step transition matrix $\mathbf{P}$, then the k -step transition probabilities are the entries of the matrix $\mathbf{P}^k$. Specifically,

$$P^{(k)}(i \rightarrow j) = \text{the } (i, j) \text{ entry of } \mathbf{P}^k$$

6.1.3 Specifying an Initial Distribution

Thus far, every probability we have considered in this section (i.e., all the one-, two-, and higher-step transition probabilities) has been conditional. For example the entries of any one-step transition matrix indicate $P(X_{n+1} = j \mid X_n = i)$. In this subsection, we briefly explore unconditional probabilities, which result from specifying a distribution for the rv X_0 , the initial state of the chain. We will consider two cases: modeling the initial state X_0 as a random variable, and treating X_0 as having a fixed/known value.

Example 6.1.6 (Example 6.1.5 continued). The never-ending saga of the taxi driver continues! Imagine this poor fellow sleeps in his taxi, so from his perspective each new day starts in a “random” zone. Specifically, suppose for now that he has a 20% chance of waking up in zone 1, a 50% chance of waking up in zone 2, and a 30% chance of waking up in zone 3. That is, we have assigned the following *initial distribution* to the Markov chain:

i	1	2	3
$P(X_0 = i)$	.2	.5	.3

(3)

Notice that, unlike the conditional probabilities that comprise the transition matrix of the Markov chain, this initial distribution (3) specifies the *unconditional* (aka marginal) distribution for the rv X_0 . In what follows, we will sometimes refer to the bottom row (6.3) as the “initial probability vector” of X_0 .

Now consider the rv X_1 , the destination of the taxi driver's first fare. The probability his first fare wants to go somewhere in zone 3 can be determined via the Law of Total Probability:

$$\begin{aligned} P(X_1 = 3) &= P(X_0 = 1)P(X_1 = 3 \mid X_0 = 1) + P(X_0 = 2)P(X_1 = 3 \mid X_0 = 2) \\ &\quad + P(X_0 = 3)P(X_1 = 3 \mid X_0 = 3) \\ &= \sum_{i=1}^3 [P(X_0 = i)P(i \rightarrow 3)] = \sum_{i=1}^3 [P(X_0 = i)p_{i3}] \\ &= (.2)(.5) + (.5)(.1) + (.3)(.2) = .21 \end{aligned}$$

As indicated in the intermediate step, this unconditional probability can be computed by taking the product of the initial probability vector $[.2 \ .5 \ .3]$, regarded as a 1×3 matrix, with the third column of the one-step transition matrix $\mathbf{P}$. Similarly, the (unconditional) probability that his first fare wants to be dropped off in zone 2 is

$$\begin{aligned} P(X_1 = 2) &= \sum_{i=1}^3 [P(X_0 = i)P(i \rightarrow 2)] = \sum_{i=1}^3 [P(X_0 = i)p_{i2}] \\ &= (.2)(.2) + (.5)(.8) + (.3)(.4) = .56 \end{aligned}$$

The foregoing computation is the product of the initial probability vector with the second column of $\mathbf{P}$. Finally, the probability that the first fare is taken to zone 1 equals .23, which can be computed either as a similar product or by observing that $1 - (.21 + .56) = .23$. All together, the unconditional pmf of the rv X_1 is

$ i $	1	2	3
$ P(X_1 = i) $	.23	.56	.21

Clearly, the most efficient way to determine the distribution of X_1 is to compute all three products simultaneously through matrix multiplication. If we multiply the transition matrix $\mathbf{P}$ on the left by a 1×3 row vector containing the initial probabilities for X_0 , we obtain

$$[.2 \ .5 \ .3] \mathbf{P} = [.2 \ .5 \ .3] \begin{bmatrix} .3 & .2 & .5 \\ .1 & .8 & .1 \\ .4 & .4 & .2 \end{bmatrix} = [.23 \ .56 \ .21]$$

The method illustrated in the preceding example can be generalized to find the unconditional distribution of the state X_n in the chain after any number of transitions n , starting with a specified initial distribution for X_0 .

Definition 6.1.4. Let $X_0, X_1, \dots, X_n, \dots$ be a Markov chain with state space $\{1, \dots, s\}$ and one-step transition matrix $\mathbf{P}$. Let $\mathbf{v}_0 = [v_{01} \ \dots \ v_{0s}]$ be a $1 \times s$ vector specifying the initial distribution of the chain, i.e., $v_{0k} = P(X_0 = k)$ for $k = 1, \dots, s$. If $\mathbf{v}_1$ denotes the vector of marginal (i.e., unconditional) probabilities associated with X_1 , then

$$\mathbf{v}_1 = \mathbf{v}_0 \mathbf{P}$$

More generally, if $\mathbf{v}_n$ denotes the $1 \times s$ vector of marginal probabilities for X_n , then

$$\mathbf{v}_n = \mathbf{v}_0 \mathbf{P}^n$$

Proof. The formula $\mathbf{v}_1 = \mathbf{v}_0 \mathbf{P}$ can be established using the same computational approach displayed in Example 6.15. Now consider $\mathbf{v}_2$, the vector of unconditional probabilities for X_2 . By the same reasoning as in Example 6.1.6, we have

$$\mathbf{v}_2 = \mathbf{v}_1 \mathbf{P}$$

The substitution $\mathbf{v}_1 = \mathbf{v}_0 \mathbf{P}$ then yields $\mathbf{v}_2 = (\mathbf{v}_0 \mathbf{P}) \mathbf{P} = \mathbf{v}_0 \mathbf{P}^2$. Continuing by induction, we have for general n that $\mathbf{v}_n = \mathbf{v}_{n-1} \mathbf{P} = (\mathbf{v}_0 \mathbf{P}^{n-1}) \mathbf{P} = \mathbf{v}_0 \mathbf{P}^n$, as claimed. $\square$

A Fixed Initial State

The case in which the initial state X_0 is fixed or known rather than random can be handled by forming a “degenerate” initial probability distribution.

Example 6.1.7 (Example 6.1.6 continued). Suppose that our taxi driver lives in zone 3 and always goes home at night, which means that he starts each new day in zone 3. Starting with certainty in zone 3 means that $P(X_0 = 3) = 1$, while $P(X_0 = 1) = P(X_0 = 2) = 0$. Written as a pmf, the distribution of X_0 is

i	1	2	3
$P(X_0 = i)$	0	0	1

Equivalently, the probability vector for X_0 is $\mathbf{v}_0 = [0 \ 0 \ 1]$. From the original description of the Markov chain (Example 6.1.1), the initial state being zone 3 implies that $X_1 = 1$ with probability .4, $X_1 = 2$ with probability .4, and $X_1 = 3$ with probability .2. This same result can be obtained by applying the theorem of this section:

$$\mathbf{v}_1 = \mathbf{v}_0 \mathbf{P} = [0 \ 0 \ 1] \begin{bmatrix} .3 & .2 & .5 \\ .1 & .8 & .1 \\ .4 & .4 & .2 \end{bmatrix} = [.4 \ .4 \ .2]$$

Notice that left-multiplying $\mathbf{P}$ by the vector $[0 \ 0 \ 1]$ simply extracts the third row of $\mathbf{P}$. Similarly, the pmf of X_5 , the destination of the fifth passenger, is given by

$$\begin{aligned}\mathbf{v}_5 = \mathbf{v}_0 \mathbf{P}^5 &= [0 \ 0 \ 1] \begin{bmatrix} .3 & .2 & .5 \\ .1 & .8 & .1 \\ .4 & .4 & .2 \end{bmatrix}^5 = [0 \ 0 \ 1] \begin{bmatrix} .2115 & .5767 & .2118 \\ .1938 & .6125 & .1938 \\ .2073 & .5858 & .2070 \end{bmatrix} \\ &= [.2073 \ .5858 \ .2070]\end{aligned}$$

The matrix $\mathbf{P}^5$ was computed by Matlab. The row vector $\mathbf{v}_5$ is simply the third row of $\mathbf{P}^5$, because the chain begins in zone 3 with probability 1. So, starting the day at home in zone 3, the taxi driver finds himself in zone 1, 2, or 3 after five fares with probabilities .2073, .5858, and .2070, respectively.

6.1.4 Simulation of Markov chains

A typical Markov chain simulation requires two elements: the one-step transition matrix, $\mathbf{P}$, and an indication of the initial state X_0 (either as a fixed state value or as a rv with a probability distribution). The actual simulation of a single realization of the chain $X_0, X_1, X_2, \dots$ then amounts to repeated selections from the transitional probability distributions specified by elements of $\mathbf{P}$. Simulation of Markov chains allow us to confirm theoretical results and, more importantly, determine properties of Markov chains that are not covered by the theorems of this section or other theoretical results.

The main step in any Markov chain simulation is to simulate a value for the next step, X_{n+1} , based on the transition probabilities coming out of the current step X_n . Let's start with the initial state X_0 . Suppose for one particular run of the simulation, X_0 has been assigned the state i , either because that's the fixed initial state or because a single draw from some initial distribution $\mathbf{v}_0$ yielded i . Conditional on $X_0 = i$, the distribution of X_1 is determined by the transition probabilities $P(i \rightarrow j)$ for $j = 1, 2, 3, \dots$, which appear in the i th row of $\mathbf{P}$. Thus, one needs to extract the i th row of $\mathbf{P}$ and use it as the basis for a single discrete simulation. If the result of this simulation is $X_1 = j$, then the j th row of $\mathbf{P}$ can be accessed to simulate X_2 , and so on.

Example 6.1.8 (Simulating a Taxi Driver's Day). Let's simulate a typical day in the life of our taxi driver. Although a real taxi driver does not take exactly 25 fares in 1 day, for purposes of this first simulation we'll assume that he takes the same 25 fares each day.

Suppose first that the driver begins each day in a random zone X_0 , as in Example 6.1.6, specifically with the initial distribution $p(1) = .2, p(2) = .5, p(3) = .3$. We begin by simulating a single value from this initial distribution. Once that

is determined, our program should then simulate a single value of X_1 using the row of $\mathbf{P}$ corresponding to the value of X_0 , then do the same for X_2 based on the simulated value of X_1 , and so on. Figure 6.4 shows Matlab and R code for such a simulation.

Matlab

```
P = [.3 .2 .5; .1 .8 .1; .4 .4 .2];
states = [1 2 3];
v0 = [.2 .5 .3];
X = randsample(states,1,true,v0);
current = X;
for i=1:25
    nextstate = randsample(states,1,true,P(current,:));
    X = [X nextstate];
    current = nextstate;
end
```

Figure 6.4: Matlab Code for Example 6.1.8

In Matlab, $\mathbf{P}(\text{current}, :)$ calls for the row of $\mathbf{P}$ specified by the numerical index `current`; the code $\mathbf{P}[\text{current},]$ performs the same task in R. The output of both of these programs is a vector, X , containing the sequence of states for the Markov chain (beginning with X_0). For example, one run of the above program in R yielded the following output:

```
> X
[1] 2 1 3 1 2 2 2 2 1 1 1 1 3 1 3 3 1 3 2 2 2 2 3 3 1 3
```

The randomly selected initial state was $X_0 = 2$, followed by $X_1 = 1, X_2 = 3, \dots$, and finally $X_{25} = 3$. (The symbol `[1]` at the left is not the initial state; this is just R's way of denoting the beginning of X .) If we weren't interested in the initial state of the chain, the code could easily be modified not to store X_0 , in which case the indices of the output vector would match the time indices of the Markov chain (i.e., the subscripts on $X_1, X_2, \dots, X_{25}$).

To make X_0 a fixed initial state instead of a true random variable, one need simply replace the two lines of code specifying the initial state probability vector with the first random selection. In the Matlab code, the third and fourth lines could be replaced by the statement `X = 3`; to fix the taxi driver's initial state as zone 3. A similar comment applies to the R code. And, again, one could choose whether or not to store the initial state as part of the output vector.

6.2 Random Processes

In this section, we look at random processes, also called stochastic processes (“stochastic” is a synonym for “random”): time-dependent functions resulting from random phenomena.

6.2.1 Definitions and Basic Properties

Any single realization of this experiment results in a specific number, a sample value of X . We define random processes analogously.

Definition 6.2.1. For a given sample space $\mathcal{S}$ of some experiment, a **random process** is any rule that associates a time-dependent function with each outcome in $\mathcal{S}$. Any such function that may result is a **sample function** of the random process. The collection of all possible sample functions is called the **ensemble** of the random process.

Example 6.2.1 (Apple Stock Price Fluctuation). Imagine the fluctuation in the value of Apple Inc. stock (symbol: AAPL) during the next 8-h trading day, measuring time from the opening bell on Wall Street. Since that fluctuation cannot be predicted precisely, we may reasonably model the stock’s value by an appropriate random process $X(t)$; the ensemble of $X(t)$ would be subject to the constraint $X(0) = \text{yesterday’s closing value}$. Two examples of possible sample functions appear in Fig. 6.5, where we have assumed a previous day’s closing value of \$580. The ensemble of $X(t)$ consists of all possible paths that the price of Apple stock could hypothetically take tomorrow, starting at \$580 per share. Economists and statisticians use a variety of *time series* models to forecast the behaviors of such random processes.

More simply (and perhaps more usefully) put, we make the following observation: At any **fixed** time point t_0 , the ensemble of a random process $X(t)$ forms a probability distribution; that is, $X(t_0)$ is a **random variable**.

Mean and Variance Functions

At any particular time t , the random variable $X(t)$ has a probability distribution and thus has both a mean value and variance. Since $X(t)$ for fixed t is a random variable, we should be able to calculate its mean using the standard techniques. Such a mean value exists for every time t , and the mean might not be the same at every time t , i.e., the mean of $X(t)$ may vary with t . Thus, considering all values of t gives a **mean function**. Similar comments apply to the variance and standard deviation of t .

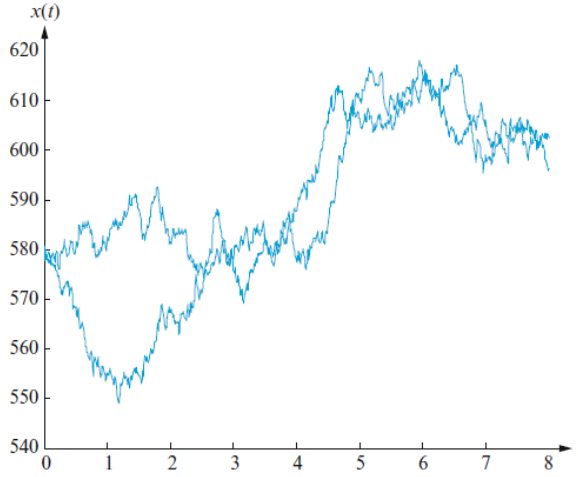


Figure 6.5: Two sample functions for a stock price's fluctuation

Definition 6.2.2. The **mean function** of a random process $X(t)$ is given by

$$\mu_X(t) = E[X(t)],$$

where $E[X(t)]$ is the expected value of the random variable $X(t)$ for the fixed time point t .

Similarly, we define the **variance function** of $X(t)$ by

$$\sigma_X^2(t) = \text{Var}(X(t)) = E[(X(t) - \mu_X(t))^2] = E[X^2(t)] - [\mu_X(t)]^2$$

and the **standard deviation function** of $X(t)$ by $\sigma_X(t) = \sqrt{\text{Var}(X(t))}$.

Notice that the mean, variance, and standard deviation functions are **nonrandom** functions of the time variable t , just as the mean, variance, and standard deviation of a random variable are numbers and not random quantities. It's vital to keep in mind that the mean function of a random process is taking an average with *respect to the ensemble* (i.e., in the “random direction”) and not with respect to time. Then we now extend the notion of covariance to a random process.

Definition 6.2.3. The **autocovariance function** of a random process $X(t)$ is defined by

$$C_{XX}(t, s) = \text{Cov}(X(t), X(s)) = E[(X(t) - \mu_X(t))(X(s) - \mu_X(s))]$$

Notice that the autocovariance function is a nonrandom function of two time points, t and s .

The Joint Distribution of Two Random Processes

Some applications involve the consideration of two random processes $X(t)$ and $Y(t)$. We may then be concerned not only with their individual distributions but also their **joint** behavior. This is especially true when $Y(t)$ is the result of some action taken on $X(t)$, such as passing the random signal $X(t)$ through an appropriate filter. To quantify their relationship, we define the **cross-covariance function** of $X(t)$ with $Y(t)$ by $C_{XY}(t, s) = \text{Cov}(X(t), Y(s))$ and the **cross-correlation function** of $X(t)$ with $Y(t)$ by $R_{XY}(t, s) = E[X(t)Y(s)]$. These two functions are, not surprisingly, connected by the formula $C_{XY}(t, s) = R_{XY}(t, s) - \mu_X(t)\mu_Y(s)$.

Definition 6.2.4. Two random processes $X(t)$ and $Y(t)$ are **independent** if, for all fixed t and s , the random variables $X(t)$ and $Y(s)$ are independent rvs as defined before. $X(t)$ and $Y(t)$ are **uncorrelated** if, for all t and s , $C_{XY}(t, s) = 0$. Finally, $X(t)$ and $Y(t)$ are **orthogonal** if $R_{XY}(t, s) = 0$ for all t and s .

Notice in these definitions that properties must hold for all times t and s . For example, the independence of $X(t)$ and $Y(t)$ requires that the random variables $X(2)$ and $Y(10)$ be independent, as must $X(2)$ and $Y(2)$ be, and so on. A similar comment applies to being uncorrelated or orthogonal.

The independence is a stronger condition than zero correlation:

$$X(t) \text{ and } Y(t) \text{ independent} \implies X(t) \text{ and } Y(t) \text{ uncorrelated,}$$

but the converse is false. If $X(t)$ and $Y(t)$ are uncorrelated, it follows from the definition of covariance that $E[X(t)Y(s)] = E[X(t)]E[Y(s)]$ for all t and s . Thus, being uncorrelated does not imply being orthogonal (nor vice versa); however, if either random process has mean identically equal to zero, then the properties of being uncorrelated and orthogonal are equivalent.

6.2.2 Stationary and Wide-Sense Stationary Processes

Firstly, we give the definitions for these two concepts:

Definition 6.2.5. A random process $X(t)$ is **(strict-sense) stationary** if all of its statistical properties are invariant with respect to time. More precisely, $X(t)$ is stationary if, for any time points $t_1, \dots, t_r$ and any value τ , the joint distribution of $X(t_1), \dots, X(t_r)$ is the same as the joint distribution of $X(t_1 + \tau), \dots, X(t_r + \tau)$.

A random process $X(t)$ is **wide-sense stationary (WSS)** if the following two conditions hold: 1. The mean function of $X(t)$, $\mu_X(t)$, is a constant. 2. The autocovariance function of $X(t)$, $C_{XX}(t, s)$, depends only on $s - t$.

Properties of Wide-Sense Stationary Processes

By definition, a WSS random process has a constant mean (function). In fact, such a process also has constant variance, because $\sigma_X^2(t) = C_{XX}(t, t)$ wide-sense stationarity requires that this covariance not depend on t . With that and our previous discussions in mind, we adopt the following notational conventions for WSS processes.

Suppose $X(t)$ is a wide-sense stationary process. Then we denote its statistical functions as follows:

$$\begin{aligned}\mu_X &= E[X(t)] \\ \sigma_X^2 &= \text{Var}(X(t)) \\ C_{XX}(\tau) &= \text{Cov}(X(t), X(t + \tau)) = R_{XX}(\tau) - \mu_X^2 \\ R_{XX}(\tau) &= E[X(t)X(t + \tau)] = C_{XX}(\tau) + \mu_X^2\end{aligned}$$

Next we present some important properties of these functions.

Proposition 6.2.1. *Let $X(t)$ be a wide-sense stationary process with autocovariance function $C_{XX}(\tau)$ and autocorrelation function $R_{XX}(\tau)$.*

Properties of $C_{XX}(\tau)$: 1. $C_{XX}(0) = E[X^2(t)] - \mu_X^2 = \sigma_X^2 = \text{Var}(X(t))$. 2. $C_{XX}(-\tau) = C_{XX}(\tau)$; that is, the autocovariance function is symmetric in τ . 3. $|C_{XX}(\tau)| \leq C_{XX}(0)$ for every τ ; that is, the autocovariance function achieves its largest value at $\tau = 0$. 4. If $X(t)$ is periodic, so is $C_{XX}(\tau)$, and with the same period. 5. If $X(t)$ is ergodic¹ and has no periodic component, then $C_{XX}(\tau) \rightarrow 0$ as $|\tau| \rightarrow \infty$.

Properties of $R_{XX}(\tau)$: 1. $R_{XX}(0) = E[X^2(t)]$, called the **mean square value** of $X(t)$. 2. $R_{XX}(-\tau) = R_{XX}(\tau)$; that is, the autocorrelation function is symmetric in τ . 3. $|R_{XX}(\tau)| \leq R_{XX}(0)$ for every τ ; that is, the autocorrelation function achieves its largest value at $\tau = 0$. 4. If $X(t)$ is periodic, so is $R_{XX}(\tau)$, and with the same period. 5. If $X(t)$ is ergodic and has no periodic component, then $R_{XX}(\tau) \rightarrow \mu_X^2$ as $|\tau| \rightarrow \infty$.

Proof. We begin with the properties of $C_{XX}(\tau)$. Property 1 follows from the covariance shortcut formula. To prove Property 2, recall that covariance is symmetric in its arguments:

$$C_{XX}(\tau) = \text{Cov}(X(t), X(t + \tau)) = \text{Cov}(X(t + \tau), X(t))$$

Because $X(t)$ is WSS, the right-most expression depends only on the difference in the two times; specifically, $\text{Cov}(X(t + \tau), X(t)) = C_{XX}(t - [t + \tau]) = C_{XX}(-\tau)$. This establishes the result.

Property 3 is left as an exercise. We note that Property 3 makes intuitive sense: since covariance measures the association between two variables, and τ

represents the time distance between these two variables, covariance should be largest when that time difference is as small as possible. That is, the behaviors of $X(t)$ and $X(t + \tau)$ should be more closely related when τ is small than when τ is large.

Properties 1–3 for the autocorrelation function follow automatically, since $R_{XX}(\tau)$ and $C_{XX}(\tau)$ only differ by the constant μ_X^2 .

Toward proving property 4, suppose $X(t)$ is periodic with period d , so $X(t) = X(t + d)$ for all t . Then, for any τ ,

$$\begin{aligned} R_{XX}(\tau + d) &= E[X(t)X(t + \tau + d)] \\ &= E[X(t)X(t + \tau)] \quad \text{because } X(t + \tau + d) = X(t + \tau) \\ &= R_{XX}(\tau) \end{aligned}$$

which shows that $R_{XX}(\tau)$ is also periodic with period d . The analogous property holds for autocovariance, because subtracting μ_X^2 to get C_{XX} from R_{XX} does not affect periodicity.

A formal proof of Property 5 is beyond the scope of this book; however, the paragraph above regarding Property 3 should give some intuition for why covariance should vanish as $|\tau| \rightarrow \infty$. Some further information about “ergodicity” appears at the end of this section. $\square$

Ergodic Processes

As we’ll see in this section, it is desirable to understand the statistical properties of a random process—mean, variance, and so on. But because these are properties of the ensemble, a problem arises: in practice, we generally only observe a single realization of the process, and so we must somehow reconstruct the ensemble properties from this one signal. Thankfully, many stationary random processes have the feature that their time and ensemble properties match (e.g., the time average of a single realization equals the ensemble mean). A process with this feature is called **ergodic**.

To give some intuition for this concept, imagine you have a large collection of identical dice. If you did not know how dice behave, you would have two options: roll all of the dice once, or roll one die many times. These should give us roughly the same information, but the first method illustrates **ensemble properties** (many realizations at one point in time) while the second gives **time properties** (one sampled die observed again and again over time). This process is ergodic. The benefit of ergodicity here is that we can learn about the behavior of all dice by purchasing (and repeatedly rolling) just a single die.

To make the property of ergodicity more precise, we need to introduce the time-dimension analogues of mean, autocorrelation, etc. For a fixed value $T > 0$,

the average of a function $x(t)$ over the interval $[-T, T]$ is defined in calculus by

$$\langle x(t) \rangle_T = \frac{1}{2T} \int_{-T}^T x(t) dt$$

By allowing T to approach infinity, we may define the **time average** of a function $x(t)$ by

$$\langle x(t) \rangle = \lim_{T \rightarrow \infty} \langle x(t) \rangle_T = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt$$

If the function $x(t)$ is periodic, the time average defined above is equal to the average of $x(t)$ across one period.

The time average of a **random process** $X(t)$ is defined by replacing $x(t)$ in the above expression with $X(t)$. The result is a quantity that clearly does not depend on time (since we have integrated dt) but which may vary across different members of the ensemble (i.e., the time average $\langle X(t) \rangle$ is still a random quantity).

Definition 6.2.6. A random process $X(t)$ is **mean ergodic** if its time average and ensemble average are the same, i.e., if

$$\langle X(t) \rangle = E[X(t)]$$

in the mean square sense.

Since the time average of $X(t)$ does not depend on t , a necessary condition for mean ergodicity is for $E[X(t)]$ to be a constant. Generally speaking, a random process must be stationary in order to be ergodic, but this is not always sufficient.

Example 6.2.2 (Mean Ergodicity of a Periodic Random Process). Let $X(t) = A_0 \cos(\omega_0 t + \Theta)$, where $\Theta \sim \text{Unif}(-\pi, \pi]$ and $\omega_0 \neq 0$. We readily find that $E[X(t)] = 0$. Now consider the time average:

$$\begin{aligned} \langle X(t) \rangle_T &= \langle A_0 \cos(\omega_0 t + \Theta) \rangle_T \\ &= \frac{1}{2T} \int_{-T}^T A_0 \cos(\omega_0 t + \Theta) dt = \frac{A_0}{2\omega_0 T} [\sin(\omega_0 T + \Theta) - \sin(-\omega_0 T + \Theta)] \end{aligned}$$

Since the term in brackets is bounded no matter the value of Θ , the factor of T in the denominator implies that $\langle X(t) \rangle_T \rightarrow 0$ as $T \rightarrow \infty$, whence $\langle X(t) \rangle = 0$. Therefore, by definition, $X(t)$ is mean ergodic.

The preceding example indicates the most common situation wherein a random process is stationary but not ergodic: some element of the process is random but not time-varying. See Example 6.2.3 below, for other such examples.

Similar to the definition of time average, we may define the **time autocorrelation** of a random process by

$$\langle X(t)X(t+\tau) \rangle = \lim_{T \rightarrow \infty} \langle X(t)X(t+\tau) \rangle_T = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T X(t)X(t+\tau) dt$$

$X(t)$ is said to be **autocorrelation ergodic** if $\langle X(t)X(t+\tau) \rangle = E[X(t)X(t+\tau)]$ in the mean square sense. Since the time autocorrelation does not depend on t , a random process can only be autocorrelation ergodic if its autocorrelation function $R_{XX}(t, t+\tau)$ is also free of t (the second condition for wide-sense stationarity).

Example 6.2.3 (Autocorrelation Ergodicity (Example 6.2.2 continued)). It is easily to get that the random process $X(t) = A_0 \cos(\omega_0 t + \Theta)$ has autocorrelation function $R_{XX}(\tau) = (A_0^2/2) \cos(\omega_0 \tau)$. Applying an appropriate trig identity, we can also find its time autocorrelation:

$$\begin{aligned} \langle X(t)X(t+\tau) \rangle_T &= \frac{1}{2T} \int_{-T}^T A_0 \cos(\omega_0 t) \cdot A_0 \cos(\omega_0(t+\tau)) dt \\ &= \frac{A_0^2}{2T} \int_{-T}^T \frac{1}{2} [\cos(\omega_0 \tau) + \cos(\omega_0(2t+\tau))] dt \\ &= \frac{A_0^2}{2T} \cdot \frac{1}{2} \cos(\omega_0 \tau) [T - (-T)] + \frac{A_0^2}{4T} \int_{-T}^T \cos(\omega_0(2t+\tau)) dt \\ &= \frac{A_0^2}{2} \cos(\omega_0 \tau) + \frac{A_0^2}{8\omega_0 T} [\cos(\omega_0(2T+\tau)) - \cos(\omega_0(-2T+\tau))] \end{aligned}$$

Taking the limit as $T \rightarrow \infty$, the second term above goes to zero, and we have $\langle X(t)X(t+\tau) \rangle = (A_0^2/2) \cos(\omega_0 \tau)$, the same as the autocorrelation function of $X(t)$. Hence, $X(t)$ is autocorrelation ergodic.

However, suppose we replace the constant amplitude A_0 with a random amplitude A , i.e., a random variable not varying with time. Then calculations show that

$$R_{XX}(\tau) = \frac{E[A^2]}{2} \cos(\omega_0 \tau) \quad \text{while} \quad \langle X(t)X(t+\tau) \rangle = \frac{A^2}{2} \cos(\omega_0 \tau)$$

These are *not* equal—that is, now $X(t)$ is *not* autocorrelation ergodic—unless it happens that $E[A^2] = A^2$ in the mean square sense, which is true iff $\text{Var}(A) = 0$.

6.2.3 Discrete-Time Random Processes

Much of what we have discussed in the previous two sections applies equally to discrete-time random processes (aka random sequences). After reviewing definitions and properties for the discrete-time case, we introduce a few specific examples of important discrete-time models.

Recall from subsection 6.2.1 that a random sequence is simply a list of random variables X_1, X_2 , and so on; we write X_n for the general term. (We may also define a sequence indexed by the entire set of integers: $\dots, X_{-2}, X_{-1}, X_0, X_1, X_2, \dots$) The subscript takes the place of the time index t from earlier. The sequence can also be denoted $X[1], X[2], \dots$ or $X[n]$ to mirror the continuous-time notation; the square brackets will remind us that we're working on a discrete-time scale.

Example 6.2.4 (Discrete-Time Stock Price Process). Let's return to the value of Apple Inc. stock, but now consider only recording the closing price at the end of each trading day (starting, say, January 2 of next year). If we define X_n to be the closing price of Apple stock on the n -th recorded day, then we can model $X_1, X_2, \dots$ as a random sequence. Figure 6.6 shows one possible sample function of this random sequence, assuming the closing price just prior to our day 1 was \$580. In effect, we have converted a continuous-time process (analog) into a discrete-time process (digital) by **sampling** the process from Example 6.2.1 at designated times.

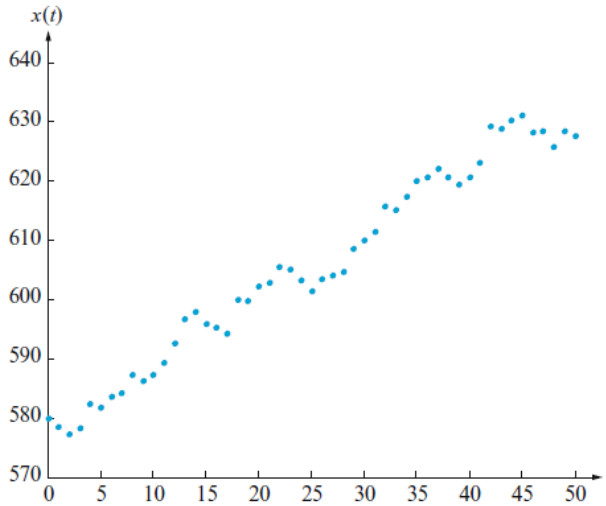


Figure 6.6: A sample function of the random sequence of Example 6.2.4

For any random sequence, we can define several statistical functions for times $n = 1, 2, 3, \dots$ as follows:

- Mean function: $\mu_X[n] = E[X_n]$
- Variance function: $\sigma_X^2[n] = \text{Var}(X_n)$

- Standard deviation function: $\sigma_X[n] = \sqrt{\text{Var}(X_n)}$
- Autocovariance function: $C_{XX}[n, m] = \text{Cov}(X_n, X_m)$
- Autocorrelation function $R_{XX}[n, m] = E[X_n X_m]$

The relationships between these functions established in the continuous-time case still hold, e.g., $\sigma_X^2[n] = C_{XX}[n, n]$, and

$$C_{XX}[n, m] = R_{XX}[n, m] - \mu_X[n]\mu_X[m].$$

A random sequence is (**strict-sense**) **stationary** if the joint distribution of $X[n_1], \dots, X[n_r]$ equals the joint distribution of $X[n_1 + k], \dots, X[n_r + k]$ for any indices $n_1, \dots, n_r$ and any integer k . A random sequence is **wide-sense stationary** if (1) $\mu_X[n]$ is a constant, μ_X , and (2) $C_{XX}[n, m]$ depends only on the difference $m - n$; if we call this difference k , we may then denote the autocovariance function as $C_{XX}[k]$. As was true for continuous-time processes, we may make an equivalent definition regarding the mean and autocorrelation functions.

Special Discrete Sequences

Perhaps the simplest type of random sequence is the **Bernoulli random sequence**: let $X_1, X_2, \dots$ be iid, with each X_n following a Bernoulli(p) distribution. A sample function of a Bernoulli random sequence with $p = .6$ appears in Fig. 6.7. By grace of the variables being iid, a Bernoulli random sequence is trivially (strict-sense) stationary; in particular, we have

$$\mu_X[n] = E[X_n] = p, \sigma_X^2[n] = \text{Var}(X_n) = p(1 - p), \text{ and}$$

$$C_{XX}[n, m] = \text{Cov}(X_n, X_m) = \begin{cases} p(1 - p) & m = n \\ 0 & m \neq n \end{cases}$$

A more general iid sequence will also be strict-sense stationary, but the formulas for the mean and variance will, of course, depend on the underlying common distribution of the X_n .

From an iid sequence, we can construct a much more interesting random sequence as follows: define $S_1 = X_1, S_2 = X_1 + X_2$, and so on, so that $S_n = S_{n-1} + X_n = \sum_{i=1}^n X_i$ for all $n \geq 2$. The resulting sequence of partial sums $S_1, S_2, S_3, \dots$ is called a **random walk**. For example, we know the sum of iid Bernoulli rvs is binomial; hence, if X_n represents a Bernoulli random sequence, then the corresponding random walk S_n has a $\text{Bin}(n, p)$ distribution for each n .

We can use the properties of iid sums to derive some general properties of random walks.

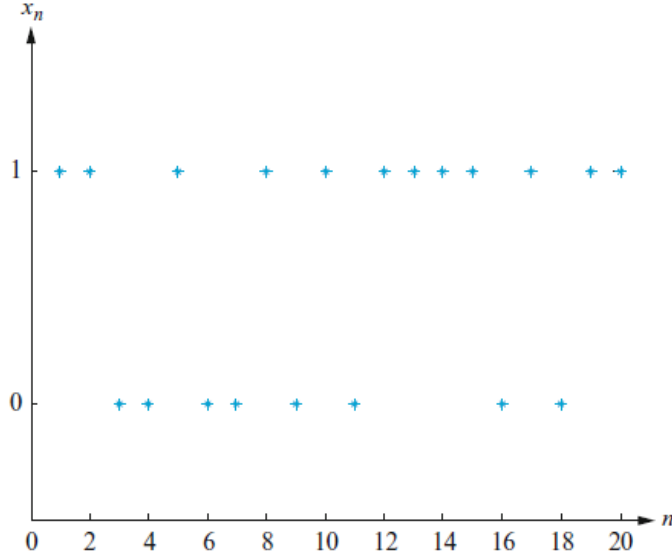


Figure 6.7: A sample function of a Bernoulli random sequence

Proposition 6.2.2. *Let $X_1, X_2, \dots$ be an iid sequence with common mean μ_X and common variance σ_X^2 . Let S_n be the associated random walk, i.e., $S_n = X_1 + \dots + X_n$ for every n . Then*

- 1 . $\mu_S[n] = E[S_n] = n\mu_X$
- 2 . $\sigma_S^2[n] = \text{Var}(S_n) = n\sigma_X^2$
- 3 . $C_{SS}[n, m] = \min(n, m)\sigma_X^2$

Proof. The proofs of properties 1 and 2 were given in advance of the Central Limit Theorem. To prove property 3, assume $m > n$ and proceed as follows:

$$\begin{aligned}
 C_{SS}[n, m] &= \text{Cov}(S_n, S_m) = \text{Cov}(X_1 + \dots + X_n, X_1 + \dots + X_m) \\
 &= \text{Cov}(X_1 + \dots + X_n, X_1 + \dots + X_n + X_{n+1} + \dots + X_m) \\
 &= \text{Cov}(X_1 + \dots + X_n, X_1 + \dots + X_n) \\
 &\quad + \text{Cov}(X_1 + \dots + X_n, X_{n+1} + \dots + X_m) \\
 &= \text{Cov}(S_n, S_n) + \text{Cov}(X_1 + \dots + X_n, X_{n+1} + \dots + X_m)
 \end{aligned}$$

In the third line, we have used the distributive property of covariance. The first term, $\text{Cov}(S_n, S_n)$, is simply $\text{Var}(S_n)$ (the covariance of S_n with itself). In the second term, the X_i s ($i = 1$ to n) in the first argument are independent of the X_j s ($j = n + 1$ to m) in the second argument; therefore, that covariance equals zero. Thus, we have $C_{SS}[n, m] = \text{Var}(S_n) = n\sigma_X^2$.

This holds for $m > n$; if $m < n$, the same argument yields $C_{SS}[n, m] = m\sigma_X^2$. Therefore, for general n and m we may write $C_{SS}[n, m] = \min(n, m)\sigma_X^2$. $\square$

6.2.4 Poisson Processes

In the previous studies, we indicated that, under rather general conditions, the Poisson distribution furnishes a probability model for the number of events of some sort (logins to a server, arrivals of radioactive pulses, flaws on the surface of a wafer, etc.) that occur in some fixed interval of time or region of space. We now present a more formal development of conditions that lead to the Poisson distribution in such contexts, and then explore properties of this event process.

Definition 6.2.7. Consider the experiment of observing randomly occurring events of some type continuously over time. Define $X(0) = 0$, and let $X(t)$ for $t > 0$ to be the number of events that occur in the time interval $(0, t]$. $X(t)$ is called a **Poisson (counting) process** if it satisfies the following two conditions:

- 1 The numbers of events occurring in nonoverlapping time intervals are independent.
- 2 There exists a parameter $\lambda > 0$, called the **rate** of the process, such that the number of events occurring in any interval of length τ has a Poisson distribution with mean $\lambda\tau$.

Later in this section, we present an alternative definition of a Poisson counting process which does not explicitly assume that the event count follows a Poisson distribution.

Condition 1 states that a Poisson counting process $X(t)$ has **independent increments**: if $(s_1, t_1]$ and $(s_2, t_2]$ are two time intervals with $t_1 \leq s_2$, so that the intervals do not overlap, then the number of events that occur in the first interval is independent of the number of events that occur in the second interval. That is, the “increment” $X(t_1) - X(s_1)$ is independent of $X(t_2) - X(s_2)$.

Condition 2 states that for any $t > 0$ and $\tau > 0$, the number of events in the interval $(t, t + \tau]$ has a Poisson distribution with mean $\lambda\tau$. Since this count is represented by $X(t + \tau) - X(t)$, we may write $X(t + \tau) - X(t) \sim \text{Poisson}(\lambda\tau)$. Since the distribution of this “increment” depends only on τ and not t , we say that a Poisson counting process has **stationary increments**. By substituting

$t = 0$ and $\tau = t$ into this expression, we have that $X(t) \sim \text{Poisson}(\lambda t)$, so that at each time t the process itself has a Poisson distribution. It follows that

$$\mu_X(t) = \lambda t \quad \text{and} \quad \sigma_X^2(t) = \lambda t \quad (\text{since mean and variance are equal for Poisson})$$

Theorem 6.2.1. *Suppose events occur in accordance with the conditions of a Poisson counting process. Define T_1 = the time until the first event occurs and, for $n \geq 2$, T_n = the time between the occurrence of the $(n-1)$ -th and n -th events. Also, define Y_n = the time until the n -th event occurs, starting at $t = 0$. Then*

1 $T_1, T_2, \dots$ are independent exponential random variables with parameter λ (mean $1/\lambda$) and

2 Y_n is a gamma random variable with parameters $\alpha = n$ and $\beta = 1/\lambda$ (aka the Erlang distribution).

Proof. Since the time intervals spanned by T_1, T_2 , etc. do not overlap, the T_n are independent by condition 1 of a Poisson process (independent increments). To find the distribution of T_1 , start with its cdf: for $t > 0$,

$$\begin{aligned} F_{T_1}(t) &= P(T_1 \leq t) = 1 - P(T_1 > t) \\ &= 1 - P(\text{no events occur in the time interval } (0, t]) \\ &= 1 - P(X(t) = 0) \quad \text{where } X(t) \sim \text{Poisson}(\lambda t) \\ &= 1 - \frac{e^{-\lambda t}(\lambda t)^0}{0!} = 1 - e^{-\lambda t} \implies \\ f_{T_1}(t) &= \frac{d}{dt} F_{T_1}(t) = \frac{d}{dt} [1 - e^{-\lambda t}] = \lambda e^{-\lambda t} \end{aligned}$$

This is the $\text{exponential}(\lambda)$ pdf, as claimed. The distribution of T_2 is also $\text{exponential}(\lambda)$ because, thanks to Condition 1 of the definition, we may “restart the clock” when the first event occurs and derive the pdf of T_2 in the exact same manner as above. Propagating this idea forward, we have that $T_n \sim \text{exponential}(\lambda)$ for all n .

As for Y_n , notice we may write $Y_n = T_1 + \dots + T_n$, which implies Y_n is the sum of n iid $\text{exponential}(\lambda)$ rvs. Using moment-generating functions, that the sum of n iid $\text{exponential}(\lambda)$ rvs has a $\text{gamma}(n, 1/\lambda)$ distribution. $\square$

We know that the Poisson distribution is additive, i.e., the sum of two independent Poisson rvs is again Poisson distributed. This result immediately generalizes to Poisson counting processes.

Proposition 6.2.3. *Let $X_1(t)$ and $X_2(t)$ be independent Poisson processes with rate parameters λ_1 and λ_2 , respectively. Define a new random process by $X(t) =$*

$X_1(t) + X_2(t)$. Then $X(t)$ is also a Poisson process, with rate parameter $\lambda_1 + \lambda_2$. (This theorem can be extended to the sum of k independent Poisson processes for $k > 2$ as well.)

The foregoing proposition shows that we can combine separate Poisson processes into a single Poisson process. Interestingly, we can also do the reverse: if we can categorize the events of a Poisson process (e.g., arrivals of people separated into women's arrivals and men's arrivals), then we can decompose the overall process into two "smaller" processes. We make this more precise in the next proposition.

Proposition 6.2.4. *Suppose events occur according to the conditions of a Poisson process, and that each event can be classified as either Type 1 or Type 2. Suppose that each event is Type 1 with probability p , independent of the types of all other events and independent of the number of events that have occurred. Define two random processes: $X_1(t)$ = number of Type 1 events up to time t , and $X_2(t)$ = number of Type 2 events up to time t . Then*

- 1 $X_1(t)$ is a Poisson process with rate parameter $p\lambda$;
- 2 $X_2(t)$ is a Poisson process with rate parameter $(1 - p)\lambda$; and
- 3 $X_1(t)$ and $X_2(t)$ are independent.

Proof. We will derive the joint distribution of $X_1(t)$ and $X_2(t)$, i.e., $P(X_1(t) = x \text{ and } X_2(t) = y)$, for arbitrary nonnegative integers x and y . The event $\{X_1(t) = x \text{ and } X_2(t) = y\}$ is equivalent to the event

$$\{X(t) = x + y \text{ and exactly } x \text{ of these } x + y \text{ events are Type 1}\} = A \cap B$$

where $X(t)$ denotes the overall Poisson process. The second part of this event follows a binomial model: we have a fixed number of trials ($x + y$), each with two basic outcomes (Type 1 or Type 2), plus independent trials and constant probability by assumption. Combining that with the known Poisson distribution of $X(t)$ and the Multiplication Rule $P(A \cap B) = P(A)P(B|A)$ gives

$$\begin{aligned} & P(X(t) = x + y \text{ and exactly } x \text{ of these } x + y \text{ events are Type 1}) \\ &= \frac{e^{-\lambda t} (\lambda t)^{x+y}}{(x+y)!} \cdot \binom{x+y}{x} p^x (1-p)^y = \frac{e^{-\lambda t} (\lambda t)^{x+y} p^x (1-p)^y}{x! y!} \\ &= \frac{e^{-[p+(1-p)]\lambda t} (\lambda t)^{x+y} p^x (1-p)^y}{x! y!} = \frac{e^{-p\lambda t} (p\lambda t)^x}{x!} \cdot \frac{e^{-(1-p)\lambda t} ((1-p)\lambda t)^y}{y!} \end{aligned}$$

We recognize these two functions as the pmfs of a Poisson($p\lambda t$) distribution and a Poisson($(1-p)\lambda t$) distribution, respectively. Moreover, since the joint pmf of $X_1(t)$ and $X_2(t)$ separates into the product of individual pmfs, $X_1(t)$ and $X_2(t)$ are independent. $\square$

The definition of a Poisson process at the beginning of this section almost seems a tautology, since we said $X(t)$ is a Poisson process if it has a Poisson distribution. The following theorem provides an alternative way to define a Poisson counting process.

Theorem 6.2.2. *Consider the experiment of observing randomly occurring events of some type along continuous time. Let $X(0) = 0$, and let $X(t)$ for $t > 0$ to be the number of events that occur in the time interval $(0, t]$. Suppose that $X(t)$ has the following properties:*

- 1 $X(t)$ has independent and stationary increments.
- 2 There exists $\lambda > 0$ such that, for any time interval of length h , the probability that exactly one event occurs is $\lambda h + o(h)$.
- 3 The probability of more than one event occurring in an interval of length h is $o(h)$.

Then $X(t)$ is a Poisson counting process with rate parameter λ .

Proof. Because of the stationarity assumption, it suffices to consider a time interval beginning at time 0. Let $P_k(t)$ denote the probability that exactly k events occur in the interval $[0, t]$. First consider $P_0(t+h)$, the probability that no events occur during the first $t+h$ units of time. In order for this to happen, no events must occur in $[0, t]$ and also no events must occur during the next h units of time. Since these two time intervals are nonoverlapping, the number of events that occur in the first interval is independent of the number that occur in the second interval. Thus

$$\begin{aligned} P_0(t+h) &= P_0(t) \cdot P(\text{no events in an interval of length } h) \\ &= P_0(t) \cdot [1 - P(\text{exactly one event}) - P(\text{at least two events})] \\ &= P_0(t) \cdot [1 - (\lambda h + o(h)) - o(h)] \\ &= P_0(t) \cdot [1 - \lambda h - o(h) - o(h)] = P_0(t) \cdot [1 - \lambda h + o(h)] \end{aligned}$$

Rearranging this expression gives

$$\frac{P_0(t+h) - P_0(t)}{h} = -\lambda P_0(t) + \frac{o(h)}{h}$$

Now taking the limit as $h \rightarrow 0$ gives the derivative of $P_0(t)$:

$$P'_0(t) = -\lambda P_0(t)$$

This differential equation has the unique solution $P_0(t) = ce^{-\lambda t}$, where the constant c is determined by the initial condition $P_0(0) = 1$. This implies that

$c = 1$ and thus that $P_0(t) = e^{-\lambda t}$, which sure enough is the probability for no events when the distribution is Poisson with parameter λt .

Now consider the probability of general k . In order to have k events occur in the interval $[0, t + h]$, it must be the case that either (1) k events occur in $[0, t]$ and none in the next h time units, or (2) $k - 1$ events occur in $[0, t]$ and one occurs in the next h time units, or (3) For $l \geq 2$, $k - l$ occur in $[0, t]$ and l occur in the next h time units. By condition 3 in the theorem, the probability of the event in (3) is $o(h)$. Writing $P_k(t + h)$ as a sum of probabilities corresponding to cases (1), (2), and (3), rearranging, dividing by h , and taking the limit as $h \rightarrow 0$ gives the following system of differential equations:

$$P'_k(t) = -\lambda P_k(t) + \lambda P_{k-1}(t) \quad k = 1, 2, 3, \dots$$

Letting $Q_k(t) = P_k(t)e^{\lambda t}$, the above differential equation becomes $Q'_k(t) = \lambda Q_{k-1}(t)$. This system can be solved recursively starting with $Q_0(t) = 1$ (because $P_0(t) = e^{-\lambda t}$) to give $Q_k(t) = \lambda^k t^k / k!$, whence

$$P_k(t) = \frac{(\lambda t)^k e^{-\lambda t}}{k!} \quad k = 0, 1, 2, \dots$$

□

6.2.5 Gaussian Processes

We have introduced the normal or Gaussian distribution before. Here, we consider the extension of the normal distribution to random processes. Engineers commonly use such models for noise in audio signals and the (seemingly) random motion of small particles. We'll explore both of these applications shortly.

Definition 6.2.8. A random process $X(t)$ is a **Gaussian process** if for all time points, $t_1, \dots, t_n$ the random variables $X(t_1), \dots, X(t_n)$ have a multivariate normal distribution. In particular, the distribution of $X(t)$ at any time point t is normal.

As discussed before, we can also characterize a joint Gaussian distribution by requiring that all linear combinations of the random variables be Gaussian. Applying that characterization here, we have an alternate definition of a Gaussian process: $X(t)$ is a Gaussian process iff all linear combinations of $X(t_1), \dots, X(t_n)$ have a normal distribution for $n = 1, 2, 3, \dots$, and all time-points $t_1, \dots, t_n$.

In subsection 6.2.2, we distinguished strict-sense stationary processes from wide-sense stationary processes. We noted that a strict-sense stationary process is automatically WSS, but not vice versa. However, the opposite is true for a Gaussian process $X(t)$ is WSS: this implies the mean and covariance structure

of $X(t)$ are time-invariant. But we know that mean and covariance completely characterize a joint Gaussian distribution; all other statistical properties can be derived from these two. Thus, if a Gaussian process is WSS, *all* its statistical properties must be time-invariant.

Proposition 6.2.5. *Suppose $X(t)$ is a Gaussian process. Then $X(t)$ is wide-sense stationary if, and only if, $X(t)$ is strict-sense stationary.*

Brownian Motion

In Example 7.4, we introduced the idea of Brownian motion, a model for the seemingly random behavior of a dust particle on a liquid surface. Physicists also use the Brownian motion model to describe a variety of physical processes, including the motion of some celestial bodies in response to gravitational forces. A precise mathematical construction of Brownian motion is beyond the scope of this book; however, we characterize Brownian motion in the following definition.

Definition 6.2.9. A (one-dimensional) **Brownian motion process** (also called a **Wiener process**) with parameter $\alpha > 0$, denoted $B(t)$, is a Gaussian process with the following properties:

- 1 . $\mu_B(t) \equiv 0$; that is, Brownian motion is a mean-zero process.
- 2 . $\sigma_B^2(t) = \alpha t$, so $\sigma_B(t) = \sqrt{\alpha t}$.
- 3 . $B(t)$ has stationary and independent increments.

If $\alpha = 1$, $B(t)$ is called **standard Brownian motion**.

Brownian Motion as a Limit

The Brownian motion process described above can actually be constructed as the limit of a discrete-time random process. We will shrink both the time increment and the size of a jump in this random walk as follows: for some $h > 0$ and $\Delta x > 0$, suppose at times $h, 2h, 3h$, etc. the walk moves $+\Delta x$ or $-\Delta x$ with probability .5 each. Then, with $\lfloor \cdot \rfloor$ denoting the greatest-integer function, the random process $B(t)$ defined by

$$B(t) = (\Delta x)X_1 + \cdots + (\Delta x)X_{\lfloor t/h \rfloor} = (\Delta x)S_{\lfloor t/h \rfloor}$$

indicates the location of the random walk at time t . The coefficient Δx changes the motion increment from ± 1 to $\pm \Delta x$; the time index $n = \lfloor t/h \rfloor$ equals the number of moves (equivalently, the number of h -second time intervals) in the interval $[0, t]$.

From the properties of the random walk,

$$\begin{aligned}\mu_B(t) &= E[(\Delta x)S_{\lfloor t/h \rfloor}] = (\Delta x)E[S_{\lfloor t/h \rfloor}] \\ &= (\Delta x) \cdot 0 = 0\end{aligned}$$

and

$$\begin{aligned}\sigma_B^2(t) &= \text{Var}((\Delta x)S_{\lfloor t/h \rfloor}) = (\Delta x)^2 \text{Var}(S_{\lfloor t/h \rfloor}) \\ &= (\Delta x)^2 \cdot \lfloor t/h \rfloor\end{aligned}$$

Moreover, the Central Limit Theorem tells us that, for large values of $\lfloor t/h \rfloor$, the distribution of $B(t)$ is approximately normal.

Up to now, the choices of h and Δx have been arbitrary. But suppose we choose $\Delta x = \sqrt{\alpha h}$ for some $\alpha > 0$. Then, if we shrink $h \rightarrow 0$ (effectively moving from discrete time to continuous time), $B(t)$ will be normally distributed with mean 0 and variance

$$\lim_{h \rightarrow 0} \alpha h \cdot \lfloor t/h \rfloor = \alpha t$$

The properties of independent and stationary increments clearly follow as consequences of the iid steps in the random walk. Thus, $B(t)$ becomes a Brownian motion process as $h \rightarrow 0$.

Further Properties of Brownian Motion

Proposition 6.2.6. *Let $B(t)$ be Brownian motion and, for $t_0 > 0$, let $M = \max_{0 \leq t \leq t_0} B(t)$. Then*

$$P(M > x_0) = 2P(B(t_0) > x_0) = 2 \left[1 - \Phi \left(\frac{x_0}{\sqrt{\alpha t_0}} \right) \right] \quad (4)$$

The second equality in the proposition comes from the fact that $B(t_0) \sim N(0, \sqrt{\alpha t_0})$. Replacing x_0 with m on both sides of Eq. (4) and differentiating with respect to m , we can determine the pdf of this rv:

$$f_M(m) = \frac{2}{\sqrt{2\pi\alpha t_0}} e^{-m^2/(2\alpha t_0)} \quad m > 0$$

The foregoing proposition also allows us to obtain the distribution of the random variable T = the first time which the process hits level x_0 . To see this, note that a sample path will first hit level x_0 by time t_0 iff the maximum level of the path during the time interval $[0, t_0]$ is at least x_0 . In symbols, $T \leq t_0$ iff $M \geq x_0$. Since the probability of the latter event is what appears in the last proposition box, we immediately have the following result.

Proposition 6.2.7. *Let T be the first time that a Brownian motion process reaches level x_0 . Then*

$$F_T(t) = P(T \leq t) = 2 \left[1 - \Phi \left(\frac{x_0}{\sqrt{\alpha t}} \right) \right],$$

from which it follows that

$$f_T(t) = \frac{x_0}{\sqrt{2\pi\alpha}} t^{-3/2} e^{-x_0^2/(2\alpha t)} \quad t > 0$$

Variations on Brownian Motion

Let $B(t)$ denote a standard Brownian motion process (i.e., with $\alpha = 1$), and let μ and $\sigma > 0$ be constants. **Brownian motion with drift** is the process $X(t) = \mu t + \sigma B(t)$. $X(t)$ also has stationary and independent increments; an increment $X(t+\tau) - X(t)$ is normally distributed with mean $\mu\tau$ and variance $\sigma^2\tau$. Brownian motion with drift has many interesting applications and properties. For example, suppose the drift parameter μ is negative. Then over time $X(t)$ will tend toward ever lower values. It can be shown that M , the maximum level attained over all time $t \geq 0$, has an exponential distribution with parameter $2|\mu|/\sigma^2$.

Standard Brownian motion and Brownian motion with drift both allow for positive and negative values of the process. Thus they are not typically acceptable models for the behavior of the price of some asset over time. A stochastic process $Z(t)$ is called **geometric Brownian motion with drift parameter α** if $X(t) = \ln[Z(t)]$ is a Brownian motion process with drift having mean parameter $\mu = \alpha - \sigma^2/2$ and standard deviation parameter σ . Since $Z(t) = \exp(X(t))$, a geometric Brownian motion process will be nonnegative. Any particular sample path will show random fluctuations about a long-term exponential decay or growth curve.

Geometric Brownian motion is a popular model for the pricing of assets. Let $X(t)$ be the price of an asset at time t . The ratio $X(t)/X(0)$ is the proportion by which the price has increased or decreased between time 0 and time t . In the same way that we obtained Brownian motion as a limit of a simple symmetric random walk, geometric Brownian motion can be obtained as a limit in which the price at each time point either increases by a multiplicative factor u or goes down by another particular multiplicative factor d . The limit is taken as the number of price changes during $(0, t]$ gets arbitrarily large, while the factors u and d get closer to 1 and the two probabilities associated with u and d approach .5. Geometric Brownian motion is the basis for the famous Black-Scholes option pricing formula that is used extensively in quantitative finance. This formula specifies a fair price for a contract allowing an investor to purchase an asset at some future time point for a particular price (e.g., a contract permitting an investor to purchase 100 shares of Facebook stock at a price of \$20 per share 3 months from now).

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